Microeconometrics

Matthew Tan

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Contents

1 Basic Asymptotic Theory

1.1 Scalar Limit Theorems

Definition 1.1 (Convergence in Probability). A sequence of random variables $\{X_n\}$ converges in probability to a random variable X if for all $\varepsilon > 0$,

$$\lim_{n\to\infty} \mathrm{P}(|X_n - X| < \varepsilon) = 1$$

Alternatively, we can represent convergence in probability as for any $c > 0$

$$\lim_{n\to\infty} \mathrm{P}(|X_n - X| > c) = 0$$

Remark. Note that a constant is a special case of a random variable with probability 1

Theorem 1.2 (Weak Law of Large Numbers). For $i = 1, \ldots, n$, let x_i be identically and independently distributed with finite mean μ and variance σ^2. Then, as $n \to \infty$,

$$\bar{x}_n = \frac{1}{n}\sum_{i=1}^{n} x_i \xrightarrow{p} \mu$$

Proof. Define $\bar{X}_n := \frac{1}{n}\sum_{i=1}^{n} X_i$. Then, by the assumption of finite variance and the fact that $\{X_n\}$ is iid

$$\mathrm{var}(\bar{X}_n) = \frac{1}{n^2} n\mathrm{var}(X_1) = \frac{\sigma^2}{n}$$

Since $\mathrm{E}[\bar{X}_n] = \mu$, by Chebyshev's inequality we have

$$\mathrm{P}(|\bar{X}_n - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2}$$

Then, we have

$$\mathrm{P}(|\bar{X}_n - \mu| < \varepsilon) = 1 - \mathrm{P}(|\bar{X}_n - \mu| \geq \varepsilon) \leq 1 - \frac{\sigma^2}{n\varepsilon^2}$$

Taking the limit

$$\lim_{n\to\infty} \mathrm{P}(|\bar{X}_n - \mu| < \varepsilon) \geq 1$$

and the fact that for any X, $\mathrm{P}(X) \leq 1$, yields

$$\lim_{n\to\infty} \mathrm{P}(|\bar{X}_n - \mu| < \varepsilon) = 1$$

■

Definition 1.3 (Convergence in Distribution). We say that $\bar{x}_n \xrightarrow{d} \mathcal{N}(\mu, \sigma^2)$ i.e. $\bar{x}_n$ converges to a normal distribution just when

$$\lim_{n\to\infty} F_n(\bar{x}_n) = \Phi(x)$$

where $\Phi(x)$ is the normal CDF

Remark. In most cases, $\Phi(x)$ is used to denote the standard normal CDF

Proposition 1.4 (Convergence in probability implies convergence in distribution). Let $\{X_n\}$ be a sequence of random variables. If $X_n \xrightarrow{p} X$ then $X_n \xrightarrow{d} X$

Theorem 1.5 (Central Limit Theorem (Lindenberg-Levy)). Suppose $X_1, X_2, X_3, \ldots$ is a sequence of identically and independently distributed random variables with finite mean μ and variance σ^2. Then, as $n \to \infty$, the random variables $\sqrt{n}(\bar{X}_n - \mu)$ converge in distribution to the normal $\mathcal{N}(0, \sigma^2)$:

$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \xrightarrow{d} \mathcal{N}(0, 1)$$

where $\bar{X}_n := \frac{1}{n}\sum_{i=1}^{n} X_n$ i.e. the sample mean

To prove this we first need to use moment generating functions which determine the distribution

Definition 1.6 (Moment Generating Function). The moment generating function of an r.v. X is $M(t) = \mathrm{E}(e^{tX})$, as a function of t, if this is finite on some open interval or neighbourhood $(-a, a)$ containing 0. Otherwise, we say that the MGF of X does not exist.

Remark. There is a more general form of the MGF known as the characteristic function of a random variable defined as

$$\mathrm{E}(e^{i\lambda x}) = \int_{-\infty}^{\infty} e^{i\lambda x} d\mathrm{F}(x)$$

Lemma 1.7. Let $M_X(t)$ and $M_Y(t)$ be the moment generating functions of random variables X and Y respectively. If both moment generating functions exist, and $M_X(t) = M_Y(t)$ for all values of t, then X and Y have the same distribution

Then, we begin the proof of the main claim

Proof. Suppose that the MGF of X_j exists (which will be true if it has some known distribution). WLOG, let $M(t) = \mathrm{E}(e^{tX_j})$ and let $\mathrm{E}(X_j) = \mathrm{E}(\bar{X}_n) = 0$ and $\mathrm{Var}(X_j) = 1$ (since we standardise $\bar{X}_n$ later). Then, $M(0) = 1, M'(0) = \mathrm{E}(X_j) = 0, M''(0) = \mathrm{Var}(X_j) = 1$. We want to show that the MGF of $\frac{\bar{X}_n - \mathrm{E}(\bar{X}_n)}{\sqrt{\mathrm{Var}(\bar{X}_n}}$ converges to the MGF of $\mathcal{N}(0,1)$ i.e. $e^{\frac{t^2}{2}}$. By the properties of MGF, we have

$$\mathrm{E}(e^{t(X_1+\cdots+X_n)/\sqrt{n}}) = \mathrm{E}(e^{\frac{tX_1}{\sqrt{n}}} \cdots \mathrm{E}(e^{\frac{tX_n}{\sqrt{n}}} = \left(M(\frac{t}{\sqrt{n}}\right)^n$$

Because $\lim_{n\to\infty} \left(M(\frac{t}{\sqrt{n}}\right)^n$ diverges, we first take the natural log of the MGF. So,

$$\begin{aligned}
\lim_{n\to\infty} \ln M(\frac{t}{\sqrt{n}}) &= \lim_{y\to 0} \frac{\ln M(yt)}{y^2}, && \text{where } y = \frac{1}{\sqrt{n}} \\
&= \lim_{y\to 0} \frac{tM'(yt)}{2yM(yt)}, && \text{by L'Hopital's rule} \\
&= \frac{t}{2} \lim_{y\to 0} \frac{M'(yt)}{y}, && \text{since } M(yt) \to 1 \\
&= \frac{t^2}{2} \lim_{y\to 0} M''(yt), && \text{by L'Hopital's rule} \\
&= \frac{t^2}{2}, && \text{since } M''(yt) \to 1
\end{aligned}$$

Therefore, $\left(M(\frac{t}{\sqrt{n}}\right)^n \to e^{\frac{t^2}{n}}$ as desired.

Since the MGF determines the distribution by the lemma, $\frac{\bar{X}_n - \mathrm{E}(\bar{X}_n)}{\sqrt{\mathrm{Var}(\bar{X}_n}} \xrightarrow{d} \mathcal{N}(0,1)$. ■

The below is useful in some cases where we don't have *identically distributed* random variables, such as under heteroskedasticity.

Definition 1.8 (Mean Square Convergence). Consider a sequence of random variables $\{X_n\}$. Assuming $\mathrm{E}[X^2]$ and $\mathrm{E}[X_n^2]$ exist, then X_n converges to X in mean square if

$$\lim_{n\to\infty} \mathrm{E}[(X_n - X)^2] = 0$$

Remark. Specifically, when we're considering a sample mean $\bar{X}_n$, we want to show that

$$\lim_{n\to\infty} \mathrm{E}[(\bar{X}_n - \mu)^2] = 0$$

where $\mu = \mathrm{E}[X]$. Notice then that in the case of an estimator $\hat{\beta}$, to establish

$$\lim_{n\to\infty} \mathrm{E}[(\hat{\beta} - \beta)^2] = 0$$

it suffices to show

- Unbiasedness: $\mathrm{E}[\hat{\beta}] = \beta$
- Variance tends to zero: $\lim_{n\to\infty} \mathrm{Var}(\hat{\beta}) = 0$

This is intuitively obvious because $\mathrm{Var}(\hat{\beta}) = \mathrm{E}[(\hat{\beta} - \mathrm{E}[\hat{\beta}])^2] = \mathrm{E}(\hat{\beta} - \beta)^2$ by unbiasedness

Proposition 1.9 (Convergence in Mean Square implies Convergence in Probability). Suppose

$$\lim_{n\to\infty} \mathrm{E}[(\hat{\beta} - \beta)^2] = 0$$

Then, $\hat{\beta} \xrightarrow{p} \beta$

Proof. By Chebyshev's Inequality (or in fact Markov's inequality), write

$$\mathrm{P}(|\hat{\beta} - \beta| > \varepsilon) = \mathrm{P}((\hat{\beta} - \beta)^2 > \varepsilon^2) \leq \frac{\mathrm{E}[(\hat{\beta} - \beta)^2]}{\varepsilon^2}$$

Then, taking the limits both sides gives us

$$\lim_{n\to\infty} \mathrm{P}(|\hat{\beta} - \beta| > \varepsilon) \leq \lim_{n\to\infty} \frac{\mathrm{E}[(\hat{\beta} - \beta)^2]}{\varepsilon^2} = 0$$

By the definition of a probability measure,

$$\lim_{n\to\infty} \mathrm{P}(|\hat{\beta} - \beta| > \varepsilon) = 0$$

Hence, $\hat{\beta} \xrightarrow{p} \beta$ ■

The proof here is quite instructive since we're not technically taught MSC convergence and so its easy to show convergence in probability from it.

1.2 New Limit Theorems

Theorem 1.10 (Slutsky Theorem). Suppose Y, X are (scalar or vector-valued) random variables. Let $Y_n \xrightarrow{p} c$ and $X_n \xrightarrow{d} X$, then:

(a) $Y_n + X_n \xrightarrow{d} c + X$ (i.e. $c + X_n$ has the same asymptotic distribution as $Y_n + X_n$)

(b) $Y_n X_n \xrightarrow{d} cX$ (i.e. cX_n has the same asymptotic distribution as $Y_n X_n$)

(c) $Y_n^{-1} X_n \xrightarrow{d} c^{-1} X$ if $c \neq 0$

(d) if $c = 0$, then $Y_n X_n \xrightarrow{p} 0$

Remark. Notice that (a) and (b) imply (c) and (d), and so (a) and (b) are the crux of Slutsky's Theorem. For (d), this is because converging in distribution to a constant just is converging in probability to that constant.
Furthermore, everything holds if we replace d with p i.e. convergence in distribution with convergence in probability. NOTE: THIS DOES NOT MEAN THAT CONVERGENCE IN PROBABILITY IMPLIES CONVERGENCE IN DISTRIBUTION

Theorem 1.11 (The Continuous Mapping Theorem). Suppose Y, X are (scalar or vector-valued) random variables. Let g be a continuous function, then:

(a) if $Y_n \xrightarrow{a} c$, then $g(Y_n) \xrightarrow{a} g(c)$

(b) if $X_n \xrightarrow{d} X$, then $g(X_n) \xrightarrow{d} g(X)$

Theorem 1.12 (Multivariate Central Limit Theorem). Let Z_i for $i, \dots, n$ be independent and identically distributed m-dimensional random vectors with finite mean vector $\mu_Z = \mathrm{E}(Z_i)$ and finite positive definite covariance matrix $\Sigma_Z = \mathrm{E}\{(Z_i - \mu_Z)(Z_i - \mu_Z)'\}$. Then,

$$\sqrt{n}(\bar{Z}_n - \mu_Z) \xrightarrow{d} \mathcal{N}(\mathbf{0}_m, \Sigma_Z)$$

where $\bar{Z}_n = n^{-1} \sum_{i=1}^{n} Z_i$ and $\mathcal{N}(\mathbf{0}_m, \Sigma_Z)$ is multivariate normal.

MIGHT BE USEFUL BUT OPTIONAL

Theorem 1.13 (δ-method (Single Variable)). If $\sqrt{n}(Z_n - \mu) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$ and if $g(Z_n)$ is a continuous and continuously differentiable function with $g'(\mu) \neq 0$ and not involving n, then

$$\sqrt{n}(g(Z_n) - g(\mu)) \xrightarrow{d} \mathcal{N}(0, \{g'(\mu)\}^2 \sigma^2)$$

Remark. Notice that the mean and variance of the limiting distribution are the mean and variance of the linear Taylor Series approximation

$$g(Z_n) = g(\mu) + g'(\mu)(Z_n - \mu)$$

We also have the multivariate version

Theorem 1.14 (δ-method (Multivariable)). If $\mathbf{Z}_n$ is an $K \times 1$ sequence of vector valued random variables such that $\sqrt{n}(\mathbf{Z}_n - \mu) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \mathbf{\Sigma})$ and if $\mathbf{c}(\mathbf{Z}_n)$ is a set of J continuous and continuously differentiable functions of $\mathbf{z}_n$ with $\mathbf{C}(\mu) \neq \mathbf{0}$ and not involving n, then

$$\sqrt{n}(\mathbf{c}(\mathbf{Z}_n) - \mathbf{c}(\mu)) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \mathbf{C}(\mu)\mathbf{\Sigma}\mathbf{C}(\mu)')$$

where $\mathbf{C}(\mu)$ is the $J \times K$ matrix $\partial \mathbf{c}(\mu)/\partial \mu'$. The j-th row of $\mathbf{C}(\mu)$ is the vector of partial derivatives of the j-th function with respect to μ'

2 Review of Linear/Matrix Algebra

2.1 Simple Mechanics of Matrices

Definition 2.1 (Matrix). An $n \times K$ matrix $\mathbf{A}$ is an array with n rows and K columns

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1K} \\ a_{21} & a_{22} & \cdots & a_{2K} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nK} \end{pmatrix}$$

Definition 2.2 (Special Matrices). We have the following matrices with special properties

- **Square Matrix**: A matrix $\mathbf{A}$ is a square matrix when $n = K$ i.e.

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}$$

- **Symmetric Matrix**: a square matrix $\mathbf{A}$ that has $a_{ik} = a_{ki}$ for all i, k is called a symmetric matrix. For example

$$\mathbf{A} = \begin{pmatrix} 3 & 2 & 9 \\ 2 & 1 & 6 \\ 9 & 6 & 5 \end{pmatrix}$$

- **Diagonal Matrix**: a square matrix that has only non-zero entries on the main diagonal (and hence is necessarily symmetric)

$$\mathbf{A} = \begin{pmatrix} a_{11} & 0 & \cdots & 0 \\ 0 & a_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{pmatrix}$$

- **Scalar Matrix**: a diagonal matrix with the same value in all diagonal elements

$$\mathbf{A} = \begin{pmatrix} a_{11} & 0 & \cdots & 0 \\ 0 & a_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{pmatrix}$$

  where $a_{11} = a_{22} = \cdots = a_{nn}$

- **Identity Matrix**: a scalar matrix with ones on the diagonal

$$\mathbf{I} = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}$$

Remark. A useful way to think about matrices is that they are linear transformations of vectors. For instance, the identity matrix transforms a vector $\mathbf{x}$ back to itself while the scalar matrix transforms the vector $\mathbf{x}$ to $\lambda\mathbf{x}$ if the scalar matrix contains λ on its main diagonal

Definition 2.3 (Transpose). The transpose of an $n \times K$ matrix $\mathbf{A}$, denoted $\mathbf{A}'$ is obtained by creating the matrix whose kth column is the kth row of $\mathbf{A}$ (and so is a $K \times n$ matrix. An equivalent formulation of this is for a matrix $\mathbf{A}$, we have

$$\mathbf{B} = \mathbf{A}' \Leftrightarrow b_{ik} = a_{ki}, \ \ \forall i, k$$

Proposition 2.4. If $\mathbf{A}$ is symmetric, then $\mathbf{A}' = \mathbf{A}$

Example 2.5. Suppose we have

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$$

then,

$$\mathbf{A}' = \begin{pmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{pmatrix}$$

Definition 2.6 (Algebra of Matrices). We have the following for addition and multiplication

- **Addition**: Let $\mathbf{A}$ and $\mathbf{B}$ be two matrices of the same dimension, then

$$\mathbf{A} + \mathbf{B} = (a_{ij} + b_{ij})$$

  For subtraction we similarly have

$$\mathbf{A} - \mathbf{B} = (a_{ij} - b_{ij})$$

  though technically we have to define scalar multiplication first and multiply $\mathbf{B}$ by a scalar -1

- **Scalar multiplication**: Let $c \in \mathbb{R}$, then $c\mathbf{A} = (ca_{ij})$ i.e. you multiply every entry in $\mathbf{A}$ by c

- **Matrix Multiplication**: Let $\mathbf{A}$ be $n \times K$ and $\mathbf{B}$ be $K \times m$. Then, $\mathbf{C} = \mathbf{AB} = (c_{ij}) == (\sum_{k=1}^{K}(a_{ik}b_{kj})$ (i.e. the c_{ij} entry is the inner product of the ith row of $\mathbf{A}$ and the kth row of $\mathbf{B}$) where $\mathbf{AB}$ is an $n \times m$ matrix. For example, if

$$\mathbf{A} = \begin{pmatrix} 3 & 2 & 9 \\ 2 & 1 & 6 \\ 9 & 6 & 5 \end{pmatrix} \mathbf{B} = \begin{pmatrix} 2 & 1 & 5 \\ 3 & 2 & 7 \\ 1 & 2 & 9 \end{pmatrix}$$

then

$$\mathbf{AB} = \begin{pmatrix} 21 & 25 & 110 \\ 13 & 16 & 71 \\ 41 & 31 & 132 \end{pmatrix}$$

Proposition 2.7 (Properties of algebra of Matrices). We have the following properties

- **Addition**: for addition we have
  1. Adding zero or the null matrix: $\mathbf{A} + \mathbf{0} = \mathbf{A}$
  2. Commutative: $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$
  3. Associative: $(\mathbf{A} + \mathbf{B}) + \mathbf{C} = \mathbf{A} + (\mathbf{B} + \mathbf{C})$
- **Scalar Multiplication**: for scalar multiplication we have
  1. $0\mathbf{A} = \mathbf{0}$
  2. $-1\mathbf{A} = -\mathbf{A}$
- **Matrix Multiplication**: for matrix multiplication we have
  1. Not commutative: $\mathbf{AB} \neq \mathbf{BA}$
  2. Associative: $(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC})$
  3. Distributive: $\mathbf{A}(\mathbf{B} + \mathbf{C}) = \mathbf{AB} + \mathbf{AC}$
  4. Transpose of a product: $(\mathbf{AB})' = \mathbf{B}'\mathbf{A}'$
  5. Transpose of an extended product: $(\mathbf{ABC})' = \mathbf{C}'\mathbf{B}'\mathbf{A}'$
  6. Identity: $\mathbf{AI}_K = \mathbf{I}_n\mathbf{A} = \mathbf{A}$

2.2 Properties: Rank, Trace, Determinant, Inverse

Definition 2.8 (Linear Dependence). A set of $k \geq 2$ vectors are said to be linearly dependent if at least one of the vectors in the set can be written as a linear combination of the others

Definition 2.9 (Linear Independence). Let $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k \in \mathbb{R}^n$ and $a_1, a_2, \ldots, a_k \in \mathbb{R}$. Then, $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k$ is said to be linearly dependent if the only solution $(a_1, \ldots, a_k)$ to

$$a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \cdots + a_k\mathbf{v}_k = \mathbf{0}$$

is $a_1 = a_2 = \cdots = a_k = 0$

Definition 2.10 (Norm). For a vector $\mathbf{v} \in \mathbb{R}^n$, the Euclidean norm of a vector $||\mathbf{v}||$ is defined as

$$||\mathbf{v}||^2 = \mathbf{v}'\mathbf{v}$$

i.e. the square root of the inner product of a vector with itself

Proposition 2.11. For any vector $\mathbf{v} \in \mathbb{R}^n \backslash \{\mathbf{0}\}$,

$$||\mathbf{v}||^2 > 0$$

Proof. Trivial. ■

Definition 2.12 (Determinant). Let $\mathbf{A}$ be a $K \times K$ square matrix. The determinant of $\mathbf{A}$, denoted $|\mathbf{A}|$, is such that for any row i,

$$|\mathbf{A}| = \sum_{k=1}^{K} a_{ik}(-1)^{i+k}|A_{ik}|$$

where $|A_{ik}|$ is a minor of $\mathbf{A}$ i.e. the determinant of A_{ik} where A_{ik} is a matrix obtained after deleting row i and column k.

We call $C_{ik} = (-1)^{i+k}|A_{ik}|$ the **cofactor**

Remark. For a 2×2 matrix $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, a useful rule is $|\mathbf{A}| = ad - bc$

Example 2.13. If $\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$, then

$$|\mathbf{A}| = a_{11}a_{22}a_{33} + a_{21}a_{32}a_{13} + a_{12}a_{23}a_{31} - a_{31}a_{22}a_{13} - a_{32}a_{23}a_{11} - a_{21}a_{12}a_{33}$$

This is known as the *Rule of Sarrus*

Definition 2.14 (Inverse of a Matrix). Let $\mathbf{A}$ be a square $n \times n$ matrix. The inverse matrix, denoted by $\mathbf{A}^{-1}$, is defined such that

$$\mathbf{A}^{-1}\mathbf{A} = \mathbf{A}\mathbf{A}^{-1} = \mathbf{I}_n$$

In general, we have $\mathbf{A}^{-1} = \frac{1}{|\mathbf{A}|}\mathbf{C}'$, where

$$\mathbf{C} = \begin{pmatrix} c_{11} & c_{12} & \cdots & c_{1K} \\ c_{21} & c_{22} & \cdots & c_{2K} \\ \vdots & \vdots & \ddots & \vdots \\ c_{n1} & c_{n2} & \cdots & c_{nK} \end{pmatrix}$$

is the matrix of cofactors: $C_{ik} = (-1)^{i+k}|A_{ik}|$

Example 2.15. Let $\mathbf{A}$ be a 2×2 matrix

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

then,

$$A^{-1} = \frac{1}{a_{11}a_{22} - a_{12}a_{21}} \begin{pmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{pmatrix}$$

where $\mathbf{A}^{-1}\mathbf{A} = \mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$. Notice that the denominator of the fraction is precisely the determinant of $\mathbf{A}$

Definition 2.16 (Rank). The rank of a matrix, denoted $rank(\mathbf{A})$, is the maximum number of linearly independent columns (or rows). A matrix $\mathbf{A}$ is full rank if its rank is equal to the number of columns it contains

Proposition 2.17 (Properties of Rank). We have the following

- $rank(\mathbf{A}) = rank(\mathbf{A}')$
- $rank(\mathbf{A}) = rank(\mathbf{A}'\mathbf{A}) = rank(\mathbf{A}\mathbf{A}')$
- $|\mathbf{A}| \neq 0$ if and only if it has full rank

Example 2.18. Let $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$ which clearly $rank(\mathbf{A}) = 1$. Then $|\mathbf{A}| = 1 \cdot 4 - 2 \cdot 2 = 0$

Definition 2.19 (Trace). The race of a $n \times n$ matrix $\mathbf{A}$, denoted $tr(\mathbf{A})$, is the sum of its diagonal elements

$$tr(\mathbf{A}) = \sum_{i=1}^{n} a_{ii}$$

Proposition 2.20 (Properties of Trace). For a $n \times n$ matrix $\mathbf{A}$,

- $tr(c\mathbf{A}) = c \cdot tr(\mathbf{A})$
- $tr(\mathbf{A}') = tr(\mathbf{A})$
- $tr(\mathbf{A} + \mathbf{B}) = tr(\mathbf{A}) + tr(\mathbf{B})$
- $tr(\mathbf{I}_n) = n$
- $tr(\mathbf{AB}) = tr(\mathbf{BA})$
- $tr(\mathbf{xy}') = tr(\mathbf{y}'\mathbf{x})$, for $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$
- $tr(\mathbf{a}'\mathbf{a}) = tr(\mathbf{aa}') = \mathbf{a}'\mathbf{a}$, for $\mathbf{a} \in \mathbb{R}^n$

We also have the following special matrices.

Definition 2.21 (Symmetric and Idempotent Matrices). Let $\mathbf{A}$ be a $n \times n$ matrix. We say that $\mathbf{A}$ is *symmetric* just when $\mathbf{A} = \mathbf{A}'$ and *idempotent* just when $\mathbf{A}^2 = \mathbf{A}$

It is useful to know the following relationships between the inverse, idempotent, and symmetric matrices.

Lemma 2.22. Let $\mathbf{A}$ be an $n \times m$ matrix. Then, $\mathbf{A}'\mathbf{A}$ is symmetric.

Proof. Let $(\mathbf{A}'\mathbf{A})' = \mathbf{A}'(\mathbf{A}')' = \mathbf{A}'\mathbf{A}$ since $(\mathbf{A}')' = \mathbf{A}$ ■

Lemma 2.23. Let $\mathbf{A}$ be an $n \times n$ symmetric matrix. Then, if the inverse $\mathbf{A}^{-1}$ exists, it is symmetric.

Proof. Let $\mathbf{A}$ be an invertible symmetric matrix. Then, $\mathbf{AA}^{-1} = \mathbf{I}$. Taking the transpose, $(\mathbf{AA}^{-1})' = (\mathbf{I})' = \mathbf{I}$ since $\mathbf{I}$ is symmetric. So, $\mathbf{I} = (\mathbf{A}^{-1})'\mathbf{A}'$. So, we have

$$(\mathbf{AA}^{-1})' = \mathbf{I} = (\mathbf{A}^{-1})'\mathbf{A}'$$

Since $\mathbf{A}$ is symmetric, we know that $\mathbf{A}' = \mathbf{A}$. Hence, $(\mathbf{A}^{-1})' = \mathbf{A}^{-1}$ ■

We want to show that the Projection and Annihilator matrix have the above properties.

Proposition 2.24. Define $\mathbf{P} := \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ (the projection matrix) and $\mathbf{M} := \mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ (the annihilator matrix). Then, $\mathbf{P}$ and $\mathbf{M}$ are symmetric and idempotent

Proof. For the projection matrix $\mathbf{P} := \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$

- $\mathbf{P} = \mathbf{P}'$: $(\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}')' \underset{(1)}{=} (\mathbf{X}')'(\mathbf{X}'\mathbf{X})^{-1\prime}\mathbf{X}' \underset{(2)}{=} \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$.

  (1) follows by the transpose of the extended product and (2) follows by **Lemma 2.21**.

- $\mathbf{P}^2 = \mathbf{P}$:

$$\begin{aligned}
\mathbf{P}^2 &= \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \\
&= \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}[\mathbf{X}'\mathbf{X}](\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \\
&= \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{I}\mathbf{X}' \\
&= \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \\
&= \mathbf{P}
\end{aligned}$$

For the annihilator matrix $\mathbf{M} := \mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$

- $\mathbf{M} = \mathbf{M}'$: Consider

$$\begin{aligned}\mathbf{M}' &= \{\mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\}' \\ &= \mathbf{I}' - [\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}']' \\ &= \mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \\ &= \mathbf{M}\end{aligned}$$

  This follows by symmetry of $\mathbf{I}$ and $\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ which follows from the results of $\mathbf{P}$

- $\mathbf{M}^2 = \mathbf{M}$: Consider

$$\begin{aligned}\mathbf{M}^2 &= [\mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'][\mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'] \\ &= [\mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'] - [\mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}']\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \\ &= \mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \\ &= \mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' = \mathbf{M}\end{aligned}$$

■

2.3 Quadratic Forms and Definiteness

Definition 2.25 (Quadratic Form). A quadratic form is $\mathbf{x}'\mathbf{A}\mathbf{x}$ where $\mathbf{A}$ is an $n \times n$ symmetric matrix and $\mathbf{x}$ is $n \times 1$

Example 2.26. It is worth memorising for $n = 2, 3$ cases:

$$\begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} a & u \\ u & b \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = ax^2 + by^2 + 2uxy$$

and

$$\begin{pmatrix} x & y & z \end{pmatrix} \begin{pmatrix} a & u & v \\ u & b & w \\ v & w & c \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = ax^2 + by^2 + cz^2 + 2uxy + 2vxz + 2wyz$$

Definition 2.27 (Definiteness). For a $n \times n$ symmetric matrix $\mathbf{A}$, we say that it is

- Positive Definite if $\mathbf{x}'\mathbf{A}\mathbf{x} > 0$ for all nonzero $\mathbf{x}$
- Negative Definite if $\mathbf{x}'\mathbf{A}\mathbf{x} < 0$ for all nonzero $\mathbf{x}$
- Positive Semidefinite if $\mathbf{x}'\mathbf{A}\mathbf{x} \geq 0$ for all nonzero $\mathbf{x}$
- Negative Semidefinite if $\mathbf{x}'\mathbf{A}\mathbf{x} \leq 0$ for all nonzero $\mathbf{x}$

Remark. Note that variance-covariance matrices are positive semi-definite by definition

Remark. Notice that if we have a matrix and vector product such as $\mathbf{X}\mathbf{v}$, if we can guarantee that $\mathbf{v} \in \mathbf{R}^n \backslash \{\mathbf{0}\}$ and $\mathbf{X}$ is full rank, then note that

$$\mathbf{v}'\mathbf{X}'\mathbf{X}\mathbf{v} = (\mathbf{X}\mathbf{v})'\mathbf{X}\mathbf{v} = ||\mathbf{X}\mathbf{v}||^2 > 0$$

which suffices to show that $\mathbf{X}'\mathbf{X}$ is positive definite

2.4 Calculus and Matrix Algebra

Proposition 2.28 (Differentiation of Matrices). We have the following rules for differentiating matrices

- First rule:
$$\frac{\partial(\mathbf{Ax})}{\partial \mathbf{x}} = \mathbf{A}'$$
- Second rule:
$$\frac{\partial(\mathbf{x}'\mathbf{Ax})}{\partial \mathbf{x}} = (\mathbf{A}+\mathbf{A}')\mathbf{x}$$

Example 2.29. Suppose we have $\mathbf{A} = \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix}$ and $x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$. Then,

$$\frac{\partial(\mathbf{Ax})}{\partial \mathbf{x}} = \frac{\partial \begin{pmatrix} x_1 + 3x_2 \\ 2x_1 + 4x_2 \end{pmatrix}}{\partial \mathbf{x}} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \mathbf{A}'$$

and

$$\begin{aligned} \frac{\partial(\mathbf{x}'\mathbf{Ax})}{\partial \mathbf{x}} = \frac{\partial(x_1^2 + 5x_1x_2 + 4x_2^2)}{\partial \mathbf{x}} &= \begin{pmatrix} 2x_1 + 5x_2 \\ 5x_1 + 8x_2 \end{pmatrix} \\ &= \left\{ \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix} + \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \right\} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \\ &= (\mathbf{A}+\mathbf{A}')\mathbf{x} \end{aligned}$$

2.5 Useful: Eigenvalues and the Spectral Theorem

Theorem 2.30 (Cauchy-Schwarz Inequality). For $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$,

$$|\mathbf{x} \cdot \mathbf{y}| \leq ||\mathbf{x}|| \cdot ||\mathbf{y}||$$

Definition 2.31 (Upper/Lower Triangular Matrices). A $n \times n$ matrix $\mathbf{A}$ is *upper triangular* if all entries "below the main diagonal" are 0 i.e. $a_{ij} = 0$ for $j < i$; it is *lower triangular* if all entries "above the main diagonal" are 0 i.e. $a_{ij} = 0$ for $i < j$

Remark. A upper/lower triangular matrix with all nonzero entries on the diagonal is invertible; conversely, a triangular matrix with at least one zero entry on the diagonal is not invertible

Definition 2.32 (Eigenvalues and Eigenvectors). Let $\mathbf{A}$ be an $n \times n$ matrix. A vector $\mathbf{v} \in \mathbb{R}^n$ is called an eigenvector of $\mathbf{A}$ if it is nonzero and if there is a scalar $\lambda \in \mathbb{R}$ for which $\mathbf{Av} = \lambda\mathbf{v}$. We call this scalar λ the eigenvalue

Example 2.33. Let $\mathbf{A} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$. Then, $\mathbf{u} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$. We can see this since

$$\begin{aligned} \mathbf{Au} &= \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} \\ \mathbf{Av} &= \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 3 \\ -3 \end{pmatrix} = 3 \begin{pmatrix} 1 \\ -1 \end{pmatrix} \end{aligned}$$

Hence $\mathbf{u}$ is an eigenvector of $\mathbf{A}$ with eigenvalue 1 and $\mathbf{v}$ is an eigenvector of $\mathbf{A}$ with eigenvalue 3

Proposition 2.34. Let $\mathbf{M}$ be an $n \times n$ upper or lower triangular matrix. The eigenvalues of $\mathbf{M}$ are exactly the diagonal entries. For each eigenvalue λ, the corresponding eigenvectors are nonzero vectors in the null space $N(\mathbf{M} - \lambda\mathbf{I}_n)$ i.e. the nonzero solutions $\mathbf{x}$ to the upper or lower triangular system

$$(\mathbf{M} - \lambda\mathbf{I}_n)\mathbf{x} = \mathbf{0}$$

of n linear equations in n-variables which can be solved via back-substitution

Proposition 2.35. Let $\mathbf{A}$ be a 2×2 matrix. We have the following:

1. The eigenvalues of $\mathbf{A}$ in $\mathbb{R}$ are exactly the roots of the characteristic polynomial $P_A(\lambda) = \lambda^2 - tr(\mathbf{A})\lambda + \det(\mathbf{A})$
2. For each such root λ, the corresponding eigenvectors are nonzero vectors in $N(\mathbf{A} - \lambda\mathbf{I}_2)$

Proposition 2.36. Let $\mathbf{A}$ be an $n \times n$ matrix and $\mathbf{v}_1, \ldots, \mathbf{v}_r$ a collection of eigenvectors for $\mathbf{A}$ with respective eigenvalues $\lambda_1, \ldots, \lambda_r$, so $\mathbf{A}\mathbf{v}_j = \lambda_j \mathbf{v}_j$ for all j.

1. If the r eigenvalues are pairwise different then the collection of r vectors $\mathbf{v}_1, \ldots, \mathbf{v}_r$ is linearly independent, so $r \leq n$. Thus, an $n \times n$ matrix cannot have more than n different eigenvalues
2. If $\mathbf{A}$ is symmetric and $\lambda_i \neq \lambda)j$, then $\mathbf{v}_i \cdot \mathbf{v}_j = 0$ i.e. eigenvectors for different eigenvalues for a symmetric matrix are orthogonal to one another.

Proposition 2.37. A symmetric $n \times n$ matrix $\mathbf{A}$ is

- positive definite when its eigenvalues are all positive
- negative definite when its eigenvalues are all negative
- indefinite when some eigenvalue is positive and some eigenvalue is negative
- positive semidefinite when all eigenvalues are ≥ 0 and 0 is an eigenvalue
- negative semidefinite when all eigenvalues are ≤ 0 and 0 is an eigenvalue

Remark. For a diagonal matrix $\mathbf{D}$, its eigenvalues are precisely its diagonal entries.

Theorem 2.38 (Spectral Theorem via matrix decomposition). Let $\mathbf{A}$ be a symmetric $n \times n$ matrix with orthogonal eigenvectors $\mathbf{w}_1, \ldots, \mathbf{w}_n$ with corresponding eigenvalues $\lambda_1, \ldots, \lambda_n$. Let $\mathbf{W}$ be the $n \times n$ matrix whose columns are the unit vectors $\mathbf{w}_j / ||\mathbf{w}_j||$. Then, $\mathbf{W}^{-1} = \mathbf{W}'$ (i.e. $\mathbf{W}$ is orthogonal) and

$$\mathbf{A} = \mathbf{W}\mathbf{D}\mathbf{W}' = \mathbf{W}\mathbf{D}\mathbf{W}^{-1}$$

where $\mathbf{D}$ is the diagonal matrix $\mathbf{D} = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix}$

Proposition 2.39 (Second Order Conditions). For a function $f(\mathbf{x})$, consider the matrix of second derivatives

$$\mathcal{H} = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 x_n} \\ \frac{\partial^2 f}{\partial x_2 x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n x_1} & \frac{\partial^2 f}{\partial x_n x_2} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{pmatrix}$$

Then,

- If $\mathcal{H}$ (evaluated at $\mathbf{a}$ where $\nabla f(\mathbf{a}) = \mathbf{0}$) is positive definite, then the solution satisfying $\nabla f(\mathbf{x}) = \mathbf{0}$ is a minimum
- If $\mathcal{H}$ (evaluated at $\mathbf{a}$ where $\nabla f(\mathbf{a}) = \mathbf{0}$) is negative definite, then the solution satisfying $\nabla f(\mathbf{x}) = \mathbf{0}$ is a maximum
- If $\mathcal{H}$ evaluated at $\mathbf{a}$ where $\nabla f(\mathbf{a}) = \mathbf{0}$) is indefinite, then the solution satisfying $\nabla f(\mathbf{x}) = \mathbf{0}$ is a saddle point

Proposition 2.40. Consider an $n \times n$ diagonal matrix $\mathbf{A}$ such that

$$\mathbf{A} = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix}$$

Then, $\lambda_1, \ldots, \lambda_n$ are precisely its eigenvalues

Proof. Consider the diagonal $n \times n$ matrix $\mathbf{A}$, and set of $n \times 1$ vectors

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \cdots, \mathbf{v}_n = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}$$

Notice that these vectors all satisfy

$$\mathbf{A}\mathbf{v}_i = \lambda_i \mathbf{A}$$

where λ_i are precisely the i-th diagonal entry and so are eigenvalues of $\mathbf{A}$. Since we know that the number of non-zero eigenvalues are at most the rank of the square $n \times n$ matrix, the set of eigenvalues obtained $\lambda_1, \ldots, \lambda_n$ are exactly all the eigenvalues of $\mathbf{A}$. ■

3 Ordinary Least Squares Regression

3.1 Single Variable OLS

Definition 3.1 (Simple Regression Model). The simple regression model is

$$y_i = \beta_1 + \beta_1 x_i + u_i$$

where y_i and x_i are observable random scalars, u_i is the unobservable random disturbance or error and β_1, β_2 are the parameters we would like to estimate.

Proposition 3.2 (OLS Estimators). Consider the OLS problem

$$(\hat{\beta}_1, \hat{\beta}_2) = \underset{\beta_1, \beta_2}{\operatorname{argmin}} \sum_{i=1}^{n} (y_i - (\beta_1 + \beta_2 x_i))^2$$

Then,

$$\hat{\beta}_1 = \bar{y} - \hat{\beta}_2 \bar{x}$$
$$\hat{\beta}_2 = \frac{\sum_{i=1}^{n} (y_i - \bar{y})(x_i - \bar{x})}{\sum_{i=1}^{n} (x_i - \bar{x})^2}$$

uniquely solves our OLS problem.

Proof. Let $L = \sum_{i=1}^{n} (y_i - (\beta_1 + \beta_2 x_i))^2$. Then, our FOCs are

$$\frac{\partial L}{\partial \beta_1} = -2 \sum_{i=1}^{n} (y_i - (\beta_1 + \beta_2 x_i) = 0$$
$$\frac{\partial L}{\partial \beta_1} = -2 \sum_{i=1}^{n} x_i (y_i - (\beta_1 + \beta_2 x_i)) = 0$$

We can solve the first condition as

$$\frac{1}{n}\sum_{i=1}^{n}(y_i - (\beta_1 + \beta_2 x_i) = 0 \implies \beta_1 = \bar{y} - \beta_2\bar{x}$$

where $\bar{y} = n^{-1}\sum_{i=1}^{n} y_i$ and $\bar{x} = n^{-1}\sum_{i=1}^{n} x_i$. Then, substituting this result into the second condition

$$\begin{aligned}
&\frac{1}{n}\sum_{i=1}^{n} x_i(y_i - (\bar{y} - \beta_2\bar{x} + \beta_2 x_i)) = 0\\
&\frac{1}{n}\sum_{i=1}^{n} x_i(y_i - \bar{y}) - \beta_2\frac{1}{n}\sum_{i=1}^{n} x_i(x_i - \bar{x}) = 0\\
&\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x} + \bar{x})(y_i - \bar{y}) = \beta_2\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x} + \bar{x})(x_i - \bar{x})\\
&\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y}) + \frac{1}{n}\sum_{i=1}^{n}\bar{x}(y_i - \bar{y}) = \beta_2\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2 + \beta_2\frac{1}{n}\sum_{i=1}^{n}\bar{x}(x_i - \bar{x})
\end{aligned}$$

since $n^{-1}\sum\bar{x}(y_i - \bar{y}) = n^{-1}\sum_{i=1}^{n}\bar{x}(x_i - \bar{x}) = 0$. Then, we yield

$$\hat{\beta}_2 = \frac{n^{-1}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{n^{-1}\sum_{i=1}^{n}(x_i - \bar{x})^2} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

Then, we also have

$$\hat{\beta}_1 = \bar{y} - \hat{\beta}_2\bar{x}$$

■

Remark. We may rewrite the OLS estimators as

$$\begin{aligned}
\hat{\beta}_1 &= \beta_1 + \bar{u} - (\hat{\beta}_2 - \beta_2)\bar{x}\\
\hat{\beta}_2 &= \beta_2 + \frac{n^{-1}\sum_{i=1}^{n}(x_i - \bar{x})u_i}{n^{-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}
\end{aligned}$$

Corollary 3.3. Let $\hat{u}_i = y_i - \hat{\beta}_1 - \hat{\beta}_2 x_i$ be the OLS residuals. Then,

$$\begin{aligned}
\frac{1}{n}\sum_{i=1}^{n}\hat{u}_i &= 0\\
\frac{1}{n}\sum_{i=1}^{n}\hat{u}_i x_i &= 0
\end{aligned}$$

Furthermore, $\hat{y}_i = \hat{\beta}_1 + \hat{\beta}_2 x_i$ be the predicted values for y_i. Then,

$$\frac{1}{n}\sum_{i=1}^{n}\hat{y}_i = \bar{y}$$

Proof. The first two results follow from the FOCs for our OLS problem. Then, write

$$\frac{1}{n}\sum_{i=1}^{n}(\hat{\beta}_1 + \hat{\beta}_2 x_i) = \frac{1}{n}\sum_{i=1}^{n}(\bar{y} - \hat{\beta}_2\bar{x} + \hat{\beta}_2 x_i)$$

Notice that $\hat{\beta}_2\frac{1}{n}\sum_{i=1}^{n}\bar{x} = \hat{\beta}_2\frac{1}{n}\sum_{i=1}^{n}x_i$. So,

$$\frac{1}{n}\sum_{i=1}^{n}\bar{y} = \bar{y}$$

■

We also have the following concepts

Definition 3.4. Consider the simple regression model. Then,

- Total Sum of Squares (SST): $\sum_{i=1}^{n}(y_i - \bar{y})^2$
- Regression Sum of Squares (SSR): $\sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2$
- Sum of Squared Errors (SSE): $\sum_{i=1}^{n} u_i^2$

We also have R^2 such that

$$R^2 \equiv \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$$

Proposition 3.5. Consider the simple regression model. Then,

$$SST = SSR + SSE$$

Proof. See QE Notes. ■

3.1.1 Finite Sample Properties

Assumption 3.6. We assume

(i) $\mathrm{E}(u_i|x_1, \ldots, x_n) = 0$ (Conditional Mean 0)

(ii) $\mathrm{Var}(u_i|x_i, \ldots, x_n) = \sigma^2$ with $0 < \sigma^2 < \infty$ (Homoskedasticity)

(iii) $\mathrm{E}[u_i u_j|x_i, \ldots, x_n) = 0$ for $i \neq j$ (no serial autocorrelation in errors)

Theorem 3.7 (Simple OLS Finite Sample Properties). Let **Assumption 3.3** hold. Then,

(a) The OLS estimators $\hat{\beta}_1$ and $\hat{\beta}_2$ are unbiased, that is,

$$\mathrm{E}(\hat{\beta}_1) = \beta_1 \quad \text{and} \quad \mathrm{E}(\hat{\beta}_2) = \beta_2$$

(b) The conditional variances are

$$\begin{aligned}
\mathrm{Var}(\hat{\beta}_1|x_1, \ldots, x_n) &= \frac{n^{-1}\sum_{i=1}^{n} x_i^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sigma^2 \\
\mathrm{Var}(\hat{\beta}_2|x_1, \ldots, x_n) &= \frac{\sigma^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}
\end{aligned}$$

(c) The OLS estimators $\hat{\beta}_1$ and $\hat{\beta}_2$ are BLUE (Best Linear Unbiased Estimators)

(d) $\mathrm{E}(\hat{\sigma}^2) = \sigma^2$, where the estimator of the error variance is defined as

$$\hat{\sigma}^2 = \frac{1}{n-2}\sum_{i=1}^{n} \hat{u}_i^2$$

Remark. We say that an estimator is BLUE just when it has the smallest variance (and hence most efficient) out of all estimators that are linear.

Proof. **Proof of Unbiasedness**: We can write

$$\begin{aligned}
\hat{\beta}_1 &= \beta_1 + \bar{u} - (\hat{\beta}_2 - \beta_2)\bar{x} \\
\hat{\beta}_2 &= \beta_2 + \frac{n^{-1}\sum_{i=1}^{n}(x_i - \bar{x})u_i}{n^{-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}
\end{aligned}$$

Then, by LIE,

$$\mathrm{E}(\hat{\beta}_1) = \mathrm{E}\{\mathrm{E}(\hat{\beta}_1|x_1,\ldots,x_n)\}$$
$$\mathrm{E}(\hat{\beta}_2) = \mathrm{E}\{\mathrm{E}(\hat{\beta}_2|x_1,\ldots,x_n)\}$$

Begin with considering $\hat{\beta}_2$ since $\hat{\beta}_1$ is in terms of $\hat{\beta}_2$ By the properties of conditional expectation,

$$\mathrm{E}(\hat{\beta}_2|x_1,\ldots,x_n) = \beta_2 + \frac{\sum_{i=1}^n (x_i-\bar{x})\mathrm{E}(u_i|x_1,\ldots,x_n)}{\sum_{i=1}^n (x_i-\bar{x})^2}$$

By Assumption 3.6 (i), $\mathrm{E}(u_i|x_1,\ldots,x_n) = 0$. Hence, $\mathrm{E}(\hat{\beta}_2|x_1,\ldots,x_n) = \beta_2$. By LIE, we yield

$$\mathrm{E}(\hat{\beta}_2) = \mathrm{E}\{\mathrm{E}(\hat{\beta}_2|x_1,\ldots,x_n)\} = \beta_2$$

For $\hat{\beta}_1$, we have

$$\mathrm{E}(\hat{\beta}_1|x_1,\ldots,x_n) = \beta_1 + \mathrm{E}(\bar{u}|x_1,\ldots,x_n) - \mathrm{E}((\hat{\beta}_2-\beta_2)\bar{x}|x_1,\ldots,x_n)$$

By properties of LIE and Assumption 3.6 (i) where $\mathrm{E}(u_i|x_1,\ldots,x_n) = 0$ and our result for $\hat{\beta}_2$, we have

$$\mathrm{E}(\hat{\beta}_1|x_1,\ldots,x_n) = \beta_1 + -\bar{x}\mathrm{E}(\hat{\beta}_2-\beta_2|x_1,\ldots,x_n) = \beta_1 - \bar{x}(\beta_2-\beta_2) = \beta_1$$

By LIE, we have

$$\mathrm{E}(\hat{\beta}_1) = \mathrm{E}\{\mathrm{E}(\hat{\beta}_1|x_1,\ldots,x_n)\} = \beta_1$$

Proof of Conditional Variances: To start with $\hat{\beta}_2$, we have by the properties of conditional variance

$$\mathrm{Var}(\hat{\beta}_2|x_1,\ldots,x_n) = \frac{\mathrm{Var}\{\sum_{i=1}^n (x_i-\bar{x})u_i|x_1,\ldots,x_n\}}{\{\sum_{i=1}^n (x_i-\bar{x})^2\}^2}$$

where $\mathrm{Var}\{\sum_{i=1}^n (x_i-\bar{x})u_i|x_1,\ldots,x_n\}$ equals

$$\sum_{i=1}^n\sum_{j=1}^n (x_i-\bar{x})(x_j-\bar{x})\mathrm{Cov}(u_i,u_j|x_1,\ldots,x_n)$$

Since $\mathrm{Cov}(u_i,u_j|x_1,\ldots,x_n) = \mathrm{E}[u_iu_j|x_1,\ldots,x_n] - \mathrm{E}[u_i|x_1,\ldots,x_n]\mathrm{E}[u_j|x_1,\ldots,x_j]$, by Assumption 3.6 (i) and (iii), we have $\mathrm{E}[u_i|x_1,\ldots,x_n] = 0$ and $\mathrm{E}[u_iu_j|x_1,\ldots,x_n] = 0$ for all $i \neq j$. Therefore,

$$\mathrm{Var}(\hat{\beta}_2|x_1,\ldots,x_n) = \frac{\sum_{i=1}^n (x_i-\bar{x})^2\mathrm{Var}(u_i|x_1,\ldots,x_n)}{\{\sum_{i=1}^n (x_i-\bar{x})^2\}^2}$$

Finally, by Assumption 3.6 (ii), $\mathrm{Var}(u_i|x_1,\ldots,x_n) = \sigma^2$. Therefore,

$$\mathrm{Var}(\hat{\beta}_2|x_1,\ldots,x_n) = \frac{\sum_{i=1}^n (x_i-\bar{x})^2}{\{\sum_{i=1}^n (x_i-\bar{x})^2\}^2} = \frac{\sigma^2}{\sum_{i=1}^n (x_i-\bar{x})^2}$$

For $\hat{\beta}_1$, we have

$$\mathrm{Var}(\hat{\beta}_1|x_1,\ldots,x_n) = \mathrm{Var}(\bar{u}|x_1,\ldots,x_n) + \left[n^{-1}\sum_{i=1}^n x_i\right]^2 \mathrm{Var}(\hat{\beta}_2|x_1,\ldots,x_n)$$

since $\mathrm{Cov}(\bar{u},\hat{\beta}_2-\beta_2|x_1,\ldots,x_n) = \mathrm{Cov}(\bar{u},\hat{\beta}_2|x_1,\ldots,x_n) = \mathrm{E}[\bar{u}\hat{\beta}_2|x_1,\ldots,x_n] = \frac{1}{n^{-1}\sum_{i=1}^n (x_i-\bar{x})^2}\mathrm{E}[\sum_{j=1}^n u_i \sum_{i=1}^n (x_i-\bar{x})u_i|x_1,\ldots,x_n]$. By Assumption 3.6 (iii), $\frac{1}{n^{-1}\sum_{i=1}^n (x_i-\bar{x})^2}\sum_{i=1}^n (x_i-\bar{x})\mathrm{E}[u_i^2|x_1,\ldots,x_n] = 0$ since the term $\sum_{i=1}^n (x_i-\bar{x})\mathrm{E}[u_i^2|x_1,\ldots,x_n] = n\sigma^2\bar{x} - n\sigma^2\bar{x}$. Then, Assumption 3.6 (ii) and substituting the result for the conditional variance of $\hat{\beta}_2$ gives

$$\mathrm{Var}(\hat{\beta}_1|x_1,\ldots,x_n) = n^{-1}\sigma^2 + \left[n^{-1}\sum_{i=1}^n x_i\right]^2 \frac{\sigma^2}{\sum_{i=1}^n (x_i-\bar{x})^2}$$

Then,

$$\text{Var}(\hat{\beta}_1|x_1,\ldots,x_n) = \frac{\sum_{i=1}^n (x_i-\bar{x})^2 + n\bar{x}^2}{n\sum_{i=1}^n (x_i-\bar{x})^2}\sigma^2$$

Since we know that $\sum_{i=1}^n x_i^2 = \sum_{i=1}^n (x_i-\bar{x})^2 + n\bar{x}$, we yield

$$\text{Var}(\hat{\beta}_1|x_1,\ldots,x_n) = \frac{n^{-1}\sum_{i=1}^n x_i^2}{\sum_{i=1}^n (x_i-\bar{x})^2}\sigma^2$$

Proof of BLUE: We need to show three things about $\hat{\beta}_1$ and $b\hat{e}ta_2$ – that they are (i) unbiased (proven already), (ii) linear and (iii) has the smallest variance of all other linear estimators. For (ii), both $\hat{\beta}_1$ and $\hat{\beta}_2$ are clearly linear in y_i. Then, consider
Proof of Unbiasedness for variance estimator: We define $\hat{\sigma}^2 := \frac{1}{n-2}\sum_{i=1}^n \hat{u}_i^2$. Then, substituting $\hat{u}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i$, we yield

$$\begin{aligned}
\hat{\sigma}^2 &= \frac{1}{n-2}\sum_{i=1}^n (y_i - \hat{\beta}_1 - \hat{\beta}_2 x_i)^2 \\
&= \frac{1}{n-2}\sum_{i=1}^n (\beta_1 + \beta_2 x_i + u_i - \hat{\beta}_1 - \hat{\beta}_2 x_i)^2 \\
&= \frac{1}{n-2}\sum_{i=1}^n [u_i - (\hat{\beta}_1 - \beta_1) - (\hat{\beta}_2 - \beta_2)x_i]^2 \\
&= \frac{1}{n-2}\sum_{i=1}^n [(u_i - \bar{u}) + (\hat{\beta}_2 - \beta_2)\bar{x} - (\hat{\beta}_2 - \beta_2)x_i]^2 \\
&= \frac{1}{n-2}\sum_{i=1}^n [(u_i - \bar{u}) - (\hat{\beta}_2 - \beta_2)(x_i - \bar{x})]^2
\end{aligned}$$

Now, just consider

$$\sum_{i=1}^n \hat{u}_i^2 = \sum_{i=1}^n (u_i - \bar{u})^2 - 2(\hat{\beta}_2 - \beta_2)\sum_{i=1}^n (x_i - \bar{x})(u_i - \bar{u}) + (\hat{\beta}_2 - \beta_2)^2 \sum_{i=1}^n (x_i - \bar{x})^2$$

We consider each term in turn: for the first term

$$\begin{aligned}
\sum_{i=1}^n (u_i - \bar{u}) &= \sum_{i=1}^n (u_i^2 - 2u_i\bar{u} + \bar{u}^2) \\
&= \sum_{i=1}^n u_i^2 - 2n\bar{u} + n\bar{u} \\
&= \sum_{i=1}^n u_i^2 - n\bar{u} \\
&= \sum_{i=1}^n u_i^2 - \frac{\left(\sum_{i=1}^n u_i\right)^2}{n}
\end{aligned}$$

For the second term, note that $\sum_{i=1}^n (x_i - \bar{x}) = 0$. So, $\sum_{i=1}^n (x_i - \bar{x})(u_i - \bar{u}) = \sum_{i=1}^n (x_i - \bar{x})u_i$. Then, we have since $(\hat{\beta}_2 - \beta_2) = \sum_{i=1}^n (x_i - \bar{x})u_i / \sum_{i=1}^n (x_i - \bar{x})^2$,

$$-2(\hat{\beta}_2 - \beta_2)\sum_{i=1}^n (x_i - \bar{x})(u_i - \bar{u}) = -2\frac{(\sum_{i=1}^n (x_i - \bar{x})u_i)^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

Likewise, the third term goes to $(\hat{\beta}_2 - \beta_2)^2 \sum_{i=1}^n (x_i - \bar{x})^2 = \left(\frac{\sum_{i=1}^n (x_i - \bar{x}) u_i}{\sum_{i=1}^n (x_i - \bar{x})^2}\right)^2 \sum_{i=1}^n (x_i - \bar{x})^2 = \frac{[\sum_{i=1}^n (x_i - \bar{x}) u_i]^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$. Taking all this together, we get

$$\begin{aligned}
\sum_{i=1}^n \hat{u}_i^2 &= \sum_{i=1}^n u_i^2 - \frac{\left(\sum_{i=1}^n u_i\right)^2}{n} - 2\frac{(\sum_{i=1}^n (x_i - \bar{x}) u_i)^2}{\sum_{i=1}^n (x_i - \bar{x})^2} + \frac{[\sum_{i=1}^n (x_i - \bar{x}) u_i]^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \\
&= \sum_{i=1}^n u_i^2 - \frac{\left(\sum_{i=1}^n u_i\right)^2}{n} - \frac{(\sum_{i=1}^n (x_i - \bar{x}) u_i)^2}{\sum_{i=1}^n (x_i - \bar{x})^2}
\end{aligned}$$

Taking conditional expectations, we have by linearity,

$$\begin{aligned}
\mathrm{E}[\sum_{i=1}^n \hat{u}_i^2 | x_1, \ldots, x_n] &= \mathrm{E}\left[\sum_{i=1}^n u_i^2 - \frac{\left(\sum_{i=1}^n u_i\right)^2}{n} - \frac{(\sum_{i=1}^n (x_i - \bar{x}) u_i)^2}{\sum_{i=1}^n (x_i - \bar{x})^2} | x_1, \ldots, x_n\right] \\
&= \mathrm{E}\left[\sum_{i=1}^n u_i^2 | x_1, \ldots, x_n\right] - \frac{1}{n}\mathrm{E}\left[\sum_{i=1}^n u_i^2 | x_1, \ldots, x_n\right] - \frac{1}{\sum_{i=1}^n (x_i - \bar{x})^2}\mathrm{E}\left[\left(\sum_{i=1}^n (x_i - \bar{x}) u_i\right)^2 | x_1, \ldots, x_n\right]
\end{aligned}$$

Note that the second term is

$$\frac{1}{n}\mathrm{E}\left[\left(\sum_{i=1}^n u_i\right)^2 | x_1, \ldots, x_n\right] = \frac{1}{n}\mathrm{E}\left[\sum_{i=1}^n u_i^2 | x_1, \ldots, x_n\right]$$

obtains because

$$\mathrm{E}[\sum_{i \neq j} a_i u_i u_j | x_1, \ldots, x_n] = 0$$

By assumption, $\mathrm{E}(u_i^2 | x_1, \ldots, x_n) = \sigma^2$. So the first term goes to $n\sigma^2$ and the second term goes to σ^2. The third term is far more complicated. Define $a_i := x_i - \bar{x}$ we have

$$\begin{aligned}
\mathrm{E}[(\sum_{i=1}^n a_i u_i)^2 | x_1, \ldots, x_n] &= \mathrm{E}[\sum_{i=1}^n a_i^2 u_i^2 + \sum_{i<j} a_i a_j u_i u_j | x_1, \ldots, x_n] \\
&= \sum_{i=1}^n a_i \mathrm{E}[u_i^2 | x_1, \ldots, x_n] + \sum_{i<j} a_i a_j \mathrm{E}[u_i u_j | x_1, \ldots, x_n] \\
&= \sigma^2 \sum_{i=1}^n (x_i - \bar{x})^2
\end{aligned}$$

since $\mathrm{E}[u_i u_j | x_1, \ldots, x_n] = 0$ for $i \neq j$. So, the third term becomes

$$\frac{1}{\sum_{i=1}^n (x_i - \bar{x})^2}\mathrm{E}\left[\left(\sum_{i=1}^n (x_i - \bar{x}) u_i\right)^2 | x_1, \ldots, x_n\right] = \frac{1}{\sum_{i=1}^n (x_i - \bar{x})^2} \cdot \sigma^2 \sum_{i=1}^n (x_i - \bar{x})^2 = \sigma^2$$

Then,

$$\mathrm{E}[\sum_{i=1}^n \hat{u}_i^2 | x_1, \ldots, x_n] = n\sigma^2 - \sigma^2 - \sigma^2 = (n-2)\sigma^2$$

Finally, we have

$$\mathrm{E}\left[\frac{1}{n-2}\sigma^2\right] = \frac{1}{n-2}\mathrm{E}\left[\sum_{i=1}^n \hat{u}_i^2 | x_1, \ldots, x_n\right] = \frac{(n-2)\sigma^2}{n-2}$$

This completes the proof. ■

Remark. When you want to prove things, show that the assumptions hold for the theorems to be applied and also re-express stuff to make it easier

Assumption 3.8. We assume

(i) $u_1, \ldots, u_n | x_1, \ldots, x_n \sim \mathcal{N}(0, \sigma^2)$ (jointly)

(ii) $\mathrm{E}[u_i u_j | x_1, \ldots, x_n) = 0$ for $i \neq j$

Theorem 3.9 (Normality). Let **Assumption 3.8** hold. Then,

(a) For $k = 1, 2$

$$\hat{\beta}_k | x_1, \ldots, x_n \sim \mathcal{N}\{\beta_k, \mathrm{Var}(\hat{\beta}_k | x_1, \ldots, x_n)\}$$

(b) The estimator of the error variance

$$\hat{\sigma}^2 = \frac{1}{n-2} \sum_{i=1}^{n} \hat{u}_i^2$$

satisfies

$$(n-2)\frac{\hat{\sigma}^2}{\sigma^2} \sim \chi^2_{n-2}$$

(c) Under the null hypothesis that $\mathrm{H}_0 : \beta_j = \beta_j^0$ for $j = 1, 2$

$$t_{\beta_j} = \frac{\hat{\beta}_j - \beta_j^0}{\mathrm{se}(\hat{\beta}_j)} \sim t_{n-2}$$

where

$$\mathrm{se}(\hat{\beta}_1) = \sqrt{\frac{n^{-1}\sum_{i=1}^{n} x_i^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}\hat{\sigma}^2}, \quad \mathrm{se}(\hat{\beta}_2) = \sqrt{\frac{\hat{\sigma}^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}}$$

Proof. **Proof of (a)**: We express the OLS estimators as

$$\hat{\beta}_1 = \beta_1 + \bar{u} - (\hat{\beta}_2 - \beta_2)\bar{x}$$
$$\hat{\beta}_2 = \beta_2 + \frac{n^{-1}\sum_{i=1}^{n}(x_i - \bar{x})u_i}{n^{-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

Then, $\hat{\beta}_2 | x_1, \ldots, x_n$ is a linear combination of all u_i since x_i's are taken as given. Since $u_1, \ldots, u_n | x_1, \ldots, x_n$ are jointly normal, a linear combination of jointly normal variables are likewise normal. Hence $\hat{\beta}_2 | x_1, \ldots, x_n$ is normal. Likewise, since $\hat{\beta}_1$ is in terms of $\hat{\beta}_2$, $\hat{\beta}_1 | x_1, \ldots, x_n$ is also a linear combination of jointly normal variables $u_1, \ldots, u_n$. Hence, $\hat{\beta}_1 | x_1, \ldots, x_n$ is a normal random variable. It suffices to find the conditional mean and variance of $\hat{\beta}_1$ and $\hat{\beta}_2$. Appealing to **Theorem 3.7**, we know that

$$\hat{\beta}_1 | x_1, \ldots, x_n \sim \mathcal{N}\left(\beta_1, \frac{n^{-1}\sum_{i=1}^{n} x_i^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sigma^2\right)$$
$$\hat{\beta}_2 | x_1, \ldots, x_n \sim \mathcal{N}\left(\beta_2, \frac{\sigma^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}\right)$$

since we have derived the conditional mean and variance of $\hat{\beta}_1$ and $\hat{\beta}_2$.
Proof of (b): Defining $\hat{\sigma}^2 = \frac{1}{n-2}\sum_{i=1}^{n} \hat{u}_i^2$, we want to show that $(n-2)\hat{\sigma}^2/\sigma^2 \sim \chi^2_{n-2}$. I think this proof is not easy so let's take some time to go through the intuition.

1. First is to note that we are treating x_is as **fixed** and so all the assumptions over u_i conditioning on x_is hold

2. The main aim of the proof is to leverage the decomposition in **Theorem 3.7 (d)** and show that the three components are independent χ^2

To start the proof, note that

$$\sum_{i=1}^n \hat{u}_i^2 = \sum_{i=1}^n u_i^2 - \frac{\left(\sum_{i=1}^n u_i\right)^2}{n} - \frac{(\sum_{i=1}^n (x_i - \bar{x})u_i)^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

as per **Theorem 3.7 (d)**. Then, by the definition of $\hat{\sigma}^2 = \frac{1}{n-2}\sum_{i=1}^n \hat{u}_i^2$, we get

$$(n-2)\frac{\hat{\sigma}^2}{\sigma^2} = \sum_{i=1}^n (u_i/\sigma)^2 - \left(\sum_{i=1}^n u_i/(\sqrt{n}\sigma)\right)^2 - \left(\sum_{i=1}^2 c_i u_i/\sigma\right)^2$$

where $c_i = \frac{x_i - \bar{x}}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2}}$. SINCE WE TREAT x_i's as fixed, by **Assumption 3.8 (i)** (we implicitly condition on x_i), $u_i \sim \mathcal{N}(0, \sigma^2)$. Furthermore, since u_i's are jointly normal, covariance 0 implies independence which is the case by **Assumption 3.8 (ii)**. So,

$$\sum_{i=1}^n (u_i/\sigma)^2 = \sum_{i=1}^n Z_i^2 \sim \chi_n^2$$

where $Z_i \sim \mathcal{N}(0,1)$ i.i.d. Similarly, notice that $\sum_{i=1}^n u_i/(\sqrt{n}\sigma) \sim \mathcal{N}(0,1)$, so $\left(\sum_{i=1}^n u_i/(\sqrt{n}\sigma)\right)^2 \sim \chi_1^2$. Lastly, consider the term

$$\sum_{i=1}^2 c_i u_i/\sigma^2$$

We know this is normal, since it is a linear combination of indep. normal u_i. So, find the (conditional) mean and variance. By **Assumption 3.6 (i)**, $\mathrm{E}[u_i|x_1, \ldots, x_n] = 0$. Consider the variance

$$\begin{aligned}
\mathrm{Var}\left[\sum_{i=1}^n c_i u_i/\sigma | x_1, \ldots, x_n\right] &= \sum_{i=1}^n \frac{c_i^2}{\sigma^2}\mathrm{Var}(u_i|x_1, \ldots, x_n) + 2\sum_{i<j}\frac{c_i c_j}{\sigma^2}\mathrm{Cov}(u_i, u_j|x_1, \ldots, x_n) \\
&= \sum_{i=1}^n c_i^2 = 1
\end{aligned}$$

by **Assumption 3.6 (ii)** and the fact that $\sum_{i=1}^n c_i^2 = 1$ (this is easy to verify). So, $\sum_{i=1}^2 c_i u_i/\sigma^2 \sim \mathcal{N}(0,1)$. And so

$$\left(\sum_{i=1}^2 c_i u_i/\sigma\right)^2 \sim \chi_1^2$$

It remains to show that each term are independent. They are because they are jointly normal (since they are linear combinations of independent normals themselves, any linear combination of these three terms would yield a normal too), and the covariance between them is obviously 0 conditional on x_i's. Hence,

$$(n-2)\frac{\hat{\sigma}^2}{\sigma} \sim \chi_n^2 - \chi_1^2 - \chi_1^2 = \chi_{n-2}^2$$

Proof of (c): Define

$$\frac{\hat{\beta}_j - \beta_j^0}{\mathrm{s.d.}(\hat{\beta}_j)} = \frac{\hat{\beta}_j - \beta_j^0}{\sqrt{\mathrm{Var}(\hat{\beta}_j|x_1, \ldots, x_n)}}$$

Notice that the denominator is the s.d. of $\hat{\beta}_j$ not the s.e. of $\hat{\beta}_j$. Under **Assumption 3.8**, $\hat{\beta}_j - \beta_j^0 \sim \mathcal{N}\{0, \mathrm{Var}(\hat{\beta}_j|x_1, \ldots, x_n)\}$. Hence,

$$\frac{\hat{\beta}_j - \beta_j^0}{\mathrm{s.d.}(\hat{\beta}_\mathrm{j})} \sim \mathcal{N}(0,1)$$

Notice that **Theorem 3.9(c)** uses s.e.$(\hat{\beta}_j)$ formed using $\hat{\sigma}^2$ and **Theorem 3.9(b)** says that

$$(n-2)\frac{\hat{\sigma}^2}{\sigma^2} \sim \chi^2_{n-2}$$

Consider $\hat{\beta}_2$ where our object of interest is

$$t_{\beta_2} = \frac{\hat{\beta}_2 - \beta_2^0}{\text{s.e.}(\hat{\beta}_2)} = \frac{\hat{\beta}_2 - \beta_2^0}{\sqrt{\frac{\hat{\sigma}^2}{\sum_{i=1}^n (x_i - \bar{x})^2}}}$$

We can write this as

$$t_{\beta_2} = \frac{\hat{\beta}_2 - \beta_2^0}{\sqrt{\frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}}} \frac{\sqrt{\frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}}}{\sqrt{\frac{\hat{\sigma}^2}{\sum_{i=1}^n (x_i - \bar{x})^2}}} = \frac{\hat{\beta}_2 - \beta_2^0}{\sqrt{\frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}}} \left\{ \frac{(n-2)\frac{\hat{\sigma}^2}{\sigma^2}}{n-2} \right\}^{-1/2}$$

Notice that

$$t_{\beta_2} == \frac{\hat{\beta}_2 - \beta_2^0}{\sqrt{\frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}}} \left\{ \frac{(n-2)\frac{\hat{\sigma}^2}{\sigma^2}}{n-2} \right\}^{-1/2} \sim \frac{\mathcal{N}(0,1)}{\sqrt{\frac{\chi^2_{n-2}}{n-2}}} \sim t_{n-2}$$

where the denominator and numerator are independent. ■

3.1.2 Asymptotic Properties

Assumption 3.10 (Assumptions for Large Samples). We assume for large samples

(i) (x_i, u_i) for $i = 1, \ldots, n$ is independently and identically distributed with finite fourth moment and $\text{Var}(x_i) > 0$

(ii) $\text{E}(u_i | x_1, \ldots, x_n) = 0$

(iii) $\text{Var}(u_i | x_1, \ldots, x_n) = \sigma^2$, $0 < \sigma^2 < \infty$

Theorem 3.11. Under **Assumption 3.7** and let $\mu_x = \text{E}(x_i)$ and $\sigma_x^2 = \text{Var}(x_i)$, then

(a) Consistency: $\hat{\beta}_1$ and $\hat{\beta}_2$ are consistent for β_1 and β_2 i.e.

$$\hat{\beta}_1 \xrightarrow{p} \beta_1 \text{ and } \hat{\beta}_2 \xrightarrow{p} \beta_2$$

(b) Asymptotic Normality I: $\sqrt{n}(\hat{\beta}_1 - \beta_1) \xrightarrow{d} \mathcal{N}\{0, \sigma^2(1 + \mu_x^2/\sigma_x^2)\}$

(c) Asymptotic Normality II: $\sqrt{n}(\hat{\beta}_2 - \beta_2) \xrightarrow{d} \mathcal{N}\{0, \sigma^2(\sigma_x^2)^{-1}\}$

(d) Asymptotic Normality of Variance Estimator: The error variance estimator

$$\hat{\sigma^2} = \frac{1}{n-2}\sum_{i=1}^n \hat{u}_i^2$$

is consistent for σ^2 and

$$\sqrt{n}(\hat{\sigma}^2 - \sigma^2) \xrightarrow{d} \mathcal{N}\{0, \text{Var}(u_i^2)\}$$

(e) Asymptotic Normality of the t-statistic: Under the null hypothesis that $\text{H}_0 : \beta_j = \beta_j^0$, for $j = 1, 2$

$$t_{\beta_j} = \frac{\hat{\beta}_j - \beta_j^0}{\text{s.e.}(\hat{\beta}_j)} \xrightarrow{d} \mathcal{N}(0,1)$$

Proof. **Proof of (a)**: Since $\hat{\beta}_1$ depends on $\hat{\beta}_2$, we focus on $\hat{\beta}_2$ first. We can write

$$\hat{\beta}_2 = \beta_2 = \frac{n^{-1}\sum_{i=1}^{n}(x_i - \bar{x})u_i}{n^{-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

. We need to see the limiting behaviour of the denominator and numerator. For the denominator, note that

$$n^{-1}\sum_{i=1}^{n}(x_i\bar{x})^2 = n^{-1}\sum_{i=1}^{n}(x_i^2 - 2x_i\bar{x} + \bar{x}^2) = n^{-1}\sum_{i=1}^{n}x_i^2 - \bar{x}^2$$

. Since x_i is assumed to have a fourth moment and is i.i.d., by LLN, Continuous Mapping Theorem and Slutsky,

$$n^{-1}\sum_{i=1}^{n}x_i^2 - \bar{x}^2 \xrightarrow{p} \mathrm{E}(x_i^2) - \{\mathrm{E}(x_i)\}^2 = \mathrm{Var}(x_i)$$

. The numerator can be written as

$$n^{-1}\sum_{i=1}^{n}(x_i - \bar{x})u_i = n^{-1}\sum_{i=1}^{n}x_iu_i - \bar{x}n^{-1}\sum_{i=1}^{n}u_i$$

. By **Assumption 3.10 (i)**, (x_i, u_i) is i.i.d. and so is x_iu_i. Moreover, by LIE and **Assumption 3 (ii, iii)**,

$$\begin{aligned}\mathrm{E}(x_iu_i) &= \mathrm{E}\{\mathrm{E}(x_iu_i|x_i)\} = \mathrm{E}\{x_i\mathrm{E}(u_i|x_1,\ldots,x_n)\} = 0\\ \mathrm{Var}(x_iu_i) &= \mathrm{E}\{\mathrm{E}(x_i^2u_i^2|x_i)\} = \mathrm{E}\{x_i^2\mathrm{E}(u_i^2|x_1,\ldots,x_n)\} = \sigma^2\mathrm{E}(x_i^2) < \infty\end{aligned}$$

Then by LLN,

$$n^{-1}\sum_{i=1}^{n}x_iu_i \xrightarrow{p} \mathrm{E}(u_ix_i) = 0$$

By **Assumption 3.10 (i)** again and LLN,

$$\bar{x} \xrightarrow{p} \mathrm{E}(x_i), \quad n^{-1}\sum_{i=1}^{n}u_i \xrightarrow{p} \mathrm{E}(u_i)$$

where again by **Assumption 3(ii)** and LIE $\mathrm{E}(u_i) = \mathrm{E}\{\mathrm{E}(u_i|x_1,\ldots,x_n)\} = 0$. Hence,

$$\hat{\beta}_2 = \beta_2 = \frac{n^{-1}\sum_{i=1}^{n}(x_i - \bar{x})u_i}{n^{-1}\sum_{i=1}^{n}(x_i - \bar{x})^2} \xrightarrow{p} \beta_2 + \frac{0}{\mathrm{Var}(x_i)} = \beta_2$$

For $\hat{\beta}_1$, we can rewrite the estimator as

$$\hat{\beta}_1 = \beta_1 + \bar{u} - (\hat{\beta}_2 - \beta_2)\bar{x}$$

By **Assumption 3.10 (ii)**, LIE and LLN,

$$\bar{u} \xrightarrow{p} \mathrm{E}(u_i) = 0$$

Furthermore, we know that $\hat{\beta}_2 \xrightarrow{p} \beta_2$. By Slutsky,

$$\hat{\beta} = \beta_1 + \bar{u} - (\hat{\beta}_2 - \beta_2)\bar{x} \xrightarrow{p} \beta_1 + 0 - (\beta_2 - \beta_2)\bar{x} = \beta_1$$

Proof of (b): This uses the result of (c). Write

$$\sqrt{n}(\hat{\beta}_1 - \beta_1) = \sqrt{n}\bar{u} - \sqrt{n}(\hat{\beta}_2 - \beta_2)\bar{x}$$

Looking at the first term, by **Assumption 3.10(i)** and (c),

$$\bar{x} \xrightarrow{p} \mu_x, \quad \sqrt{n}(\hat{\beta}_2 - \beta_2 \xrightarrow{d} \mathcal{N}(0, \sigma^2\{\text{Var}(x_i)\}^{-1})$$

By Slutsky,

$$\sqrt{n}(\hat{\beta}_2 - \beta_2)\bar{x} \xrightarrow{d} \mathcal{N}(0, \sigma^2(\mu_x^2/\sigma_x^2))$$

Then, by CLT and **Assumption 3.10(i,ii,iii)**,

$$\sqrt{n}\bar{u} \xrightarrow{d} \mathcal{N}(0, \sigma^2)$$

We need to show independence of the two terms which we will do by showing that there is covariance 0 (and that they are jointly normal). It suffices to consider the i.i.d. pair $(u_i, (x_i - \mu_x)u_i)$ (NOTICE THAT WE USE μ_x INSTEAD of $\bar{x}$ because we consider asymptotics) so

$$\text{Cov}(u_i, (x_i - \bar{x})u_i) = \text{E}[(x_i - \bar{x})u_i^2] = \text{E}[(x_i - \mu_x)\text{E}\{u_i^2|x_1, \ldots, x_n)\}] = \sigma^2\text{E}(x_i - \mu_x) = 0$$

Notice that because (i) $(u_i, (x_i - \mu_x)u_i)$ is i.i.d. and (ii) they have a finite mean vector and (iii) the covariance matrix

$$\Sigma = \begin{pmatrix} \sigma^2 & 0 \\ 0 & \sigma^2\sigma_x^2 \end{pmatrix}$$

is positive definite, the pair $(u_i, (x_i - \mu_x)u_i)$ is bivariate normal. Covariance 0 implies independence and so they are independent. The variance of the two terms is thus a simple summation of their individual variances i.e. $\sigma^2(1 + \mu_x^2/\sigma_x^2)$. Consequently,

$$\sqrt{n}(\hat{\beta}_1 - \beta_1) = \sqrt{n}\bar{u} - \sqrt{n}(\hat{\beta}_2 - \beta_2)\bar{x} \xrightarrow{d} \mathcal{N}(0, \sigma^2(1 + \mu_x^2/\sigma_x^2))$$

Proof of (c): Add and subtract $\mu_x = \text{E}(x_i)$ as follows

$$\begin{aligned} n^{-1/2}\sum_{i=1}^{n}(x_i - \bar{x})u_i &= n^{-1/2}\sum_{i=1}^{n}(x_i - \mu_x + \mu_x - \bar{x})u_i \\ &= n^{-1/2}\sum_{i=1}^{n}(x_i - \mu_x)u_i - (\bar{x} - \mu_x)n^{-1/2}\sum_{i=1}^{n}u_i \end{aligned}$$

Want to show that the first term converges in distribution to a normal and the second term converges in probability to zero.

Second term: by **Assumption 3.10(i)** and LLN, $\bar{x} - \mu_x \xrightarrow{p} 0$ while by the CLT

$$n^{-1/2}\sum_{i=1}^{n}u_i \xrightarrow{d} \mathcal{N}(0, \sigma^2)$$

By the Slutsky theorem,

$$(\bar{x} - \mu_x)n^{-1/2}\sum_{i=1}^{n}u_i \xrightarrow{p} 0$$

First term

$$n^{-1/2}\sum_{i=1}^{n}(x_i - \mu_x)u_i$$

By **Assumption 3.10(i)**, $(x_i - \mu_x)u_i$ is i.i.d. Moreover, by **Assumption 3.10(i, ii)**, and LIE

$$\text{E}\{(x_i - \mu_x)u_i\} = 0$$

and the variance is

$$\text{Var}\{(x_i - \mu_x)u_i\} = \text{E}\{(x_i - \mu_x)^2u_i^2\}$$

By LIE,

$$\begin{aligned}\mathrm{E}\{(x_i-\mu_x)^2u_i^2\} &= \mathrm{E}[\mathrm{E}\{(x_i-\mu_x)^2u_i^2|x_1,\dots,x_n\}]\\ &= \mathrm{E}[(x_i-\mu_x)^2\mathrm{E}(u_i^2|x_1,\dots,x_n)]\end{aligned}$$

By **Assumption 3.10 (ii, iii)**, $\mathrm{E}(u_i^2|x_1,\dots,x_n)=\sigma^2$ s.t.

$$\mathrm{Var}\{(x_i-\mu_x)u_i\}=\sigma^2\mathrm{E}\{(x_i-\mu_x)^2\}=\sigma^2\mathrm{Var}(x_i)$$

Therefore, the quantity of interest satisfies the CLT so that

$$n^{-1/2}\sum_{i=1}^n(x_i-\mu_x)u_i \xrightarrow{d} \mathcal{N}(0,\sigma^2\mathrm{Var}(x_i)\}$$

In turn,

$$\sqrt{n}(\hat\beta_2-\beta_2)=\frac{n^{-1/2}\sum_{i=1}^n(x_i-\bar x)u_i}{n^{-1}\sum_{i=1}^n(x_i-\bar x)^2}\xrightarrow{d}\frac{\mathcal{N}\{0,\sigma^2\mathrm{Var}(x_i)\}}{\mathrm{Var}(x_i)}\sim\mathcal{N}[0,\sigma^2\{\mathrm{Var}(x_i)\}^{-1}]$$

Proof of (d): To first prove consistency (very tedious). $\hat\sigma^2=\frac{1}{n-2}\sum_{i=1}^n\hat u_i^2$. Then,

$$\begin{aligned}\hat\sigma^2 &= \frac{n}{n-2}\frac{1}{n}\sum_{i=1}^n\hat u_i^2\\ &= \frac{n}{n-2}\frac{1}{n}\sum_{i=1}^n(y_i-\hat\beta_1-\hat\beta_2x_i)^2\\ &= \frac{n}{n-2}\frac{1}{n}\sum_{i=1}^n(u_i-(\hat\beta_1-\beta_1)-(\hat\beta_2-\beta_2)x_i)^2\\ &= \frac{n}{n-2}\frac{1}{n}\sum_{i=1}^n(u_i^2-2u_i(\hat\beta_1-\beta_1)-2u_i(\hat\beta_2-\beta_2)x_i+2(\hat\beta_1-\beta_1)(\hat\beta_2-\beta_2)x_i+(\hat\beta_1-\beta_1)^2+(\hat\beta_2-\beta_2)^2x_i^2)\\ &= \frac{n}{n-2}\frac{1}{n}\sum_{i=1}^n u_i^2-2(\hat\beta_1-\beta_1)\frac{n}{n-2}\frac{1}{n}\sum_{i=1}^n u_i-2(\hat\beta_2-\beta_2)\frac{n}{n-2}\frac{1}{n}\sum_{i=1}^n u_ix_i\\ &+2(\hat\beta_1-\beta_1)(\hat\beta_2-\beta_2)\frac{n}{n-2}\frac{1}{n}\sum_{i=1}^n x_i+(\hat\beta_1-\beta_1)^2+(\hat\beta_2-\beta_2)^2\frac{n}{n-2}\frac{1}{n}\sum_{i=1}^n x_i^2\end{aligned}$$

Now, notice that every term except the first converges to 0 in probability, since $\hat\beta_1\xrightarrow{p}\beta_1$ and $\hat\beta_2\xrightarrow{p}\beta_2$. By the Continuous Mapping Theorem, we know that $(\hat\beta_j-\beta_j)^2\xrightarrow{p}0$ for $j=1,2$ and that by LLN and **Assumption 3.9**, the first and second moments of u_i and x_i are finite. Using the fact that $n/(n-2)\longrightarrow 1$ in the limit, since u_i is i.i.d. by LLN

$$\frac{n}{n-2}\frac{1}{n}\sum_{i=1}^n u_i^2\xrightarrow{p}\sigma^2$$

Hence,

$$\hat\sigma^2=\frac{1}{n-2}\sum_{i=1}^n\hat u_i^2\xrightarrow{p}\sigma^2$$

To prove normality in the limit, we can use the last decomposition. Since we have shown

$$\frac{n}{n-2}\frac{1}{n}\sum_{i=1}^n u_i^2\xrightarrow{p}\sigma^2$$

we just need to consider the other terms and their limit distribution (also showing they are jointly normal so we might add their variances up – we already know their mean is 0 since they converge in probability to

0).
Proof of (e): Consider

$$t_{\beta_2} = \frac{\hat{\beta}_2 - \beta_2^0}{\text{s.e.}(\hat{\beta}_2)}$$

with

$$\text{s.e.} = \sqrt{\frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}}$$

We can write

$$t_{\beta_2} = \frac{\sqrt{n}(\hat{\beta}_2 - \beta_2^0)}{\sqrt{\frac{\sigma^2}{n^{-1}\sum_{i=1}^n (x_i - \bar{x})^2}}}$$

By (c) we showed that

$$\sqrt{n}(\hat{\beta}_2 - \beta_2^0) \xrightarrow{d} \mathcal{N}(0, \sigma^2\{\text{Var}(x_i)\}^{-1})$$

By (d)

$$\hat{\sigma}^2 \xrightarrow{p} \sigma^2$$

By LLN

$$n^{-1}\sum_{i=1}^n (x_i - \bar{x})^2 \xrightarrow{p} \text{Var}(x_i)$$

Hence by Slutsky,

$$t_{\beta_2} = \frac{\sqrt{n}(\hat{\beta}_2 - \beta_2^0)}{\sqrt{\frac{\sigma^2}{n^{-1}\sum_{i=1}^n (x_i - \bar{x})^2}}} \xrightarrow{d} \frac{\mathcal{N}(0, \sigma^2\{\text{Var}(x_i)\}^{-1})}{\sqrt{\sigma^2/\text{Var}(x_i)}}$$

and so

$$t_{\beta_2} \xrightarrow{d} \mathcal{N}(0, 1)$$

■

3.2 Multivariate OLS

Consider the following regression model

$$y_i = \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + \cdots + \beta_K x_{Ki} + ui$$

where we let $x_{1i} = 1$ for all $i = 1, \ldots, n$ if we want a constant in the model. We may represent this in matrix form

$$\mathbf{y} = \mathbf{X}\beta + \mathbf{U}$$

where

$$\mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_n \end{pmatrix}, \quad \mathbf{X} = \begin{pmatrix} x_{11} & x_{21} & \cdots & x_{K1} \\ x_{12} & x_{22} & \cdots & x_{K2} \\ x_{13} & x_{23} & \cdots & x_{K3} \\ \vdots & \vdots & \ddots & \vdots \\ x_{1n} & x_{2n} & \cdots & x_{Kn} \end{pmatrix}, \quad \mathbf{U} = \begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ \vdots \\ u_n \end{pmatrix}$$

and $\beta = (\beta_1\ \beta_2\ \cdots\ \beta_K)'$ i.e. $\mathbf{y}$ is a $n \times 1$ matrix/vector, $\mathbf{X}$ is a $n \times K$ matrix and $\mathbf{U}$ is a $n \times 1$ matrix/vector and β is a $n \times 1$ matrix/vector.

Definition 3.12 (OLS Problem). Let $\mathbf{U}(\beta) = \mathbf{y} - \mathbf{X}\beta$. Then, the OLS problem in matrix form is

$$\min_{\beta \in \mathbb{R}^K} \mathbf{U}(\beta)'\mathbf{U}(\beta) = \min_{\beta \in \mathbb{R}^K} (\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta)$$

Proposition 3.13 (OLS Estimator). Consider the OLS problem

$$\min_{\beta \in \mathbb{R}^K} \mathbf{U}(\beta)'\mathbf{U}(\beta) = \min_{\beta \in \mathbb{R}^K} (\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta)$$

Then, our estimator that solves the OLS problem is

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\mathbf{y})$$

Proof. First, write

$$\min_{\beta \in \mathbb{R}^K} \mathbf{U}(\beta)'\mathbf{U}(\beta) = \min_{\beta \in \mathbb{R}^K} (\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta)$$

Using the transposition and product rules we write

$$\min_{\beta \in \mathbb{R}^K} \mathbf{U}(\beta)'\mathbf{U}(\beta) = \min_{\beta \in \mathbb{R}^K} \mathbf{y}'\mathbf{y} - \mathbf{y}'\mathbf{X}\beta - \beta'\mathbf{X}'\mathbf{y} + \beta'\mathbf{X}'\mathbf{X}\beta$$

We note that $\mathbf{y}'\mathbf{X}\beta = \beta'\mathbf{X}'\mathbf{y}$ (because it is a 1×1 matrix/scalar and so trivially symmetric) and hence

$$\min_{\beta \in \mathbb{R}^K} \mathbf{U}(\beta)'\mathbf{U}(\beta) = \min_{\beta \in \mathbb{R}^K} \mathbf{y}'\mathbf{y} - 2\beta'\mathbf{X}'\mathbf{y} + \beta'\mathbf{X}'\mathbf{X}\beta$$

Then,

$$L_\beta = \mathbf{y}'\mathbf{y} - 2\beta'\mathbf{X}'\mathbf{y} + \beta'\mathbf{X}'\mathbf{X}\beta$$

Our FOC is

$$\frac{\partial L_\beta}{\partial \beta} = -2\mathbf{X}'\mathbf{y} + 2\mathbf{X}'\mathbf{X}\beta = 0$$

where we have used the first rule $\frac{\partial(\mathbf{A}\mathbf{x})}{\partial \mathbf{x}} = \mathbf{A}'$

$$\frac{\partial(-2\mathbf{y}'\mathbf{X})\beta}{\partial \beta} = -2(\mathbf{y}'\mathbf{X})' = -2\mathbf{X}'\mathbf{y}$$

and the second rule $\frac{\partial(\mathbf{x}'\mathbf{A}\mathbf{x})}{\partial \mathbf{x}} = (\mathbf{A} + \mathbf{A}')\mathbf{x}$

$$\frac{\partial(\beta'\mathbf{X}'\mathbf{X}\beta)}{\partial \mathbf{x}} = (\mathbf{X}'\mathbf{X} + (\mathbf{X}'\mathbf{X})')\beta = 2\mathbf{X}'\mathbf{X}\beta$$

■

Example 3.14 (Matrix Simple Regression Model). Consider the simple model

$$\mathbf{y} = \mathbf{X}\beta + \mathbf{U}$$

where

$$\mathbf{X} = \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix} \text{ and } \mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$$

The OLS estimator is

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\mathbf{y})$$

and in the simple model, we have

$$\mathbf{X}'\mathbf{X} = \begin{pmatrix} 1 & 1 & \cdots & 1 \\ x_1 & x_2 & \cdots & x_n \end{pmatrix} \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix} = \begin{pmatrix} n & \sum_{i=1}^n x_i \\ \sum_{i=1}^n x_i & \sum_{i=1}^n x_i^2 \end{pmatrix}$$

and

$$\mathbf{X}'\mathbf{y} = \begin{pmatrix} 1 & 1 & \cdots & 1 \\ x_1 & x_2 & \cdots & x_n \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} \sum_{i=1}^n y_i \\ \sum_{i=1}^n x_i y_i \end{pmatrix}$$

We yield

$$\hat{\beta} = \begin{pmatrix} n & \sum_{i=1}^n x_i \\ \sum_{i=1}^n x_i & \sum_{i=1}^n x_i^2 \end{pmatrix}^{-1} \begin{pmatrix} \sum_{i=1}^n y_i \\ \sum_{i=1}^n x_i y_i \end{pmatrix} = \frac{1}{n\sum_{i=1}^n x_i^2 - (\sum_{i=1}^n x_i)^2} \begin{pmatrix} \sum_{i=1}^n x_i^2 & -\sum_{i=1}^n x_i \\ -\sum_{i=1}^n x_i & n \end{pmatrix} \begin{pmatrix} \sum_{i=1}^n y_i \\ \sum_{i=1}^n x_i y_i \end{pmatrix}$$

since the inverse of $\mathbf{X}'\mathbf{X}$ is

$$\mathbf{X}'\mathbf{X} = \frac{1}{n\sum_{i=1}^n x_i^2 - (\sum_{i=1}^n x_i)^2} \begin{pmatrix} \sum_{i=1}^n x_i^2 & -\sum_{i=1}^n x_i \\ -\sum_{i=1}^n x_i & n \end{pmatrix}$$

Hence

$$\hat{\beta} = \frac{1}{n\sum_{i=1}^n x_i^2 - (\sum_{i=1}^n x_i)^2} \begin{pmatrix} \sum_{i=1}^n x_i^2 \sum_{i=1}^n y_i - \sum_{i=1}^n x_i \sum_{i=1}^n x_i y_i \\ -\sum_{i=1}^n x_i \sum_{i=1}^n y_i + n\sum_{i=1}^n x_i y_i \end{pmatrix}$$

Hence,

$$\begin{aligned} \hat{\beta}_1 &= \frac{\sum_{i=1}^n x_i^2 \sum_{i=1}^n y_i - \sum_{i=1}^n x_i \sum_{i=1}^n x_i y_i}{n\sum_{i=1}^n x_i^2 - (\sum_{i=1}^n x_i)^2} \\ \hat{\beta}_2 &= \frac{-\sum_{i=1}^n x_i \sum_{i=1}^n y_i + n\sum_{i=1}^n x_i y_i}{n\sum_{i=1}^n x_i^2 - (\sum_{i=1}^n x_i)^2} \end{aligned}$$

This gives us our original estimators

$$\begin{aligned} \hat{\beta}_1 &= \bar{y} - \hat{\beta}_2 \bar{x} \\ \hat{\beta}_2 &= \frac{\sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2} \end{aligned}$$

3.2.1 Finite Sample Properties

Assumption 3.15 (FS). We assume

(i) Full rank: There is no exact linear relationship among any of the regressor in the model:

$$\mathbf{X} \text{ is an } n \times K \text{ matrix with rank } K$$

This assumption is also known as the identification condition or no perfect multicolinearity

(ii) Mean Independence: $\mathrm{E}(\mathbf{U}|\mathbf{X}) = 0$

(iii) (Conditional) Homoskedasticity and nonautocorrelation:

$$\begin{aligned} \mathrm{Var}(u_i|\mathbf{X}) &= \sigma^2 \quad \text{for all } i = 1, \ldots, n \\ \mathrm{Cov}(u_i, u_j|\mathbf{X}) &= 0 \quad \text{for all } i \neq j \end{aligned}$$

In matrix form,

$$\mathrm{E}[\mathbf{U}\mathbf{U}'|\mathbf{X}] = \sigma^2 \mathbf{I}_n$$

where $\mathbf{I}_n$ is an $n \times n$ identity matrix

Remark. Expanding the Conditional Homoskedasticity and no-autocorrelation assumption

$$
\begin{aligned}
\mathrm{E}[\mathbf{U}\mathbf{U}'|\mathbf{X}] &= \mathrm{E}\left\{\begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{pmatrix}\right\} \\
&= \mathrm{E}\left[\begin{pmatrix} u_1^2 & u_1u_2 & \cdots & u_1u_n \\ u_2u_1 & u_2^2 & \cdots & u_2u_n \\ \vdots & \vdots & \ddots & \vdots \\ u_nu_1 & u_nu_2 & \cdots & u_n^2 \end{pmatrix} \mid \mathbf{X}\right] \\
&= \begin{pmatrix} \mathrm{E}(u_1^2|\mathbf{X}) & \mathrm{E}(u_1u_2|\mathbf{X}) & \cdots & \mathrm{E}(u_1u_n|\mathbf{X}) \\ \mathrm{E}(u_2u_1|\mathbf{X}) & \mathrm{E}(u_2^2|\mathbf{X}) & \cdots & \mathrm{E}(u_2u_n|\mathbf{X}) \\ \vdots & \vdots & \ddots & \vdots \\ \mathrm{E}(u_nu_1|\mathbf{X}) & \mathrm{E}(u_nu_2|\mathbf{X}) & \cdots & \mathrm{E}(u_n^2|\mathbf{X}) \end{pmatrix}
\end{aligned}
$$

Since $\mathrm{E}(\mathbf{U}|\mathbf{X}) = \mathbf{0}$, the elements on the main diagonal are the conditional variances of u_i, whereas the elements outside the main diagonal are the conditional covariances. The matrix is obviously symmetric. Thus, the assumption that $\mathrm{E}[\mathbf{U}\mathbf{U}'|\mathbf{X}] = \sigma^2\mathbf{I}_n$ implies that

$$
\mathrm{E}(\mathbf{U}\mathbf{U}|\mathbf{X}) = \sigma^2 \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}
$$

Hence, all the (conditional) variances are equal to σ^2 and all covariances are equal to zero

Theorem 3.16 (OLS Finite Sample Properties). Let Assumption 3.15 hold. Consider the model equation $\mathbf{y} = \mathbf{X}\beta + \mathbf{U}$ and the OLS estimator $\hat{\beta}$. Then,

(a) $\hat{\beta}$ is unbiased i.e.
$$\mathrm{E}(\hat{\beta}) = \beta$$

(b) The conditional variance of $\hat{\beta}$ is
$$\mathrm{Var}(\hat{\beta}|\mathbf{X}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$$

(c) Gauss-Markov Theorem: The OLS estimator $\hat{\beta}$ of the unknown β is BLUE: the best (most efficient) linear conditionally unbiased estimator

(d) The estimator of the error variance
$$\hat{\sigma}^2 = \frac{1}{n-K}\sum_{i=1}^{n}\hat{u}_i^2$$
is unbiased, i.e. $\mathrm{E}(\hat{\sigma}^2) = \sigma^2$.

Proof. Proof of (a): WTS $\hat{\beta} = \beta$. As usual, we want to write $\hat{\beta}$ in terms of β. We may write $\hat{\beta} = \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{U}$ since

$$
\begin{aligned}
\hat{\beta} &= (\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\mathbf{y}) \\
&= (\mathbf{X}'\mathbf{X})^{-1}\{\mathbf{X}'(\mathbf{X}\beta + \mathbf{U})\} \\
&= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}\beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{U} \\
&= \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{U}
\end{aligned}
$$

where the second equality follows by substituting the model equation $\mathbf{y} = \mathbf{X}\beta + \mathbf{U}$ and the fourth equality follows by the fact that $(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X} = \mathbf{I}_K$. Then, by LIE,

$$\begin{aligned}
\mathrm{E}(\hat{\beta}) &= \mathrm{E}[\mathrm{E}(\beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{U}|\mathbf{X})] \\
&= \mathrm{E}(\beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathrm{E}(\mathbf{U}|\mathbf{X})) \\
&= \beta
\end{aligned}$$

since $\mathrm{E}[\mathbf{U}|\mathbf{X}]$ by mean independence assumption.
Proof of (b): by definition, $\mathrm{Var}(\hat{\beta}|\mathbf{X}) = \mathrm{E}\{(\hat{\beta}-\beta)(\hat{\beta}-\beta)'|\mathbf{X}\}$ with $\hat{\beta}-\beta = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{U}$ from the expression of $\hat{\beta}$ in terms of β i.e. the conditional variance $K \times K$ matrix of $\hat{\beta}$ is the conditional expectation of the outer product of $\hat{\beta} - \beta$. Substituting $\hat{\beta} - \beta = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{U}$ into $\mathrm{Var}(\hat{\beta}|\mathbf{X})$, we get

$$\mathrm{Var}(\hat{\beta}|\mathbf{X}) = \mathrm{E}[\{(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{U}\}\{(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{U}\}'|\mathbf{X}]$$

By the property that $(\mathbf{ABC})' = \mathbf{C}'\mathbf{B}'\mathbf{A}'$, we can write

$$\mathrm{Var}(\hat{\beta}|\mathbf{X}) = \mathrm{E}[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{U}\mathbf{U}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}|\mathbf{X}]$$

where in particular we know that $(\mathbf{X}'\mathbf{X})^{-1}$ is symmetric and hence equals its own transpose. Since we condition on $\mathbf{X}$, we yield

$$\mathrm{Var}(\hat{\beta}|\mathbf{X}) = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathrm{E}[\mathbf{U}\mathbf{U}'|\mathbf{X}]\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}$$

By assumption, $\mathrm{E}[\mathbf{U}\mathbf{U}'|\mathbf{X}) = \sigma^2\mathbf{I}_n$. Hence,

$$\mathrm{Var}(\hat{\beta}|\mathbf{X}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}$$

since σ^2 is a scalar and $\mathbf{I}_n$ disappears. Furthermore, since $\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X} = \mathbf{I}_K$, we yield

$$\mathrm{Var}(\hat{\beta}|\mathbf{X}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$$

Proof of (c): Here, we want to prove that $\hat{\beta}$ has the smallest variance of all linear estimators. We have already shown that $\hat{\beta}$ is unbiased. Then, since $\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} = \mathbf{A}\mathbf{y}$ where $\mathbf{A} \equiv (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$, it is linear in $\mathbf{y}$. Let $\mathbf{b}$ be an arbitrary unbiased linear estimator of β, which implies that $\mathbf{b} = \mathbf{C}\mathbf{y}$ where $\mathbf{C}$ is a $K \times n$ matrix. If $\mathbf{b}$ is unbiased, then

$$\mathrm{E}[\mathbf{C}\mathbf{y}|\mathbf{X}] = \mathrm{E}[\mathbf{C}\mathbf{X}\beta + \mathbf{C}\mathbf{U}|\mathbf{X}] = \beta$$

This implies that $\mathbf{C}\mathbf{X} = \mathbf{I}_K$, since $\mathbf{C}$ depends only on $\mathbf{X}$ and thus $\mathbf{E}[\mathbf{U}|\mathbf{X}] = 0$. Write $\mathbf{C}$ as

$$\mathbf{C} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + \mathbf{D}$$

with $\mathbf{D}\mathbf{X} = 0$. This is because we need to satisfy $\mathbf{C}\mathbf{X} = \mathbf{I}_K$, since $\mathbf{C}\mathbf{X} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X} + \mathbf{D}\mathbf{X} = \mathbf{I}_K + \mathbf{0}$. Then, we yield

$$\mathrm{Var}(\mathbf{b}|\mathbf{X}) = \sigma^2\mathbf{C}\mathbf{C}'$$

which can be found by replacing $(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ with $\mathbf{C}$. Let $\mathbf{D} = \mathbf{C} - (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$, so $\mathbf{D}\mathbf{y} = \mathbf{b} - \beta$. Then,

$$\mathrm{Var}(\mathbf{b}|\mathbf{X}) = \sigma^2[(\mathbf{D} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}')(\mathbf{D} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}')']$$

Using the fact that $\mathbf{D}\mathbf{X} = \mathbf{0}$, we yield

$$\mathbf{Var}(\mathbf{b}|\mathbf{X}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1} + \sigma^2\mathbf{D}\mathbf{D}' = \mathrm{Var}(\hat{\beta}|\mathbf{X}) + \sigma^2\mathbf{D}\mathbf{D}'$$

Notice that for any $\mathbf{q}$, $\mathbf{q}'\mathbf{D}\mathbf{D}'\mathbf{q} = (\mathbf{D}'\mathbf{q})'(\mathbf{D}'\mathbf{q}) = ||\mathbf{D}'\mathbf{q}||^2 \geq 0$. Hence, $\mathbf{D}\mathbf{D}'$ is positive semi-definite. Hence, the conditional covariance matrix of $\mathbf{b}$ equals that of $\hat{\beta}$ plus a nonnegative definite matrix. Therefore, every quadratic form in $\mathrm{Var}(\mathbf{b}|\mathbf{X})$ is larger than the corresponding quadratic form in $\mathrm{Var}(\hat{\beta}|\mathbf{X})$, which implies that $\hat{\beta}$ is BLUE.
Proof of (d): Most of the algebraic work is done in the remark below. Notice that we can write

$$\hat{\sigma}^2 = \frac{1}{n-K}\sum_{i=1}^{n}\hat{u}_i^2 = \frac{\hat{\mathbf{U}}'\hat{\mathbf{U}}}{n-K}$$

We want to show that

$$\mathrm{E}[\hat{\sigma}^2] = \mathrm{E}\left[\frac{\hat{\mathbf{U}}'\hat{\mathbf{U}}}{n-K}\right] = \sigma^2$$

By LIE, $\mathrm{E}[\hat{\mathbf{U}}'\hat{\mathbf{U}}] = \mathrm{E}[\mathrm{E}(\hat{\mathbf{U}}'\hat{\mathbf{U}}|\mathbf{X})]$. Since $\hat{\mathbf{U}}'\hat{\mathbf{U}} = tr(\mathbf{U}'\mathbf{M}\mathbf{U}) = tr(\mathbf{M}\mathbf{U}\mathbf{U}')$,

$$\mathrm{E}(\hat{\mathbf{U}}'\hat{\mathbf{U}}|\mathbf{X}) = \mathrm{E}[tr(\mathbf{M}\mathbf{U}\mathbf{U}')|\mathbf{X}]$$

Notice that the trace of a matrix is just the sum of its diagonal, and hence by linearity of expectations, we have for any matrix $\mathbf{A}$ $\mathrm{E}[tr(\mathbf{A})] = tr(\mathrm{E}[\mathbf{A}])$. Then, we can write

$$\mathrm{E}(\hat{\mathbf{U}}'\hat{\mathbf{U}}|\mathbf{X}) = \mathrm{E}[tr(\mathbf{M}\mathbf{U}\mathbf{U}')|\mathbf{X}] = tr(\mathrm{E}[\mathbf{M}\mathbf{U}\mathbf{U}'|\mathbf{X}])$$

Using the fact that $\mathbf{M}$ is a function of $\mathbf{X}$,

$$\mathrm{E}(\hat{\mathbf{U}}'\hat{\mathbf{U}}|\mathbf{X}) = tr(\mathbf{M}\mathrm{E}(\mathbf{U}\mathbf{U}'|\mathbf{X}))$$

By assumption (iii), $\mathrm{E}[\mathbf{U}\mathbf{U}'|\mathbf{X}] = \sigma^2$ which gives

$$\mathrm{E}(\hat{\mathbf{U}}'\hat{\mathbf{U}}|\mathbf{X}) = \sigma^2 tr(\mathbf{M})$$

Recall our definition of $\mathbf{M} = \{\mathbf{I}_n - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\}$. By the properties of a trace,

$$\begin{aligned} tr(\mathbf{M}) &= tr\{\mathbf{I}_n - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\} \\ &= tr(\mathbf{I}_n) = tr\{\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\} \\ &= tr(\mathbf{I}_n) - \mathrm{tr}(\mathbf{I}_K) = n - K \end{aligned}$$

Hence,

$$\mathrm{E}(\hat{\mathbf{U}}'\hat{\mathbf{U}}|\mathbf{X}) = \sigma^2 tr(\mathbf{M}) = \sigma^2(n-K)$$

By LIE,

$$\mathrm{E}[\hat{\mathbf{U}}'\hat{\mathbf{U}}] = \mathrm{E}[\mathrm{E}(\hat{\mathbf{U}}'\hat{\mathbf{U}}|\mathbf{X})] = \sigma^2(n-K)$$

This gives the desired result i.e. that the estimator of the error variance is unbiased

$$\mathrm{E}(\hat{\sigma}^2) = \mathrm{E}\left(\frac{\hat{\mathbf{U}}'\hat{\mathbf{U}}}{n-K}\right) = \sigma^2$$

■

The residuals have special properties.

Remark. We can write

$$\hat{\sigma}^2 = \frac{1}{n-K}\sum_{i=1}^{n}\hat{u}_i^2 = \frac{\hat{\mathbf{U}}'\hat{\mathbf{U}}}{n-K}$$

where $\hat{\mathbf{U}}' = \mathbf{y} - \mathbf{X}\hat{\beta}$. This can be further expressed as

$$\hat{\mathbf{U}}' = \mathbf{y} - \mathbf{X}\hat{\beta} = \mathbf{y} - \mathbf{X}\{(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}\} = \{\mathbf{I}_n - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\}\mathbf{y}$$

Let $\mathbf{M} = \{\mathbf{I}_n - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\}$ so that

$$\hat{\mathbf{U}} = \mathbf{M}\mathbf{y} = \mathbf{M}(\mathbf{X}\beta + \mathbf{U}) = \mathbf{M}\mathbf{U}$$

since $\mathbf{M}\mathbf{X} = \{\mathbf{I}_n - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\}\mathbf{X} = 0$ such that

$$\hat{\mathbf{U}} = \mathbf{M}\mathbf{U}$$

Since we want to know $\hat{\sigma}^2 = \frac{\hat{\mathbf{U}}'\hat{\mathbf{U}}}{n-K}$, and

$$\hat{\sigma}^2 = \frac{\hat{\mathbf{U}}'\hat{\mathbf{U}}}{n-K} = \frac{\mathbf{U}'\mathbf{M}'\mathbf{M}\mathbf{U}}{n-K}$$

Note that $\mathbf{M}$ is symmetric (so that $\mathbf{M} = \mathbf{M}'$) and idempotent (so that $\mathbf{M}\mathbf{M} = \mathbf{M}^2 = \mathbf{M}$). Hence,

$$\hat{\mathbf{U}}'\hat{\mathbf{U}} = \mathbf{U}'\mathbf{M}'\mathbf{M}\mathbf{U} = \mathbf{U}'\mathbf{M}\mathbf{M}\mathbf{U} = \mathbf{U}'\mathbf{M}\mathbf{U}$$

The sum of squared residuals $\hat{\mathbf{U}}'\hat{\mathbf{U}} = \mathbf{U}'\mathbf{M}\mathbf{U}$ is a scalar and is a 1×1 matrix. Therefore,

$$\hat{\mathbf{U}}'\hat{\mathbf{U}} = tr(\hat{\mathbf{U}}'\hat{\mathbf{U}}) = tr(\mathbf{U}'\mathbf{M}\mathbf{U})$$

The trace of a matrix satisfies $tr(\mathbf{U}'\mathbf{M}\mathbf{U}) = tr(\mathbf{M}\mathbf{U}\mathbf{U}')$ so that

$$\hat{\mathbf{U}}'\hat{\mathbf{U}} = tr(\mathbf{U}'\mathbf{M}\mathbf{U}) = tr(\mathbf{M}\mathbf{U}\mathbf{U}')$$

Remark. In the proof of the Gauss-Markov Theorem ($\hat{\beta}$ being BLUE), we needed to say that the conditional variance matrix for an arbitrary unbiased linear estimator is "larger" than that of $\hat{\beta}$. We usually use a Loewner order. For symmetric matrices $\mathbf{A}, \mathbf{B}$, we say $\mathbf{A} \succeq \mathbf{B}$ ("$\mathbf{A}$ is larger") iff

$$\mathbf{q}'\,\mathbf{A}\,\mathbf{q} \;\geq\; \mathbf{q}'\,\mathbf{B}\,\mathbf{q} \quad \text{for every vector } \mathbf{q}.$$

Why this definition? For any linear combination $\mathbf{q}'\boldsymbol{\theta}$,

$$\mathrm{Var}(\mathbf{q}'\boldsymbol{\theta}) \;=\; \mathbf{q}'\;\mathrm{Var}(\boldsymbol{\theta})\,\mathbf{q}.$$

So comparing quadratic forms for all $\mathbf{q}$ is exactly comparing the variances of all linear combinations of the estimator.
Now apply it to our case:

$$\mathrm{Var}(\mathbf{b} \mid \mathbf{X}) \;=\; \mathrm{Var}(\hat{\boldsymbol{\beta}} \mid \mathbf{X}) \;+\; \sigma^2\,\mathbf{D}\mathbf{D}', \qquad \sigma^2\,\mathbf{D}\mathbf{D}' \succeq \mathbf{0}.$$

For any $\mathbf{q}$,

$$\underbrace{\mathbf{q}'\;\mathrm{Var}(\mathbf{b} \mid \mathbf{X})\,\mathbf{q}}_{\mathrm{Var}(\mathbf{q}'\mathbf{b}|\mathbf{X})} \;=\; \underbrace{\mathbf{q}'\;\mathrm{Var}(\hat{\boldsymbol{\beta}} \mid \mathbf{X})\,\mathbf{q}}_{\mathrm{Var}(\mathbf{q}'\hat{\boldsymbol{\beta}}|\mathbf{X})} + \sigma^2\;\underbrace{\mathbf{q}'\,\mathbf{D}\mathbf{D}'\,\mathbf{q}}_{\geq 0}.$$

Hence $\mathrm{Var}(\mathbf{q}'\mathbf{b} \mid \mathbf{X}) \;\geq\; \mathrm{Var}(\mathbf{q}'\hat{\boldsymbol{\beta}} \mid \mathbf{X})$ for all $\mathbf{q}$, which is precisely what "the variance is larger" means in the Loewner order.

Assumption 3.17 (FSN). We assume

(i) Full rank: There is no exact linear relationship among any of the regressor in the model:

$$\mathbf{X} \text{ is an } n \times K \text{ matrix with rank } K$$

This assumption is also known as the identification condition or no perfect multicolinearity

(ii) (Conditional) Normal errors: The error term is normally distributed (conditional on $\mathbf{X}$)

$$\mathbf{U}|\mathbf{X} \sim \mathcal{N}(0, \sigma^2\mathbf{I}_n)$$

Theorem 3.18 (Finite Sample Properties Under Normality). Under Assumption FSN,

(a) $\hat{\beta}|\mathbf{X}$ is multivariate normal, specifically,

$$\hat{\beta}|\mathbf{X} \sim \mathcal{N}\{\beta, \sigma^2(\mathbf{X}\mathbf{X})^{-1}\}$$

and each element of $\hat{\beta}$ satisfies

$$\hat{\beta}_k|\mathbf{X} \sim \mathcal{N}\{\beta_k, \sigma^2(\mathbf{X}'\mathbf{X})^{-1}_{kk}\}$$

where $(\mathbf{X}'\mathbf{X})^{-1}_{kk}$ denotes the kk-element of $(\mathbf{X}'\mathbf{X})^{-1}$

(b) The estimator of the error variance

$$\hat{\sigma}^2 = \frac{\hat{\mathbf{U}}'\hat{\mathbf{U}}}{n-K}$$

where $\hat{\mathbf{U}} = \mathbf{y} - \mathbf{X}\hat{\beta}$ is unbiased and

$$(n-K)\frac{\hat{\sigma}^2}{\sigma^2} \sim \chi^2_{n-K}$$

(c) The following t-statistics satisfy, under $\mathrm{H}_0 : \beta_k = \beta_k^0$, for $k = 0, 1, \ldots, K$

$$\frac{\hat{\beta}_k - \beta_k^0}{\text{s.d.}(\hat{\beta}_k|\mathbf{X})} \sim \mathcal{N}(0,1) \text{ and } \frac{\hat{\beta}_k - \beta_k^0}{\text{s.e.}(\hat{\beta}_k|\mathbf{X})} \sim t_{n-K}$$

where $\text{s.d.}(\hat{\beta}_k|\mathbf{X}) = \sqrt{\sigma^2(\mathbf{X}'\mathbf{X})^{-1}_{kk}}$ and $\text{s.e.}(\hat{\beta}_k|\mathbf{X}) = \sqrt{\hat{\sigma}^2(\mathbf{X}'\mathbf{X})^{-1}_{kk}}$

Proof. **Proof of (a)**: Note that the OLS estimator for β can be written as

$$\hat{\beta} = \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{U}$$

Under **Assumption 3.17 (b)**, $\mathbf{U}|\mathbf{X} \sim \mathcal{N}(\mathbf{0}, \sigma^2\mathbf{I}_n)$ and so $\hat{\beta}|\mathbf{X}$ is a multivariate normal. Hence, it suffices to find the conditional mean and variance of $\hat{\beta}$. Appealing to the proofs and results of **Theorem 3.16 (a)** and **(b)**, we yield

$$\hat{\beta}|\mathbf{X} \sim \mathcal{N}(\beta, \sigma^2(\mathbf{X}'\mathbf{X})^{-1})$$

Proof of (b): As shown by the proof of **Theorem 3.16 (d)**,

$$\hat{\mathbf{U}}'\hat{\mathbf{U}} = \mathbf{U}'\mathbf{M}'\mathbf{M}\mathbf{U} = \mathbf{U}'\mathbf{M}\mathbf{M}\mathbf{U} = \mathbf{U}'\mathbf{M}\mathbf{U}$$

Hence,

$$(n-K)\frac{\hat{\sigma}^2}{\sigma^2} = \frac{\mathbf{U}'\mathbf{M}\mathbf{U}}{\sigma^2} = (\mathbf{U}/\sigma)'\mathbf{M}(\mathbf{U}/\sigma)$$

By **Assumption 3.17(ii)**,

$$(\mathbf{U}/\sigma) \sim \mathcal{N}(0,1)$$

while $\text{rank}(\mathbf{M}) = n - K$, since $\text{rank}(\mathbf{M}) = \text{tr}(\mathbf{M})$. By the properties of the χ_2 distribution,

$$(\mathbf{U}/\sigma)'(\mathbf{U}/\sigma) \sim \chi^2_n$$

while

$$(\mathbf{U}/\sigma)'\mathbf{M}(\mathbf{U}/\sigma) \sim \chi^2_{n-K}$$

which gives the desired result

$$(n-K)\frac{\hat{\sigma}^2}{\sigma^2} \sim \chi^2_{n-K}$$

Notice that $(\mathbf{U}/\sigma)'(\mathbf{U}/\sigma)$ is the case where $\mathbf{M} = \mathbf{I}_n$ which $\text{rank}(n)$
Proof of (c): First look at

$$\frac{\hat{\beta}_k - \beta_k^0}{\text{s.d.}(\hat{\beta}_k)} = \frac{\hat{\beta}_k - \beta_k^0}{\sqrt{\sigma^2(\mathbf{X}'\mathbf{X})^{-1}_{kk}}}$$

where we look at the sd of $\hat{\beta}_j$ instead of the standard error. By proof of (a), under H_0, $\hat{\beta}_j - \beta_k^0 \sim \mathcal{N}\{0, \sigma^2(\mathbf{X}'\mathbf{X})^{-1}_{kk}\}$. Hence,

$$\frac{\hat{\beta}_k - \beta_k^0}{\text{s.d.}(\hat{\beta}_k)} \sim \mathcal{N}(0,1)$$

Now, looking at

$$\frac{\hat{\beta}_k - \beta_k^0}{\text{s.e.}(\hat{\beta}_k)} = \frac{\hat{\beta}_k - \beta_k^0}{\sqrt{\hat{\sigma}^2(\mathbf{X}'\mathbf{X})^{-1}_{kk}}}$$

where the s.e. uses $\hat{\sigma}^2$ instead of σ^2 itself. Notice that (b) says that

$$(n-K)\frac{\hat{\sigma}^2}{\sigma^2} \sim \chi^2_{n-K}$$

We can rewrite the fraction as

$$\frac{\hat{\beta}_k - \beta_k^0}{\sqrt{\hat{\sigma}^2(\mathbf{X}'\mathbf{X})^{-1}_{kk}}} = \frac{\hat{\beta}_k - \beta_k^0}{\sqrt{\sigma^2(\mathbf{X}'\mathbf{X})^{-1}_{kk}}} \frac{\sqrt{\sigma^2(\mathbf{X}'\mathbf{X})^{-1}_{kk}}}{\sqrt{\hat{\sigma}^2(\mathbf{X}'\mathbf{X})kk^{-1}}}$$

Therefore,

$$\frac{\hat{\beta}_k - \beta_k^0}{\sqrt{\hat{\sigma}^2(\mathbf{X}'\mathbf{X})^{-1}_{kk}}} = \frac{\hat{\beta}_k - \beta_k^0}{\sqrt{\sigma^2(\mathbf{X}'\mathbf{X})^{-1}_{kk}}} \left\{ \frac{(n-K)\frac{\hat{\sigma}^2}{\sigma^2}}{n-K} \right\}^{-1/2}$$

Hence,

$$\frac{\hat{\beta}_k - \beta_k^0}{\text{s.e.}(\hat{\beta}_k)} = \frac{\hat{\beta}_k - \beta_k^0}{\sqrt{\sigma^2(\mathbf{X}'\mathbf{X})^{-1}_{kk}}} \left\{ \frac{(n-K)\frac{\hat{\sigma}^2}{\sigma^2}}{n-K} \right\}^{-1/2} \sim \frac{\mathcal{N}(0,1)}{\sqrt{\frac{\chi^2_{n-K}}{n-K}}} \sim t_{n-K}$$

where the denominator and numerator are independent. ■

3.2.2 Asymptotic Properties

Assumption 3.19 (A). We assume

(i) $(X_i', u_i)'$ for $i = 1, \ldots, n$ is i.i.d. with finite fourth moments

(ii) Mean independence: $\mathrm{E}(\mathbf{U}|\mathbf{X}) = \mathbf{0}$

(iii) Rank condition:

$$\mathbf{Q} = \underset{n\to\infty}{p\lim} \frac{\mathbf{X}'\mathbf{X}}{n} = \mathrm{E}(X_i X_i')$$

has (full) rank K

(iv) (Conditional) Homodskedasticity and No Autocorrelation:

$$\mathrm{E}(\mathbf{U}\mathbf{U}'|\mathbf{X}) = \sigma^2\mathbf{I}$$

Remark. The interpretation of the rank condition is that we want the population data itself to have full rank. Furthermore, since the population data generates the sample data, the sample data will have full rank as well.

Theorem 3.20. Let Assumption A hold. Then

(a) $\hat{\beta}$ is consistent, that is,

$$\hat{\beta} \xrightarrow{p} \beta$$

(b) $\hat{\beta}$ is asymptotically normal, specifically,

$$\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} \mathcal{N}(\mathbf{0}_K, \sigma^2\mathbf{Q}^{-1})$$

(c) The estimator of the error variance

$$\hat{\sigma}^2 = \frac{\hat{\mathbf{U}}'\hat{\mathbf{U}}}{n-K}$$

is consistent for σ^2 and

$$\sqrt{n}(\hat{\sigma}^2 - \sigma^2) \xrightarrow{d} \mathcal{N}\{0, \mathrm{Var}(u_i^2)\}$$

(d) Under $\text{H}_0 : \beta_k = \beta_k^0$, the following t-statistics satisfy, for $k = 1, \dots, K$

$$\frac{\hat{\beta}_k - \beta_k^0}{\text{s.e.}(\hat{\beta}_k)} \xrightarrow{d} \mathcal{N}(0,1)$$

where $\text{s.e.}(\hat{\beta}_k) = \sqrt{\hat{\sigma}^2(\mathbf{X}'\mathbf{X})^{-1}_{kk}}$

Proof. **Proof of (a)**: We can write the OLS estimator as

$$\hat{\beta} = \beta + (\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\mathbf{U})$$

Multiply and divide by n so that

$$\hat{\beta} = \beta + \left(\frac{\mathbf{X}'\mathbf{X}}{n}\right)^{-1}\left(\frac{\mathbf{X}'\mathbf{U}}{n}\right)$$

We want to show that the matrices converge in probability element by element and apply Slutsky. By **Assumption A(i, iii)** and the Slutsky Theorem

$$\left(\frac{\mathbf{X}'\mathbf{X}}{n}\right)^{-1} \xrightarrow{p} \mathbf{Q}^{-1}$$

Notice that the assumption that $\mathbf{Q}$ has full rank is important because it implies $\mathbf{Q}$ is invertible. By **Assumption A (i, ii, iv)**, LIE and LLN

$$\left(\frac{\mathbf{X}'\mathbf{U}}{n}\right) \xrightarrow{p} \mathbf{0}$$

This yields by Slutsky again

$$\hat{\beta} \xrightarrow{p} \beta + 0 = \beta$$

Proof of (b): Rewrite the OLS estimator as

$$\sqrt{n}(\hat{\beta} - \beta) = \left(\frac{\mathbf{X}'\mathbf{X})}{n}\right)^{-1}\left(\frac{\mathbf{X}'\mathbf{U}}{\sqrt{n}}\right)$$

By **Assumption A (i, iii)** and the Slutsky Theorem,

$$\left(\frac{\mathbf{X}'\mathbf{X}}{n}\right)^{-1} \xrightarrow{p} \mathbf{Q}^{-1}$$

Notice that

$$\frac{\mathbf{X}'\mathbf{U}}{\sqrt{n}} = \begin{pmatrix} n^{-1/2}\sum_{i=1}^n x_{1i}u_i \\ n^{-1/2}\sum_{i=1}^n x_{2i}u_i \\ \vdots \\ n^{-1/2}\sum_{i=1}^n x_{Ki}u_i \end{pmatrix} = n^{-1/2}\sum_{i=1}^n \mathbf{X}_i u_i$$

We cannot use the univariate CLT for each element generally due to dependence. Instead, it suffices to show that for

$$\frac{\mathbf{X}'\mathbf{U}}{\sqrt{n}} = n^{-1/2}\sum_{i=1}^n \mathbf{X}_i u_i$$

the assumptions of the Lindeberg-Levy CLT hold. By **Assumption A(i)**, $\mathbf{X}_i u_i$ is i.i.d. with

$$\text{E}(X_i u_i) = \text{E}\{E(X_i u_i|\mathbf{X})\} = \text{E}\{X_i \text{E}(u_i|\mathbf{X})\} = \mathbf{0}$$

by **Assumption A(ii)** where we used LIE. Hence, there is a finite mean vector. By the definition of variance covariance matrix

$$\text{Var}(X_i u_i) = \text{E}(X_i u_i u_i X_i') = \text{E}(u_i^2 X_i X_i') = \text{E}\{\text{E}(u_i^2 X_i X_i'|\mathbf{X})\}$$

by LIE. Since we are conditioning on $\mathbf{X}$,

$$\mathrm{Var}(X_i u_i) = \mathrm{E}\{\mathbf{X}_i\mathbf{X}_i'\mathrm{E}(u_i^2|\mathbf{X})\}$$

By **Assumption A (iv)**, $\mathrm{E}(u_i^2|\mathbf{X}) = \sigma^2$ so that

$$\mathrm{Var}(X_i u_i) = \sigma^2\mathrm{E}(X_i X_i') = \sigma^2\mathbf{Q}$$

where $\mathrm{E}(X_i X_i')$ is finite by **Assumption A(i)**. Since a var-cov matrix is positive semidefinite by construction and we know that $\mathbf{Q}$ is full rank, then $\sigma^2\mathbf{Q}$ is positive definite. So, the conditions for the Lindeberg-Levy CLT hold and

$$n^{-1/2}\sum_{i=1}^{n}\mathbf{X}_i u_i \xrightarrow{d} \mathcal{N}(\mathbf{0}_K, \sigma^2\mathbf{Q})$$

where $\mathbf{Q} = \mathrm{E}(X_i X_i')$.
Combining the two results, by Slutsky (and the sandwich formula), we get

$$\sqrt{n}(\hat{\beta} - \beta) = \left(\frac{\mathbf{X}'\mathbf{X}}{n}\right)^{-1}\left(\frac{\mathbf{X}'\mathbf{U}}{\sqrt{n}}\right) \xrightarrow{d} \mathcal{N}(\mathbf{0}_K, \sigma^2\mathbf{Q}^{-1}\mathbf{Q}\mathbf{Q}^{-1})$$

since you have to "square" the $\mathbf{Q}^{-1}$ that you multiply the multivariate normal distribution by. Since $\mathbf{Q}\mathbf{Q}^{-1} = \mathbf{I}_K$, then

$$\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} \mathcal{N}(\mathbf{0}_K, \sigma^2\mathbf{Q}^{-1})$$

A looser notation allows us to write

$$\hat{\beta} \overset{a}{\sim} \mathcal{N}\left(\beta, \frac{\sigma^2}{n}\mathbf{Q}^{-1}\right)$$

but we're not typically allowed to write the sample size n in the asymptotic formula.
Proof of (c): First, to show consistency, consider

$$\hat{\sigma}^2 = \frac{\hat{\mathbf{U}}'\mathbf{U}}{n - K}$$

where

$$\hat{\mathbf{U}} = \mathbf{y} - \mathbf{X}\hat{\beta} = \mathbf{U} - \mathbf{X}(\hat{\beta} - \beta)$$

Since $\hat{\beta} - \beta = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{U}$, the residuals can be written as

$$\hat{\mathbf{U}} = \mathbf{U} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{U}$$

The inner product (or sum of squared residuals) is

$$\hat{\mathbf{U}}'\hat{\mathbf{U}} = \{\mathbf{U} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{U}\}'\{\mathbf{U} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{U}\}$$

Expanding yields

$$\begin{aligned}\hat{\mathbf{U}}'\hat{\mathbf{U}} = \mathbf{U}'\mathbf{U} &\\ &- \mathbf{U}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{U})\\ &- \mathbf{U}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{U}\\ &+ \mathbf{U}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{U}\end{aligned}$$

and the last term simplifies to $\mathbf{U}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{U}$. Then,

$$\hat{\mathbf{U}}'\hat{\mathbf{U}} = \mathbf{U}'\mathbf{U} - \mathbf{U}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{U}$$

The quantity of interest can be rewritten as

$$\hat{\sigma}^2 = \frac{n}{n - K}\frac{\mathbf{U}'\mathbf{U}}{n} - \frac{n}{n - K}\frac{\mathbf{U}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{U}}{n}$$

rearranging the second term of the righthandside gives

$$\hat{\sigma}^2 = \frac{n}{n-K}\frac{\mathbf{U}'\mathbf{U}}{n} - \frac{n}{n-K}\frac{\mathbf{U}'\mathbf{X}}{n}\left(\frac{\mathbf{X}'\mathbf{X}}{n}\right)^{-1}\frac{\mathbf{X}'\mathbf{U}}{n}$$

. In the limit, $n/(n-K) \to 1$. By **Assumption A(i)** and LLN, we have the following

$$\frac{\mathbf{U}'\mathbf{U}}{n} \xrightarrow{p} \sigma^2$$
$$\frac{\mathbf{U}'\mathbf{X}}{n} \xrightarrow{p} \mathbf{0}$$
$$\frac{\mathbf{X}'\mathbf{X}}{n} \xrightarrow{p} \mathbf{Q}$$
$$\frac{\mathbf{X}'\mathbf{U}}{n} \xrightarrow{p} \mathbf{0}$$

So, by Slutsky,

$$\hat{\sigma}^2 \xrightarrow{p} \sigma^2$$

For asymptotic normality, note we may write

$$\hat{\sigma}^2 = \frac{n}{n-K}\frac{\mathbf{U}'\mathbf{U}}{n} - \frac{n}{n-K}\frac{\mathbf{U}'\mathbf{X}}{n}\left(\frac{\mathbf{X}'\mathbf{X}}{n}\right)^{-1}\frac{\mathbf{X}'\mathbf{U}}{n}$$

Hence,

$$\sqrt{n}(\hat{\sigma}^2 - \sigma^2) = \sqrt{n}\frac{n}{n-K}\left(\frac{\mathbf{U}'\mathbf{U}}{n} - \sigma^2\right) - \sqrt{n}\left(\frac{n}{n-K}\frac{\mathbf{U}'\mathbf{X}}{n}\left(\frac{\mathbf{X}'\mathbf{X}}{n}\right)^{-1}\frac{\mathbf{X}'\mathbf{U}}{n}\right)$$

Notice that because convergence in probability implies convergence in distribution, the RHS term converges in distribution to 0 as before. So, we want to see the asymptotic distribution of $\sqrt{n}\frac{n}{n-K}\left(\frac{\mathbf{U}'\mathbf{U}}{n} - \sigma^2\right)$. First, note that $\mathbf{U}'\mathbf{U}/n = n^{-1}\sum_{i=1}^n u_i^2$ where u_i^2 is i.i.d. by **Assumption A(i)**. By **Assumption A(iv)**, $\mathrm{E}[\mathbf{U}'\mathbf{U}] = \mathrm{E}[\mathrm{E}\{\mathbf{U}'\mathbf{U}|\mathbf{X}\}] = \mathrm{E}[n\sigma^2] = n\sigma^2$, since the inner product is just the trace of the outer product $\mathbf{U}\mathbf{U}'$. Hence,

$$\begin{aligned}
\mathrm{E}\left[\sqrt{n}\frac{n}{n-K}\left(\frac{\mathbf{U}'\mathbf{U}}{n} - \sigma^2\right)\right] &= \sqrt{n}\frac{n}{n-K}\mathrm{E}\left(\frac{\mathbf{U}'\mathbf{U}}{n} - \sigma^2\right) \\
&= \sqrt{n}\frac{n}{n-K}\left(\sigma^2 - \sigma^2\right) \\
&= 0
\end{aligned}$$

Now, consider

$$\mathrm{Var}\left[\sqrt{n}\frac{n}{n-K}\left(\frac{\mathbf{U}'\mathbf{U}}{n} - \sigma^2\right)\right]$$

Since we consider the asymptotic variance, note that $n/(n-K) \to 1$ We can write

$$\begin{aligned}
\mathrm{Var}\left[\sqrt{n}\left(\frac{\mathbf{U}'\mathbf{U}}{n} - \sigma^2\right)\right] &= n\mathrm{Var}\left[n^{-1}\sum_{i=1}^n u_i^2\right] \\
&= \frac{1}{n}\mathrm{Var}\left[\sum_{i=1}^n u_i^2\right] \\
&= \mathrm{Var}(u_i^2)
\end{aligned}$$

where the last step follows by linearity and u_i^2 being i.i.d. Hence, by CLT

$$\sqrt{n}(\hat{\sigma}^2 - \sigma^2) \xrightarrow{d} \mathcal{N}\{0, \mathrm{Var}(u_i^2)\}$$

Proof of (d): The quantity of interest is

$$\frac{\hat{\beta}_k - \beta_k^0}{\text{s.e.}(\hat{\beta}_k)} = \frac{\hat{\beta}_k - \beta_k^0}{\sqrt{\sigma^2(\mathbf{X}'\mathbf{X})^{-1}_{kk}}}$$

Multiplying and dividing by $\sqrt{n}$,

$$\frac{\hat{\beta}_k - \beta_k^0}{\text{s.e.}(\hat{\beta}_k)} = \frac{\sqrt{n}(\hat{\beta}_k - \beta_k^0)}{\sqrt{\sigma^2(\frac{\mathbf{X}'\mathbf{X}}{n})^{-1}_{kk}}}$$

By **Theorem 3.20 (b)**,

$$\sqrt{n}(\hat{\beta}_k - \beta_k^0) \xrightarrow{d} \mathcal{N}(0, \sigma^2 \mathbf{Q}^{-1}_{kk})$$

By **Theorem 3.20 (c)**, $\hat{\sigma}^2 \xrightarrow{p} \sigma^2$ while by **Assumption A** and LLN and Slutsky,

$$\left(\frac{\mathbf{X}'\mathbf{X}}{n}\right)^{-1}_{kk} \xrightarrow{p} \mathbf{Q}^{-1}_{kk}$$

Therefore,

$$\frac{\hat{\beta}_k - \beta_k^0}{\text{s.e.}(\hat{\beta}_k)} = \frac{\sqrt{n}(\hat{\beta}_k - \beta_k^0)}{\sqrt{\hat{\sigma}^2\left(\frac{\mathbf{X}'\mathbf{X}}{n}\right)^{-1}_{kk}}} \xrightarrow{d} \mathcal{N}(0,1)$$

■

4 Hypothesis Testing

We want to conduct inference on our estimate $\hat{\beta}$. In QE, we learnt how to test more hypothesis at once i.e.

$$\begin{aligned} H_0 :& \beta_1 = \cdots = \beta_K = 0 \\ H_1 :& \bigvee_{i=1}^{K} \beta_i \neq 0 \end{aligned}$$

Our goal is to generalise this. We have three ingredients to consider

1. Null and alternative hypotheses
2. Test statistic and its distribution under the null
3. Decision rule

4.1 General Linear Hypotheses

1) Null and Alternative Hypotheses: Our generalisation is to consider J null hypotheses

$$\begin{aligned} r_{11}\beta_1 + r_{12}\beta_2 + \cdots + r_{1K}\beta_K &= q_1 \\ r_{21}\beta_1 + r_{22}\beta_2 + \cdots + r_{2K}\beta_K &= q_2 \\ &\vdots \\ r_{J1}\beta_1 + r_{J2}\beta_2 + \cdots + r_{JK}\beta_K &= q_J \end{aligned}$$

against the alternative that at least one of these is not true. Our general framework becomes a system of J restrictions

$$H_0 : \mathbf{R}\beta = \mathbf{q}$$

where

$$\mathbf{R} = \begin{pmatrix} r_{11} & r_{12} & \dots & r_{1K} \\ r_{21} & r_{22} & \dots & r_{2K} \\ \vdots & \vdots & \ddots & \vdots \\ r_{J1} & r_{J2} & \dots & r_{JK} \end{pmatrix} \text{ and } \mathbf{q} = \begin{pmatrix} q_1 \\ q_2 \\ \vdots \\ q_J \end{pmatrix}$$

and our β is the vector of K coefficients for each regressor.

Example 4.1. Consider $y_i = \beta_1 + \beta_2 x_{2i} + \beta_3 x_{3i} + \beta_4 x_{4i} + \beta_5 x_{5i} + u_i$. Testing a single coefficient is zero, say

$$H_0 : \beta_3 = 0$$

then we have $\mathbf{R} = \begin{pmatrix} 0 & 0 & 1 & 0 & 0 \end{pmatrix}$ and $q = 0$.
Alternatively, if we want to test

$$H_0 : \beta_3 = \beta_4$$

then $\mathbf{R} = \begin{pmatrix} 0 & 0 & 1 & -1 & 0 \end{pmatrix}$ and $q = 0$.
Furthermore, if we want to test

$$H_0 : \beta_2 + \beta_3 + \beta_4 = 1$$

we have $\mathbf{R} = \begin{pmatrix} 0 & 1 & 1 & 1 & 0 \end{pmatrix}$ and $q = 1$.
Lastly, to test for global significance, say

$$H_0 : \beta_2 = \beta_3 = \beta_4 = \beta_5 = 0$$

our matrices are

$$\mathbf{R} = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} \text{ and } \mathbf{q} = \mathbf{0}$$

2) Test Statistic and Distribution under the null:
Case 1: Under normal errors

Theorem 4.2 (Distribution under Null for Finite Samples). Consider the null hypothesis of interest

$$H_0 : \mathbf{R}\beta = \mathbf{q}$$

where $\mathbf{R}$ is a $J \times K$ matrix. Assume that **Assumption 3.17 (FSN)** holds i.e. Full rank and conditional normal errors. The test statistic is

$$F = \frac{(\mathbf{R}\hat{\beta} - \mathbf{q})'\{\hat{\sigma}^2 \mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\}^{-1}(\mathbf{R}\hat{\beta} - \mathbf{q})}{J}$$

Then, for each sample size n,

$$F \sim \mathrm{F}_{J,n-k}$$

where $\mathrm{F}_{J,n-k}$ is the F distribution with J and $n - K$ degrees of freedom

Proof. Consider the infeasible Wald statistic defined as

$$W = (\mathbf{R}\hat{\beta} - \mathbf{q})'\{\sigma^2 \mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\}(\mathbf{R}\hat{\beta} - \mathbf{q})$$

We say that this is infeasible because it is a function of the unknown error variance σ^2. To make this feasible, we have to estimate $\hat{\sigma}^2$ later on. Under H_0, we know that $\mathbf{R}\beta = q$, so that

$$\mathbf{R}\hat{\beta} - \mathbf{q} = \mathbf{R}\hat{\beta} - \mathbf{R}\beta = \mathbf{R}(\hat{\beta} - \beta) = \mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{U}$$

where we have assumed that $\mathbf{U}|\mathbf{X} \sim \mathcal{N}(0, \sigma^2 \mathbf{I}_n)$. Therefore, $\mathbf{R}\hat{\beta} - \mathbf{q} = \mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{U}$ is also normally distributed. The conditional mean is

$$\begin{aligned} \mathrm{E}(\mathbf{R}\hat{\beta} - \mathbf{q}|\mathbf{X}) &= \mathrm{E}\{\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{U}|\mathbf{X}\} \\ &= \mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathrm{E}[\mathbf{U}|\mathbf{X}] = 0 \end{aligned}$$

by the properties of conditional expectation and assumption. Since $\mathrm{E}(\mathbf{R}\hat{\beta} - \mathbf{q}|\mathbf{X}) = 0$, we can write

$$\begin{aligned}\mathrm{Var}(\mathbf{R}\hat{\beta} - \mathbf{q}|\mathbf{X}) &= \mathrm{E}\{\mathbf{R}(\mathbf{X'X})^{-1}\mathbf{X'UU'X}(\mathbf{X'X})^{-1}\mathbf{R'}|\mathbf{X}\} \\ &= \mathbf{R}(\mathbf{X'X})^{-1}\mathbf{X'}\mathrm{E}[\mathbf{UU'}|\mathbf{X}]\mathbf{X}(\mathbf{X'X})^{-1}\mathbf{R'}\end{aligned}$$

The reason why the mean zero condition is important is because for any $\mathbf{Z}$, the conditional variance is defined as $\mathrm{Var}(\mathbf{Z}|\mathbf{X}) = \mathrm{E}[(\mathbf{Z} - \mathrm{E}[\mathbf{Z}|\mathbf{X}])(\mathbf{Z} - \mathrm{E}[\mathbf{Z}|\mathbf{X}])'|\mathbf{X}]$. Hence, if $\mathrm{E}[\mathbf{Z}|\mathbf{X}] = 0$, then $\mathrm{Var}(\mathbf{Z}|\mathbf{X}) = \mathrm{E}[\mathbf{ZZ'}|\mathbf{X}]$. Going back, since $\mathbf{E}[\mathbf{UU'}|\mathbf{X}) = \sigma^2\mathbf{I}_n$, we yield

$$\mathrm{Var}(\mathbf{R}\hat{\beta} - \mathbf{q}|\mathbf{X}) = \sigma^2\mathbf{R}(\mathbf{X'X})^{-1}\mathbf{X'X}(\mathbf{X'X})^{-1}\mathbf{R'} = \sigma^2\mathbf{R}(\mathbf{X'X})^{-1}\mathbf{R'}$$

So, $\mathbf{R}\hat{\beta} - \mathbf{q}$ is a multivariate normal

$$\mathbf{R}\hat{\beta} - \mathbf{q} = \mathbf{R}(\mathbf{X'X})^{-1}\mathbf{X'U} \sim \mathrm{MVN}\{\mathbf{0}, \sigma^2\mathbf{R}(\mathbf{X'X})^{-1}\mathbf{R'}\}$$

where the covariance matrix is the $J \times J$ matrix $\sigma^2\mathbf{R}(\mathbf{X'X})^{-1}\mathbf{R'}$. Recall that our Wald statistic is actually a quadratic form

$$W = (\mathbf{R}\hat{\beta} - \mathbf{q})'\{\sigma^2\mathbf{R}(\mathbf{X'X})^{-1}\mathbf{R'}\}(\mathbf{R}\hat{\beta} - \mathbf{q})$$

in $\sigma^2\mathbf{R}(\mathbf{X'X})^{-1}\mathbf{R'}$ (which is precisely our variance). Therefore, by the definition of the chi-squared distribution,

$$\mathbf{W} \sim \chi^2_J$$

since the rank of $\sigma^2\mathbf{R}(\mathbf{X'X})^{-1}\mathbf{R'}$ is J. We define our F statistic as

$$F = \frac{(\mathbf{R}\hat{\beta} - \mathbf{q})\{\hat{\sigma}^2\mathbf{R}(\mathbf{X'X})^{-1}\mathbf{R'}\}(\mathbf{R}\hat{\beta} - \mathbf{q})}{J}$$

Notice that this replaces σ^2 from the infeasible Wald statistic with an estimate $\hat{\sigma}^2$. So,

$$F = \frac{W}{J}\frac{\sigma^2}{\hat{\sigma}^2} = \frac{W/J}{\{(n-K)\hat{\sigma}^2/\sigma^2\}/(n-K)}$$

By **Theorem 3.18 (Finite Sample Properties Under Normality)**,

$$(n-K)\frac{\hat{\sigma}^2}{\sigma^2} \sim \chi^2_{n-K}$$

So, we yield

$$F = \frac{W}{J}\frac{\sigma^2}{\hat{\sigma}^2} = \frac{W/J}{\{(n-K)\hat{\sigma}^2/\sigma^2\}/(n-K)} \sim \frac{\chi^2_J/J}{\chi^2_{n-K}/(n-K)} \sim F_{J,n-K}$$

where the numerator and denominator are independent by covariance 0 and properties of a multivariate normal. ■

Case 2: Without normal errors, asymptotics:

Theorem 4.3. Under **Assumption 3.19 (A)**, if $H_0 : \mathbf{R}\beta - \mathbf{q} = \mathbf{0}$ holds, then

$$W^a = (\mathbf{R}\hat{\beta} - \mathbf{q})'\{\mathbf{R}\hat{\sigma}^2(\mathbf{X'X})^{-1}\mathbf{R'}\}^{-1}(\mathbf{R}\hat{\beta} - \mathbf{q}) = JF \xrightarrow{d} \chi^2_J$$

Proof. Under the null, $\mathbf{R}\beta = \mathbf{q}$, so that

$$\mathbf{R}\hat{\beta} - \mathbf{q} = \mathbf{R}\hat{\beta} - \mathbf{R}\beta = \mathbf{R}(\hat{\beta} - \beta) = \mathbf{R}(\mathbf{X'X})^{-1}\mathbf{X'U}$$

Under **Assumption 3.19 (A)**, by the Multivariate CLT,

$$\sqrt{n}\mathbf{R}(\hat{\beta} - \beta) \xrightarrow{d} \mathrm{MVN}\{\mathbf{0}, \mathbf{R}(\sigma^2\mathbf{Q}^{-1})\mathbf{R'}\}$$

Moreover, $\hat{\sigma}^2 \xrightarrow{p} \sigma^2$. Rewrite

$$W^a = \sqrt{n}(\mathbf{R}\hat{\beta} - \mathbf{q})' \left\{ \mathbf{R}\hat{\sigma}^2 \left(\frac{\mathbf{X}'\mathbf{X}}{n} \right)^{-1} \mathbf{R}' \right\}^{-1} \sqrt{n}(\mathbf{R}\hat{\beta} - \mathbf{q})$$

$$\sqrt{n}\mathbf{R}(\hat{\beta} - \beta) \xrightarrow{d} \text{MVN}\{\mathbf{0}, \mathbf{R}(\sigma^2\mathbf{Q}^{-1})\mathbf{R}'\}$$

and

$$\left\{ \mathbf{R}\hat{\sigma}^2 \left(\frac{\mathbf{X}'\mathbf{X}}{n} \right)^{-1} \mathbf{R}' \right\} \xrightarrow{p} \{\mathbf{R}(\sigma^2\mathbf{Q}^{-1})\mathbf{R}'\}^{-1}$$

Consider that under the Univariate CLT, if z_i is i.i.d. with mean 0 and variance v, then $\sqrt{n}\hat{z} \xrightarrow{d} \mathcal{N}(0, v)$. By the CMT,

$$(\sqrt{n}\hat{z}/\sqrt{v})^2 = (\sqrt{n}\hat{z})v^{-1}(\sqrt{n}\hat{z}) \xrightarrow{d} \{\mathcal{N}(0,1)\}^2 \sim \chi^2_1$$

Generalising this for the Multivariate CLT: if $\mathbf{Z}_i$ is i.i.d. MVN$(\mathbf{0}, \mathbf{V})$, then $\sqrt{n}\hat{\mathbf{Z}} \xrightarrow{d} \text{MVN}(\mathbf{0}, \mathbf{V})$ where $\mathbf{V}$ is the $J \times J$ covariance matrix. It holds that

$$(\sqrt{n}\hat{\mathbf{Z}})'\mathbf{V}^{-1}(\sqrt{n}\hat{\mathbf{Z}}) \xrightarrow{d} \chi^2_J$$

Notice that

$$W^a = \sqrt{n}(\mathbf{R}\hat{\beta} - \mathbf{q})' \left\{ \mathbf{R}\hat{\sigma}^2 \left(\frac{\mathbf{X}'\mathbf{X}}{n} \right)^{-1} \mathbf{R}' \right\}^{-1} \sqrt{n}(\mathbf{R}\hat{\beta} - \mathbf{q})$$

precisely satisfies the quadratic form above. Hence

$$W^a \xrightarrow{d} \chi^2_J$$

■

3) Decision Rule i.e. Asymptotic Inference
This is given to us by the F-statistic table of critical values etc (such as in QE)

Example 4.4. In the linear model

$$y_i = \beta_1 + \beta_2 x_{2i} + \beta_3 x_{3i} + u_i$$

consider the hypothesis $H_0 : \beta_2 = 0$ and the statistic

$$F = \frac{(\mathbf{R}\hat{\beta} - \mathbf{q})'\{\mathbf{R}\hat{\sigma}^2(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\}^{-1}(\mathbf{R}\hat{\beta} - \mathbf{q}}{J}$$

Show that $F = t^2$ where t is the t statistic for the stated hypothesis.
Let $\beta = \begin{pmatrix} \beta_1 & \beta_2 & \beta_3 \end{pmatrix}'$ and $\mathbf{R} = \begin{pmatrix} 0 & 1 & 0 \end{pmatrix}$. Therefore, $\mathbf{R}\hat{\beta} - \mathbf{q} = \hat{\beta}_2$ since there is only one restriction $J = 1$. Recall the definition of the F statistic is

$$F = \frac{(\mathbf{R}\hat{\beta} - \mathbf{q})'\{\mathbf{R}\hat{\sigma}^2(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\}^{-1}(\mathbf{R}\hat{\beta} - \mathbf{q}}{J}$$

With one restriction and $H_0 : \beta_2 = 0$,

$$F = \hat{\beta}_2\{\mathbf{R}\hat{\sigma}^2(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\}^{-1}\hat{\beta}_2$$

Main objective is to find $\mathbf{R}\hat{\sigma}^2(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'$. Recall that $\text{Var}(\hat{\beta}|\mathbf{X}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$ where σ^2 is unknown. As an estimator for $\text{Var}(\hat{\beta}|\mathbf{X})$ we can use $\widehat{\text{Var}}(\hat{\beta}|\mathbf{X}) = \hat{\sigma}^2(\mathbf{X}'\mathbf{X})^{-1}$. In particular,

$$\widehat{\text{Var}}(\hat{\beta}|\mathbf{X}) = \begin{pmatrix} \widehat{\text{Var}}(\hat{\beta}_1|\mathbf{X}) & \widehat{\text{Cov}}(\hat{\beta}_1, \hat{\beta}_2|\mathbf{X}) & \widehat{\text{Cov}}(\hat{\beta}_1, \hat{\beta}_3|\mathbf{X}) \\ \widehat{\text{Cov}}(\hat{\beta}_2, \hat{\beta}_1|\mathbf{X}) & \widehat{\text{Var}}(\hat{\beta}_2|\mathbf{X}) & \widehat{\text{Cov}}(\hat{\beta}_2, \hat{\beta}_3|\mathbf{X}) \\ \widehat{\text{Cov}}(\hat{\beta}_3, \hat{\beta}_1|\mathbf{X}) & \widehat{\text{Cov}}(\hat{\beta}_3, \hat{\beta}_2|\mathbf{X}) & \widehat{\text{Var}}(\hat{\beta}_3|\mathbf{X}) \end{pmatrix}$$

Therefore, when testing $\mathbf{R}\hat{\sigma}^2(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}' = \widehat{\text{Var}}(\hat{\beta}_2|\mathbf{X}) = \hat{\sigma}^2(\mathbf{X}'\mathbf{X})^{-1}_{22}$. Hence,

$$F = \hat{\beta}_2\{\mathbf{R}\hat{\sigma}^2(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\}^{-1}\hat{\beta}_2 = \frac{\hat{\beta}_2^2}{\widehat{\text{Var}}(\hat{\beta}_2|\mathbf{X})} = t^2$$

since the t statistic under H_0 is

$$t = \frac{\hat{\beta}_2}{\text{s.e.}(\hat{\beta})}$$

4.2 Heteroskedasticity

Heteroskedasticity is the phenomenon of economic data whereby the error variance is non-constant across non-observations. In particular, conditional heteroskedasticity is when we have a positive definite symmetric covariance matrix where the σ_i^2 terms are functions of $\mathbf{X}$ i.e.

$$\Omega = \begin{pmatrix} \sigma_1^2(\mathbf{X}) & 0 & \cdots & 0 \\ 0 & \sigma_2^2(\mathbf{X}) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_n^2(\mathbf{X}) \end{pmatrix}$$

where we keep the assumption of no autocorrelation (since $\text{Cov}(u_i, u_j) = 0$ for $i \neq j$.

4.2.1 Consequences of Heteroskedasticity for OLS

Assumption 4.5 (FSH). For finite samples with heteroskedasticity, we assume

(i) Full Rank: $\mathbf{X}$ is an $n \times K$ matrix with rank K

(ii) Mean Independence: $\text{E}[\mathbf{U}|\mathbf{X}] = \mathbf{0}$

(iii) (Conditional) Heteroskedasticity and Autocorrelation:

$$\text{E}[\mathbf{U}\mathbf{U}'|\mathbf{X}] = \Omega$$

where Ω is a positive definite and symmetric $n \times n$ matrix.

Proposition 4.6 (OLS FS properties that remain the same). Consider the regression model $\mathbf{y} = \mathbf{X}\beta + \mathbf{U}$, and the OLS estimator $\hat{\beta}$. Under **Assumption FSH**, we have

$$\text{E}[\hat{\beta}] = \beta$$

i.e. the OLS estimator remains unbiased

Proof. The OLS estimator is $\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\mathbf{y})$. We may rewrite this as

$$\hat{\beta} = \beta + (\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\mathbf{U})$$

By LIE and linearity of conditional expectations as well as mean independence,

$$\text{E}[\hat{\beta}] = \text{E}\{\text{E}(\hat{\beta}|\mathbf{X})\} = \text{E}\{\beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\text{E}[\mathbf{U}|\mathbf{X}]\} = \beta$$

■

Theorem 4.7 (OLS FS properties that differ). Let **Assumption FSH** hold, and then the following finite sample properties differ under heteroskedasticity

(a) The conditional variance is now

$$\text{Var}(\hat{\beta}|\mathbf{X}) = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\Omega\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}$$

instead of $\sigma^2(\mathbf{X}'\mathbf{X})^{-1}$

(b) Aitken's Theorem: $\hat{\beta}$ is not BLUE because it is not the most efficient estimator

(c) Under the assumption of normal errors $\mathbf{U}|\mathbf{X} \sim \text{MVN}(\mathbf{0}, \Omega)$, the OLS estimator

$$\hat{\beta}|\mathbf{X} \sim \text{MVN}\{\beta, (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\Omega\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\}$$

(d) The estimator of the error variance

$$\hat{\sigma}^2 = \frac{1}{n-K}\sum_{i=1}^{n} \hat{u}_i^2$$

is biased i.e. $\text{E}[\hat{\sigma}^2] \neq \sigma^2$

Proof. **Proof of (a)**: Consider the definition of $\text{Var}(\hat{\beta}|\mathbf{X}) = \text{E}[(\hat{\beta} - \text{E}[\hat{\beta}|\mathbf{X}])(\hat{\beta} - \text{E}[\hat{\beta}|\mathbf{X}])'|\mathbf{X}]$. Then,

$$\begin{aligned}
\text{Var}(\hat{\beta}|\mathbf{X}) &= \text{E}\{(\hat{\beta} - \beta)(\hat{\beta} - \beta)'|\mathbf{X}\} \\
&= \text{E}\{(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{U}\mathbf{U}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}|\mathbf{X}\} \\
&= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\text{E}(\mathbf{U}\mathbf{U}'|\mathbf{X})\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \\
&= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\Omega\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}
\end{aligned}$$

Proof of (b) – (d) to be followed ■

Assumption 4.8 (AH). For large samples with heteroskedasticity, we assume

(i) $(X_i', u_i)'$ for $i = 1, \ldots, n$ is i.i.d. with finite fourth moments

(ii) Mean Independence: $\text{E}[\mathbf{U}|\mathbf{X}] = \mathbf{0}$

(iii) Rank condition: The following matrix is full rank

$$\text{Q} = \underset{n\to\infty}{plim}\frac{\mathbf{X}'\mathbf{X}}{n} = \text{E}(X_i X_i')$$

(iv) Heteroskedasticity: the $K \times K$ matrix $\text{E}(u_i^2 X_i X_i') = \mathbf{Q}^*$ is finite and positive definite

Theorem 4.9 (Asymptotic Properties of OLS under Heteroskedasticity). Under **Assumption AH**, we have

$$\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} \text{MVN}(\mathbf{0}, \mathbf{Q}^{-1}\mathbf{Q}^*\mathbf{Q}^{-1})$$

Proof. Write the OLS estimator as $\hat{beta} = \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{U}$. Then,

$$\sqrt{n}(\hat{\beta} - \beta) = \left(\frac{\mathbf{X}'\mathbf{X}}{n}\right)^{-1}\frac{\mathbf{X}'\mathbf{U}}{\sqrt{n}}$$

whereby assumption ans the Slutsky theorem,

$$\left(\frac{\mathbf{X}'\mathbf{X}}{n}\right)^{-1} \xrightarrow{p} \mathbf{Q}^{-1}$$

Most of the proof is to consider the asymptotic behaviour of

$$\frac{\mathbf{X}'\mathbf{U}}{\sqrt{n}} = \begin{pmatrix} n^{-1/2}\sum_{i=1}^{n} x_{1i}u_i \\ n^{-1/2}\sum_{i=1}^{n} x_{2i}u_i \\ \vdots \\ n^{-1/2}\sum_{i=1}^{n} x_{Ki}u_i \end{pmatrix} = n^{-1/2}\sum_{i=1}^{n} X_i u_i$$

We want to apply to Multivariate CLT to $\frac{\mathbf{X}'\mathbf{U}}{\sqrt{n}} = n^{-1/2}\sum_{i=1}^{n} X_i u_i$ so that $Z_i = X_i u_i$. So we need to show that the assumptions hold. By **Assumption A(i)**, $X_i u_i$ is i.i.d. with

$$\mathrm{E}(X_i u_i) = \mathrm{E}\{\mathrm{E}(X_i u_i|\mathbf{X})\} = \mathrm{E}\{X_i \mathrm{E}(u_i|\mathbf{X})\} = \mathbf{0}$$

by **Assumption A(ii)** where we have used LIE. Hence the mean vector is finite. By definition of variance-covariance matrix,

$$\mathrm{Var}(X_i u_i) = \mathrm{E}[X_i u_i u_i X_i'] = \mathrm{E}[u_i^2 X_i X_u']$$

since $\mathrm{E}[u_i] = 0$ and X_i and u_i are independent by assumption. By Assumption, $\mathrm{E}[u_i^2 X_i X_i') = \mathbf{Q}^*$ is finite and positive definite. Hence, the conditions for the multivariate CLT hold and

$$n^{-1/2}\sum_{i=1}^{n} X_i u_i \xrightarrow{d} \mathrm{MVN}(\mathbf{0}_K, \mathbf{Q}^*)$$

where $\mathbf{Q}^* = \mathrm{E}[u_i^2 X_i X_u']$. Bringing everything together, since

$$\sqrt{n}(\hat{\beta} - \beta) = \left(\frac{\mathbf{X}'\mathbf{X}}{n}\right)^{-1} \frac{\mathbf{X}'\mathbf{U}}{\sqrt{n}}$$

and

$$\left(\frac{\mathbf{X}'\mathbf{X}}{n}\right)^{-1} \xrightarrow{p} \mathbf{Q}^{-1}$$

while

$$\frac{\mathbf{X}'\mathbf{U}}{\sqrt{n}} \xrightarrow{d} \mathrm{MVN}(\mathbf{0}_K, \mathbf{Q}^*)$$

then by the Slutsky Theorem

$$\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} \mathrm{MVN}(\mathbf{0}, \mathbf{Q}^{-1}\mathbf{Q}^*\mathbf{Q}^{-1})$$

or alternatively

$$\hat{\beta} \overset{a}{\sim} \mathrm{MVN}(\beta, \frac{1}{n}\mathbf{Q}^{-1}\mathbf{Q}^*\mathbf{Q}^{-1})$$

■

Remark. The reason why we multiply $\mathbf{Q}^{-1}$ on both sides is essentially the same reason why when we multiply a r.v. X that converges to $\mathcal{N}(\mu, \sigma^2)$ by some scalar c, $cX \xrightarrow{d} \mathcal{N}(c\mu, (c\sigma)^2$, but in an analogous way for matrices.

Corollary 4.10. The asymptotic variance of $\hat{\beta}$ is

$$\mathrm{AVar}(\hat{\beta}) = \frac{1}{n}\mathbf{Q}^{-1}\mathbf{Q}^*\mathbf{Q}^{-1}$$

and not $\mathrm{AVar}(\hat{\beta}) = \frac{\sigma^2}{n}\mathbf{Q}^{-1}$. Hence, the standard inference, such as t and F tests, based on $\mathrm{AVar}(\hat{\beta}) = (\sigma^2/n)\mathbf{Q}^{-1}$ is not valid.

4.2.2 Dealing with Heteroskedasticity

Testing for Heteroskedasticity: White's Test
Consider the following regression model

$$y_i = \beta_1 + \beta_2 x_{2i} + \beta_3 x_{3i} + u_i$$

Step 1: Estimate the model by OLS and compute the OLS residuals

$$\hat{u}_i = y_i - \hat{\beta}_1 - \hat{\beta}_2 x_{2i} - \hat{\beta}_3 x_3 i$$

Step 2: Run the following auxiliary regression

$$\hat{u}_i^2 = \alpha_1 + \alpha_2 x_{2i} + \alpha_3 x_{3i} + \gamma_2 x_{2i}^2 + \gamma_3 x_{3i}^2 + \gamma_1 x_{2i} x_{3i} + e_i$$

Step 3: Test the null (homoskedasticity) hypothesis

$$H_0 : \alpha_2 = \alpha_3 = \gamma_1 = \gamma_2 = \gamma_3 = 0$$

with the test statistic $nR^2 \overset{a}{\sim} \chi^2_{p-1}$ where n, R^2 and p are the sample size, coefficient of determination and number of parameters of the auxiliary regression respectively.

We now want to consider what to do if we detect heteroskedasticity.
Solution 1: Heteroskedastic Robust Standard Errors
White (1980) suggests to estimate the asymptotic variance by

$$\widehat{\text{AVar}}(\hat{\beta}) = \frac{1}{n}\left(\frac{1}{n}\mathbf{X}'\mathbf{X}\right)^{-1}\left(\frac{1}{n}\sum_{i=1}^{n}\hat{u}_i^2 X_i X_i'\right)\left(\frac{1}{n}\mathbf{X}'\mathbf{X}\right)^{-1}$$

Intuition: This works because we're trying to approximate

$$\text{AVar}(\hat{\beta}) = \frac{1}{n}\mathbf{Q}^{-1}\mathbf{Q}^*\mathbf{Q}^{-1}$$

where

$$\mathbf{Q} = \underset{n\to\infty}{p\text{lim}}\frac{\mathbf{X}'\mathbf{X}}{n} \quad \text{and} \quad \mathbf{Q}^* = \text{E}[u_i^2 X_i X_i']$$

White (1980) showed that

$$\frac{1}{n}\sum_{i=1}^{n}\hat{u}_i X_i X_i' \xrightarrow{p} \text{E}[u_i^2 X_i X_i'] = \mathbf{Q}^*$$

The main innovation of this is that based on the LLN, if we knew the true errors, then

$$\frac{1}{n}\sum_{i=1}^{n}u_i X_i X_i' \xrightarrow{p} \text{E}[u_i^2 X_i X_i'] = \mathbf{Q}^*$$

But White shows that even if we replace u_i with the estimated errors $\hat{u}_i$, we can still get

$$\frac{1}{n}\sum_{i=1}^{n}\hat{u}_i X_i X_i' \xrightarrow{p} \text{E}[u_i^2 X_i X_i'] = \mathbf{Q}^*$$

Remark. This still does not deliver the most efficient estimates, but the more efficient procedures like WLS and GLS require modelling heteroskedasticity which requires deeper and richer knowledge of the data. But what it does is allow for valid inferences based of the Heteroskedasticity Robust Standard Errors

For the next two, we go through the general matrix form idea and then show specific examples in scalar cases.
Solution 2: Weighted Least Squares
Suppose that $\text{E}[\mathbf{U}\mathbf{U}'|\mathbf{X}] = \Omega$ is a diagonal matrix (i.e. off diagonal entries are 0) but is dependent on $\mathbf{X}$

Proposition 4.11 (Weighted Least Squares Estimator). Consider the weighted minimum squared error problem

$$\hat{\beta}_{WLS} = \underset{\beta\in\mathbb{R}^n}{\text{argmin}}\frac{1}{n}\sum_{i=1}^{n}w_i(y_i - \mathbf{X}_i\beta_{2=n^{-1}(\mathbf{y}-\mathbf{X}\beta)'\mathbf{W}(\mathbf{y}-\mathbf{X}\beta)}$$

Our estimator is then

$$\hat{\beta}_{WLS} = (\mathbf{X}'\mathbf{W}\mathbf{X})^{-1}\mathbf{X}'\mathbf{W}\mathbf{y}$$

Proof. We may write

$$\begin{aligned} L &= n^{-1}(\mathbf{y}-\mathbf{X}\beta)'\mathbf{W}(\mathbf{y}-\mathbf{X}\beta) \\ &= n^{-1}(\mathbf{y}'\mathbf{W}\mathbf{y}-\mathbf{y}'\mathbf{W}\mathbf{X}\beta-\beta'\mathbf{X}'\mathbf{W}\mathbf{y}+\beta'\mathbf{X}'\mathbf{W}\mathbf{X}\beta) \end{aligned}$$

Differentiating, our FOC is

$$\nabla_\beta L = \frac{2}{n}(-\mathbf{X}'\mathbf{W}\mathbf{Y}+\mathbf{X}'\mathbf{W}\mathbf{X}\beta)=0$$

Hence, we yield

$$\hat{\beta}_{WLS} = (\mathbf{X}'\mathbf{W}\mathbf{X})^{-1}\mathbf{X}'\mathbf{W}\mathbf{y}$$

■

Remark. We may interpret $\hat{\beta}_{WLS}$ as simply $\hat{\beta}_{OLS}$ but weighted according to $\mathbf{W}=(w_1,\dots,w_n)'$

Proposition 4.12 (Aitken's Theorem). Suppose Heteroskedasticity holds. Then, our WLS estimator

$$\hat{\beta}_{WLS} = (\mathbf{X}'\mathbf{W}\mathbf{X})^{-1}\mathbf{X}'\mathbf{W}\mathbf{y}$$

is BLUE

Example 4.13 (Problem Set 3 Q1). Consider the following model

$$y_{i,e}=\beta_0+\beta_1 x_{i,e}+u_{i,e}$$

where $i=1,\dots,n$ denotes a particular firm and $e=1,\dots,m_i$ denotes an employee within the firm. Hence, n denotes the number of firms while m_i denotes the number of employees in firm i, which could differ among firms. The error term $u_{i,e}$ is i.i.d. $(0,\sigma_u^2)$ across both i and e, while the regressor $x_{i,e}$ is fixed, that is, non-random.

Suppose that employee-level data is not available, but we do observe the number of firms n, the number of employees m_i and the average values of $y_{i,e}$ and $x_{i,e}$ for each firm i.e.

$$\bar{y}_i=\frac{1}{m_i}\left(\sum_{e=1}^{m_i} y_{i,e}\right),\quad \bar{x}_i=\frac{1}{m_i}\left(\sum_{e=1}^{m_i} x_{i,e}\right)$$

Consider the following model

$$\bar{y}_i=\beta_0+\beta_1\bar{x}_i+\bar{u}_i \tag{1}$$

where

$$\bar{u}_i=\frac{1}{m_i}\left(\sum_{e=1}^{m_i} u_{i,e}\right)$$

Let $\hat{\beta}_1$ be the OLS estimator of β_1 in model (1). Find $\mathrm{E}(\hat{\beta}_1)$ and $\mathrm{Var}(\hat{\beta}_1)$.
Define $h_i := 1/m_i$. Consider the following model

$$\frac{\bar{y}_i}{\sqrt{h_i}}=\beta_0\frac{1}{\sqrt{h_i}}+\beta_1\frac{\bar{x}_i}{\sqrt{h_i}}+\frac{\bar{u}_i}{\sqrt{h_i}} \tag{2}$$

Define the OLS estimators $\tilde{\beta}_0,\tilde{\beta}_1$ for β_0,β_1 in (2) and find them.

Solution 3: Generalised Least Squares
Suppose that $\mathrm{E}[\mathbf{U}\mathbf{U}'|\mathbf{X}]=\Omega$ is a not diagonal matrix (i.e. some off-diagonal entries are nonzero − autocorrelation) and $\sigma_i^2(\mathbf{X})$ is possibly dependent on $\mathbf{X}$. Then we have the model

$$\begin{aligned} \mathbf{y} &= \mathbf{X}\beta+\mathbf{U} \\ \mathrm{E}[\mathbf{U}|\mathbf{X}] &= \mathbf{0} \\ \mathrm{E}[\mathbf{U}\mathbf{U}'|\mathbf{X}] &= \Sigma \end{aligned}$$

where Σ is not diagonal and the off-diagonal entires represent covariance in the noise terms, $\mathrm{Cov}(u_i,u_j)=\Sigma_{ij}$.

Proposition 4.14 (GLS Estimator). Suppose we have the aforementioned model with the GLS problem

$$\hat{\beta}_{GLS} = \underset{\beta \in \mathbf{R}^K}{\text{argmin}} \mathbf{U}(\beta)\mathbf{U}(\beta)'$$

Then,

$$\hat{\beta}_{GLS} = (\mathbf{X}'\Sigma^{-1}\mathbf{X})^{-1}\mathbf{X}'\Sigma^{-1}\mathbf{y}$$

This is because this is our WLS estimator with Σ^{-1} in place of $\mathbf{W}$.

Example 4.15 (PSet 3 Q2). Let

$$y_i = \beta + u_i \tag{1}$$

Let $\hat{\beta}_{OLS}$ be the OLS estimator of β/ Find $\text{E}[\hat{\beta}_{OLS}]$ and $\text{Var}(\hat{\beta}_{OLS})$. Is $\hat{\beta}_{OLS}$ consistent? Find the asymptotic distribution of the OLS estimator.

Consider the transformed model

$$\frac{y_i}{\sqrt{(1/2)^i}} = \frac{\beta}{\sqrt{(1/2)^i}} + \tilde{u}_i \tag{2}$$

where

$$\tilde{u}_i = \frac{u_i}{\sqrt{(1/2)^i}}$$

Derive the OLS estimator $\hat{\beta}_{GLS}$ of β in the model (2), and find $\text{E}[\hat{\beta}_{GLS}]$ and $\text{Var}(\hat{\beta}_{GLS})$. Show that $\hat{\beta}_{GLS}$ is more efficient than $\hat{\beta}_{OLS}$

5 Nonlinear models

Definition 5.1 (Nonlinear models). The general form of a nonlinear regression model is

$$y_i = h(\mathbf{x}_i, \beta) + \varepsilon_i$$

where the first order conditions for least squares estimation of the parameters are nonlinear functions of the parameters.

In other words, there is no transformation that renders this equation linear in parameters.

Example 5.2 (CES production function). Consider

$$\text{ln}y = \text{ln}\gamma - \frac{v}{\rho}[\delta K^{-\rho} + (1-\delta)L^{-\rho}] + \varepsilon$$

There is no transformation that renders this equation linear in parameters

Example 5.3 (Nonexample). Consider the model

$$y = e^{\beta_0} x_1^{\beta_1} x_2^{\beta_2} e^{\varepsilon}$$

We can take logarithms such that

$$\text{ln}y = \beta_0 + \beta_1 \text{ln}x_1 + \beta_2 \text{ln}x_2 + \varepsilon$$

Remark. The basic problem is that if we try to find the FOCs from our least-squares problem, we get a system of nonlinear equations that do not have a closed form solution in general

5.1 Non-linear Least Squares Estimation

Definition 5.4 (Non-linear least squares Estimator). The Nonlinear Least Squares (NLLS) estimator is defined as

$$\hat{\theta} = \underset{\theta\in\Theta}{\text{argmin}} Q_n(\theta) = \underset{\theta\in\Theta}{\text{argmin}} \sum_{i=1}^{n} \{y_i - g(x_i, \theta)\}^2$$

where θ is a $K \times 1$ vector of parameters. The first order conditions are

$$\dot{Q}_n(\theta) = \frac{\partial Q_n(\theta)}{\partial \theta} = -2\sum_{i=1}^{n}\{y_i - g(x_i,\theta)\}\dot{g}(x_i,\theta) = \mathbf{0}$$

where $\dot{g}(x_i, \theta) = \partial g(x_i, \theta)/\partial\theta$

Example 5.5. Consider the model $y_i = \theta_1 + \theta_2 e^{\theta_3 x_i} + u_i$. Our NLLS estimator is

$$\hat{\theta} = \underset{\theta\in\Theta}{\text{argmin}} Q_n(\theta) = \underset{\theta\in\Theta}{\text{argmin}} \sum_{i=1}^{n} (y_i - \theta_1 + \theta_2 e^{\theta_3 x_i})^2$$

The first order conditions are

$$\nabla Q_n(\theta) = \begin{pmatrix} \frac{\partial Q_n(\theta)}{\partial \theta_1} \\ \frac{\partial Q_n(\theta)}{\partial \theta_2} \\ \frac{\partial Q_n(\theta)}{\partial \theta_3} \end{pmatrix} = \begin{pmatrix} -2\sum_{i=1}^{2} u_i(\theta) \\ -2\sum_{i=1}^{n} u_i(\theta)e^{\theta_3 x_i} \\ -2\sum_{i=1}^{n} u_i(\theta)\theta_2 x_i e^{\theta_3 x_i} \end{pmatrix} = \mathbf{0}$$

We need to use numerical methods to approximate $\hat{\theta}$ since there is no closed form solution.

Gauss-Newton Method

The Gauss-Newton Method is based on Taylor's Theorem.

Theorem 5.6 (Taylor's Theorem). Let $m \in \mathbb{N}$, let $I := [a, b]$ and let $f : I \to \mathbb{R}$ be such that f and its derivatives $\dot{f}, \ddot{f}, \ldots, f^{(m)}$ are continuous on I and that $f^{(m+1)}$ exists on (a, b). If $x_0 \in I$, then for any x in I there exists a point c between x and x_0 such that

$$f(x) = f(x_0) + \frac{\dot{f}(x_0)}{1!}(x - x_0) + \frac{\ddot{f}(x_0)}{2!}(x - x_0)^2 + \cdots + \frac{f^{(m)}(x_0)}{m!}(x - x_0)^m + \frac{f^{(m+1)}(c)}{(m+1)!}(x - x_0)^{m+1}$$

Remark. In other words, we may decompose a function $f(x)$ into

$$f(x) = P_m(x) + R_m(x)$$

where there is a polynomial approximation

$$P_m(x) = f(x_0) + \frac{\dot{f}(x_0)}{1!}(x - x_0) + \frac{\ddot{f}(x_0)}{2!}(x - x_0)^2 + \cdots + \frac{f^{(m)}(x_0)}{m!}(x - x_0)^m$$

and remainder

$$R_m(x) = \frac{f^{(m+1)}(c)}{(m+1)!}(x - x_0)^{m+1}$$

We may want to use a linear approximation of a function $f(x)$ by taking Taylor's approximation to degree $m = 1$ around x_0 where

$$f(x) \approx f(x_0) + \frac{\dot{f}(x_0)}{1!}(x - x_0)$$

Algorithm: Consider the simple model $y_i = g(x_i, \theta) + u_i$ where x_i and θ are scalars.
Step 1: Take the first order Taylor approximation around an initial estimate $\hat{\theta}_{(1)}$:

$$g(x_i, \theta) \approx g(x_i, \hat{\theta}_{(1)}) + \dot{g}(x_i, \hat{\theta}_{(1)})(\theta - \hat{\theta}_{(1)}$$

This gives us a linearised model

$$y_i \approx g(x_i, \hat{\theta}_{(1)}) + \dot{g}(x_i, \hat{\theta}_{(1)})(\theta - \hat{\theta}_{(1)} + u_i)$$

Step 2: The linearlised model

$$y_i \approx g(x_i, \hat{\theta}_{(1)}) + \dot{g}(x_i, \hat{\theta}_{(1)})(\theta - \hat{\theta}_{(1)} + u_i)$$

is a linear model with unknown θ which we can estimate by least squares again

$$\begin{aligned}\hat{\theta}_{(2)} &= \underset{\theta \in \Theta}{\text{argmin}} \tilde{Q}_n(\theta) \\ &= \underset{\theta \in \Theta}{\text{argmin}} \sum_{i=1}^{n} [y_i - g(x_i, \hat{\theta}_{(1)}) + \dot{g}(x_i, \hat{\theta}_{(1)})(\theta - \hat{\theta}_{(1)})]^2\end{aligned}$$

This gives us the FOC

$$\frac{\partial \tilde{Q}_n(\theta)}{\partial \theta} = -2 \sum_{i=1}^{n} [y_i - g(x_i, \hat{\theta}_{(1)}) + \dot{g}(x_i, \hat{\theta}_{(1)})(\theta - \hat{\theta}_{(1)})] \dot{g}(x_i, \hat{\theta}_{(1)})$$

The second round estimator becomes

$$\hat{\theta}_{(2)} = \hat{\theta}_{(1)} + \left[\sum_{i=1}^{n} \dot{g}(x_i, \hat{\theta}_{(1)})^2 \right]^{-1} \sum_{i=1}^{n} [y_i - g(x_i, \hat{\theta}_{(1)})] \dot{g}(x_i, \hat{\theta}_{(1)})$$

Step 3: we do this p times until the p-th iteration where we yield

$$\hat{\theta}_{(p+1)} = \hat{\theta}_{(p)} + \left[\sum_{i=1}^{n} \dot{g}(x_i, \hat{\theta}_{(p)})^2 \right]^{-1} \sum_{i=1}^{n} [y_i - g(x_i, \hat{\theta}_{(p)})] \dot{g}(x_i, \hat{\theta}_{(p)})$$

Iteration continues until convergence is achieved; so starting with an initial value $\hat{\theta}_{(0)}$, the process can be iterated until $|\hat{\theta}_{(p+1)} - \hat{\theta}_{(p)}|$ is small enough.

Remark. We may generalise this to the case where θ is a $K \times 1$ vector of parameters.

5.2 Maximum Likelihood Estimation

Definition 5.7 (Likelihood function). Consider n independent and identically distributed observations of y_i. Then, the likelihood function is the product of the individual densities

$$f(y_1, \ldots, y_n | \theta) = \prod_{i=1}^{n} f(y_i | \theta) = L(\theta | \mathbf{y})$$

where θ is the unknown parameter vector and $\mathbf{y}$ is used to indicate the collection of sample data. $f(\cdot|\theta)$ is the probability density function for a random variable y conditioned on a vector of parameters θ

It is usually simpler to work with the log likelihood function (and since a log transform is monotonic, it preserves the ordering)

$$\ln L(\theta, \mathbf{y}) = \prod_{i=1}^{n} \ln f(y_i | \theta)$$

We denote the log likelihood function as ℓ.
We may generalise this to condition on other variables such as $\mathbf{X}$ so that

$$L(\theta, \mathbf{y}, \mathbf{X}) = \prod_{i=1}^{n} f(\mathbf{y} | \mathbf{X}, \theta)$$

Example 5.8 (Bernoulli model). In the Bernoulli model,

$$f(y_i|\theta) = \theta^{y_i}(1-\theta)^{1-y_i}$$

where $\theta = \mathrm{P}(y_i = 1)$. Since $y_i \sim$ i.i.d., the joint density function is

$$f(y_1, \dots, y_n|\theta) = f(y_1|\theta) \times f(y_2|\theta) \times \dots \times f(y_n|\theta) = \prod_{i=1}^{n} f(y_i|\theta)$$

In the Bernoulli model,

$$f(y_1, \dots, y_n|\theta) = \prod_{i=1}^{n} \theta^{y_i}(1-\theta)^{1-y_i} = \theta^{\sum_{i=1}^{n} y_i}(1-\theta)^{\sum_{i=1}^{n}(1-y_i)}$$

Let $\bar{y} = n^{-1}\sum_{i=1}^{n} y_i$, then the joint densities can be written as

$$f(y_1, \dots, y_n|\theta) = \theta^{n\bar{y}}(1-\theta)^{n(1-\bar{y})} = \{\theta^{\bar{y}}(1-\theta)^{1-\bar{y}}\}^n$$

The joint pdf gives us the probability for any value of $\bar{y}$ for a given θ. The likelihood function then uses a sample for $y_1, \dots, y_n$ (i.e. fixing it at the realised value) to predict the unknown θ via the likelihood function. So,

$$L(\theta|\mathbf{y}) = \{\theta^{\bar{y}}(1-\theta)^{1-\bar{y}}\}^n$$

where $\bar{y}$ is treated as fixed and $L(\theta|\mathbf{y})$ varies only with $\theta \in (0, 1)$

Remark. In Greene, the identification condition for θ is such that if for any other parameter vector $\theta^* \neq \theta$, for some given data $\mathbf{y}$, $L(\theta^*|\mathbf{y}) \neq L(\theta|\mathbf{y})$

Definition 5.9 (Maximum Likelihood estimator). The maximum likelihood estimator $\hat{\theta}_{ML}$ is defined as

$$\hat{\theta}_{ML} = \underset{\theta \in \Theta}{\operatorname{argmax}} L(\theta|\mathbf{y})$$

or equivalently (since log is a monotonically increasing transform),

$$\hat{\theta}_{ML} = \underset{\theta \in \Theta}{\operatorname{argmax}} \ell(\theta|\mathbf{y})$$

where $\ell(\theta|\mathbf{y}) = \ln L(\theta|\mathbf{y})$

The necessary condition for finding $\hat{\theta}_{ML}$ (i.e. maximising L) is to solve the first order conditions

$$\frac{\partial \ell(\theta|\mathbf{y})}{\partial \theta} = \mathbf{0}$$

Example 5.10 (Maximum Likelihood Estimators for different distributions). We consider both the Bernoulli and Poisson models

1. Bernoulli model: the log-likelihood function is

   $$\ell(\theta|\mathbf{y}) = \ln L(\mathbf{y}|\theta) = \ln[\{\theta^{\bar{y}(1-\theta)^{(1-\bar{y})}}\}^n]$$

   Using the properties of log,

   $$\ell(\theta|\mathbf{y}) = n\{\bar{y}\ln(\theta) + (1-\bar{y})\ln(1-\theta)\}$$

   The ML estimator is defined as

   $$\hat{\theta}_{ML} = \underset{\theta \in (0,1)}{\operatorname{argmax}} \ell(\theta|\mathbf{y})$$

   In the Bernoulli case, we have

   $$\hat{\theta}_{ML} = \underset{\theta \in (0,1)}{\operatorname{argmax}} n\{\bar{y}\ln(\theta) + (1-\bar{y})\ln(1-\theta)\}$$

Our FOC is

$$\frac{\partial \ell(\theta|\mathbf{y})}{\partial \theta} = n\left(\frac{\bar{y}}{\theta} - \frac{1-\bar{y}}{1-\theta}\right) = 0$$

Solving for θ, our ML estimator is

$$\hat{\theta}_{ML} = \bar{y}$$

Hence, in the Bernoulli model, the ML estimator of $\theta = \mathrm{P}(y=1)$ is the sample mean $\bar{y}$. To check that $\hat{\theta}_{ML}$ is indeed a maximum, we need to find the SOC

$$\frac{\partial^2 \ell(\theta|\mathbf{y})}{\partial \theta^2} = -n\left(\frac{\bar{y}}{\theta^2} - \frac{1-\bar{y}}{(1-\theta)^2}\right)$$

Since $0 \leq \bar{y} \leq 1$, the second derivative must be negative for any value of $\theta \in (0,1)$

2. For a Poisson model,

$$L(\theta|\mathbf{y}) = f(\mathbf{y}|\theta) = \prod_{i=1}^{n} \frac{e^{-\theta}\theta^{y_i}}{y_i!}$$

Since the likelihood function has a product structure, we rewrite in terms of log-likelihood

$$\ell(\theta|\mathbf{y}) = \ln \prod_{i=1}^{n} \frac{e^{-\theta}\theta^{y_i}}{y_i!} = \sum_{i=1}^{n} \ln \frac{e^{-\theta}\theta^{y_i}}{y_i!}$$

Using the properties of log,

$$\begin{aligned}\ell(\theta|\mathbf{y}) &= \sum_{i=1}^{n} \{-\theta \ln(e) + y_i \ln(\theta) - \ln(y_i!)\} \\ &= -n\theta + \ln(\theta)\sum_{i=1}^{n} y_i - \sum_{i=1}^{n} \ln(y_i!)\end{aligned}$$

The ML estimator is defined as

$$\hat{\theta}_{ML} = \underset{\theta > 0}{\mathrm{argmax}} \ell(\theta|\mathbf{y})$$

The FOC is

$$\frac{\partial \ell(\theta|\mathbf{y})}{\partial \theta} = -1 + \frac{1}{\theta}\frac{1}{n}\sum_{i=1}^{n} y_i = 0$$

Solving for θ, we yield

$$\hat{\theta}_{ML} = \bar{y}$$

Checking the SOC,

$$\frac{\partial^2 \ell(\theta|\mathbf{y})}{\partial \theta^2} = -\frac{\bar{y}}{\theta^2} < 0, \quad \text{for } \bar{y} > 0$$

for any value of θ

Definition 5.11 (Maximum Likelihood estimator with conditioning). Consider we have a model

$$y_i = g(\mathbf{x}_i, \beta) + u_i$$

The maximum likelihood estimator $\hat{\theta}_{ML}$ is defined as

$$\hat{\theta}_{ML} = \underset{\theta \in \Theta}{\mathrm{argmax}} L(\theta|\mathbf{y}, \mathbf{x}_1, \ldots, \mathbf{x}_n)$$

or equivalently (since log is a monotonically increasing transform),

$$\hat{\theta}_{ML} = \underset{\theta \in \Theta}{\mathrm{argmax}} \ell(\theta|\mathbf{y}, \mathbf{x}_1, \ldots, \mathbf{x}_n)$$

where $\ell(\theta|\mathbf{y}) = \ln L(\theta|\mathbf{y}, \mathbf{x}_1, \ldots, \mathbf{x}_n)$

Example 5.12. Consider the Poisson regression model

$$y_i = e^{\mathbf{x}_i'\beta} + u_i$$

where x_i is a $p \times 1$ vector of regressors and β is a $p \times 1$ vector of unknown parameters. Our objective is to estimate β by ML.
In this model, each y_i is drawn from a Poisson population with parameter $\theta_i(\beta)$, which is related to the regressors $\mathbf{x}_i$. In particular, we may write the conditional density as

$$f_{\theta_i(\beta)}(y_i|\mathbf{x}_i) = \frac{e^{-\theta_i(\beta)}\{\theta_i(\beta)\}^{y_i}}{y_i!}, \quad y_i = 0, 1, 2, \dots$$

The Poisson regression model set $\theta_i(\beta) = e^{\mathbf{x}_i'\beta}$. In an i.i.d. context, the conditional density is

$$f_{\theta_i(\beta)}(\mathbf{y}|\mathbf{x}_1, \dots, \mathbf{x}_n) = \prod_{i=1}^{n} \frac{e^{-\theta_i(\beta)}\{\theta_i(\beta)\}^{y_i}}{y_i!}$$

The likelihood function is

$$L_(\beta|\mathbf{y}, \mathbf{x}_1, \dots, \mathbf{x}_n) = f_{\theta_i(\beta)}(\mathbf{y}|\mathbf{x}_1, \dots, \mathbf{x}_n) = \prod_{i=1}^{n} \frac{e^{-\theta_i(\beta)}\{\theta_i(\beta)\}^{y_i}}{y_i!}$$

where $\theta_i = e^{\mathbf{x}_i'\beta}$. The conditional log-likelihood function is

$$\ell(\beta|\mathbf{y}, \mathbf{x}_1, \dots, \mathbf{x}_n) = \ln \prod_{i=1}^{n} \frac{e^{-\theta_i(\beta)}\{\theta_i(\beta)\}^{y_i}}{y_i!} = \sum_{i=1}^{n} \ln \frac{e^{-\theta_i(\beta)}\{\theta_i(\beta)\}^{y_i}}{y_i!}$$

Using properties of log, we can write

$$\begin{aligned}\ell(\beta|\mathbf{y}, \mathbf{x}_1, \dots, \mathbf{x}_n) &= \sum_{i=1}^{n} \{-\theta_i(\beta)\ln(e) + y_i \ln\{\theta_i(\beta)\} - \ln(y_i!)\} \\ &= \sum_{i=1}^{n} \{-e^{\mathbf{x}_i'\beta} + y_i \mathbf{x}_i'\beta - \ln y_i!\}\end{aligned}$$

The first order conditions are

$$\frac{\partial \ell(\beta|\mathbf{y}, \mathbf{x}_1, \dots, \mathbf{x}_n)}{\partial \beta} = n^{-1} \sum_{i=1}^{n} (y_i - e^{\mathbf{x}_i'\beta})\mathbf{x}_i = \mathbf{0}$$

This is a system of nonlinear equations with no closed form solution and therefore iterative methods need to be used to approximate the solution.

Remark. The slides write

$$\frac{\partial \ell(\beta|\mathbf{y}, \mathbf{x}_1, \dots, \mathbf{x}_n)}{\partial \beta} = n^{-1} \sum_{i=1}^{n} (y_i - e^{\mathbf{x}_i'\beta})\mathbf{x}_i' = \mathbf{0}$$

which I suspect is wrong (multiply by $\mathbf{x}_i'$ instead of $\mathbf{x}_i$) since by the rules of matrix differentiation we have

$$\frac{\partial (\mathbf{A}\mathbf{x})}{\partial \mathbf{x}} = \mathbf{A}'$$

5.3 Extremum Estimators and Asymptotics

We want to give conditions for (i) consistency and (ii) asymptotic normality

Definition 5.13 (Extremum Estimator).

Remark. The LS estimator $\hat{\theta}_{LS}$ and ML estimator $\hat{\theta}_{ML}$ (as well as the GMM estimator) are examples of extremum estimators since they are about maximising or minimising some function

Theorem 5.14 (Consistency and Asymptotic Normality). Let the following assumptions hold

(A) The parameter space Θ is compact (θ_0 is in Θ)

(B) $Q_n(\theta)$ is continuous in $\theta \in \Theta$

(C) For a nonstochastic function $Q(\theta)$,

$$\sup_{\theta \in \Theta} |Q_n(\theta) - Q(\theta)| \xrightarrow{p} 0$$

where $Q(\theta)$ attains a unique global optimum at θ

Then,

$$\hat{\theta}_n \xrightarrow{p} \theta_0$$

where θ_0 is the true parameter value we are trying to estimate. Furthermore,

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} \mathcal{N}(0, A(\theta_0)^{-1} B(\theta_0)^{-1} A(\theta_0)^{-1})$$

Therefore, to check if the NLLS or ML estimators are consistent or converges asymptotically to the normal distribution, it suffices to check if assumptions (A)-(C) hold.

Proof. To be followed ∎

5.3.1 A precis in real analysis

Definition 5.15 (Sup and Inf). Let S be a nonempty subset of $\mathbb{R}$.

1. If S is bounded above, then a number s is said to be a supremum of S if it satisfies
   (a) s is an upper bound of S, and
   (b) if v is any upper bound of S, then $s \leq v$
2. If S is bounded below, then a number w is said to be the infimum of S if it satisfies
   (a) w is a lower bound of S, and
   (b) if t is any lower bound of S, then $t \leq w$

Definition 5.16 (Continuity). Let f be a mapping from the metric space (X, d) to the metric space (Y, ρ). f is continuous at $a \in X$ if for any $\epsilon > 0$, there exists $\delta > 0$ such that $d(x, a) < \delta$ implies $\rho(f(x), f(a)) < \epsilon$. f is continuous if it is continuous at all $a \in X$

Proposition 5.17 (Continuity). f is continuous at a if ad only if for every sequence (a_n) with $a_n \to a$ as $n \to \infty$, $f(a_n) \to f(a)$ as $n \to \infty$

Proposition 5.18 (Differentiability implies continuity). Suppose a function $f(x)$ is differentiable on some interval $I \in [a, b]$. Then, $f(x)$ is continuous on I.

Remark. The converse is not true in general.

Definition 5.19 (Compactness). A subset K of metric space (X, d) is (sequentially) compact if any sequence in K has a subsequence which converges to a point in K. The space (X, d) is compact if X is compact.

Definition 5.20 (Bounded Space). A set A is a metric space (X, d) is said to be bounded if there is an $M > 0$ such that $d(a, b) \leq M$ for all $a, b \in A$.

Definition 5.21 (Closed Sets). If (X, d) is a metric space, then $A \subseteq X$ is closed (or closed in X) if A^c is open.

Proposition 5.22. The following statements are equivalent

1. A is closed
2. A contains all of its boundary points
3. For any sequence (a_n) in A, if $a_n \to a$ as $n \to \infty$, then $a \in A$

Theorem 5.23 (Heine-Borel). In $\mathbb{R}^n$, a set is compact if and only if it is closed and bounded

6 Discrete Choice models

We want to understand what contributes to the probability of each of (discrete) choices represented by the response vector $\mathbf{y}$.

6.1 Binary Choice

Definition 6.1 (Binary Choice Model). Binary choice models are characterised by a dependent variable that takes only two values

$$y = \begin{cases} 1 & \text{if event } A \text{ is observed} \\ 0 & \text{otherwise} \end{cases}$$

The objective is to model the probability that $y = 1$. Hence, consider the $K \times 1$ regressor vector $\mathbf{x}_i$, where we specify

$$\begin{aligned} \mathrm{P}_\theta(y = 1|\mathbf{x}) &= G(\mathbf{x}, \theta) \\ \mathrm{P}_\theta(y = 0|\mathbf{x}) &= 1 - G(\mathbf{x}, \theta) \end{aligned}$$

Note that in this model, the CEF is

$$\mathrm{E}[y|\mathbf{x}] = 1 \times G(\mathbf{x}, \theta) + 0 \times \{1 - G(\mathbf{x}, \theta)\} = G(\mathbf{x}, \theta)$$

6.1.1 Linear Probability Model

Definition 6.2. The linear probability model assumes that

$$\mathrm{P}_\theta(y = 1|\mathbf{x}) = G(\mathbf{x}, \theta) = \mathbf{x}'\theta = \theta_1 + \theta_2 x_2 + \cdots + \theta_K x_K$$

and specifies

$$y_i = \theta_1 + \theta_2 x_{2i} + \cdots + \theta_K x_{Ki} + u_i$$

or

$$y_i = \mathbf{x}_i'\theta + u_i$$

where x_i is a $K \times 1$ vector of regressors for an individual i and θ is a $K \times 1$ vector of unknown parameters

The problem with this is

1. The model error term is heteroskedastic by construction. Notice that

$$\mathrm{Var}(u_i|\mathbf{x}_i) = \mathbf{x}_i'\theta(1 - \mathbf{x}_i'\theta)$$

since u_i equals either $-\mathbf{x}_i'\theta$ or $1 - \mathbf{x}'\theta$ since y_i must equal 0 or 1. So the term is conditionally heteroskedastic. The consequence of this is that the OLS estimator is no longer the most efficient (the Gauss-Markov Theorem does not apply) but we can still use HAC.

2. the predicted probabilities

$$\hat{y}_i = \mathbf{x}_i'\hat{\theta} \notin [0,1]$$

which violates the assumption of a probability measure and we may not be able to interpret the linear probability model

The upshot of this is that we want to impose constraints such that

- $\lim_{\mathbf{x}'\theta \longrightarrow +\infty} \mathrm{P}_\theta(y=1|\mathbf{x}) = 1$
- $\lim_{\mathbf{x}'\theta \longrightarrow -\infty} \mathrm{P}_\theta(y=1|\mathbf{x}) = 0$

This is achieved by choosing $\mathrm{P}_\theta(y=1|\mathbf{x}) = G(\mathbf{x},\theta)$ to be a distribution function, which is what we achieve through Probit and Logit models.

6.1.2 Probit and Logit

Definition 6.3 (Probit and Logit). The probit model specifies

$$\mathrm{P}_\theta(y=1|\mathbf{x}) = \int_{-\infty}^{\mathbf{x}'\theta} \phi(t)dt = \Phi(\mathbf{x}'\theta)$$

where $\Phi(\mathbf{x}'\theta)$ denotes the standard normal distribution function and ϕ the standard normal density.
The logit chooses the logistic distribution instead of the standard normal such that

$$\mathrm{P}_\theta(y=1|\mathbf{x}) = \frac{\exp(\mathbf{x}'\theta)}{1+\exp(\mathbf{x}'\theta)} = \Lambda(\mathbf{x}'\theta)$$

In general, we have that our model

$$\mathrm{P}_\theta(y=1|\mathbf{x}) = \mathrm{E}(y|\mathbf{x}) = G(\mathbf{x}'\theta) = G(+\theta_1 + \theta_2 x_2 + \cdots + \theta_K x_K)$$

where G is nonlinear.

Definition 6.4 (Marginal effects). We define the marginal effects of explanatory variables differently for binary and continuous explanatory variables

1. Binary Regressors: suppose x_2 is a binary variable that takes 0 or 1. Then, the partial effect of x_2 (depending on all other values of x_k) is

   $$G(\theta_1 + \theta_2 + \theta_3 x_3 + \cdots + \theta_K x_K) - G(\theta_1 + \theta_3 x_3 + \cdots + \theta_K x_K)$$

   The intuition behind this is that on the LHS we set $x_2 = 1$ and subtract $G(\cdot)$ when $x_2 = 0$ in which case θ_2 also disappears

2. Continuous regressors: For a continuous variable x_j, the marginal effect is

   $$\frac{\partial \mathrm{E}[y|\mathbf{x}]}{\partial x_j} = \frac{\partial G(\mathbf{x}'\theta)}{\partial x_j}$$

Example 6.5 (Probit Marginal Effect). Consider the Probit model

$$\mathrm{P}_\theta(y=1|\mathbf{x}) = \int_{-\infty}^{\mathbf{x}'\theta} \phi(t)d(t) = \Phi(\mathbf{x}'\theta)$$

Then, the marginal effect is

$$\frac{\partial \mathrm{E}[y|\mathbf{x}]}{\partial x_j} = \frac{d\Phi(\mathbf{x}'\theta)}{d(\mathbf{x}'\theta)}\frac{\partial(\mathbf{x}'\theta)}{\partial x_j} = \frac{d\Phi(\mathbf{x}'\theta)}{d(\mathbf{x}'\theta)}\theta_j = \phi(\mathbf{x}'\theta)\theta_j$$

where $\phi(\mathbf{x}'\theta)$ is known as the scale factor. The marginal effect has the be evaluated at particular values of x (such as the mean)

$$\frac{\partial \mathrm{E}[y|\mathbf{x}]}{\partial x_j}\Big|_{\bar{\mathbf{x}}} = \phi(\bar{\mathbf{x}}'\theta)\theta_j$$

Example 6.6 (Logit Marginal Effect). Consider the Logit model

$$\mathrm{P}_\theta(y=1|\mathbf{x}) = \frac{\exp(\mathbf{x}'\theta)}{1+\exp(\mathbf{x}'\theta)} = \Lambda(\mathbf{x}'\theta)$$

Marginal effect:

$$\begin{aligned}\frac{\partial \mathrm{E}[y|\mathbf{x}]}{\partial x_j} &= \frac{d\Lambda(\mathbf{x}'\theta)}{d(\mathbf{x}'\theta)}\frac{\partial(\mathbf{x}'\theta)}{\partial x_j} \\ &= \frac{d\Lambda(\mathbf{x}'\theta)}{d(\mathbf{x}'\theta)}\theta_j \\ &= \frac{\exp(\mathbf{x}'\theta)\{1+\exp(\mathbf{x}'\theta)\} - \exp(\mathbf{x}'\theta)\exp(\mathbf{x}'\theta)}{\{1+\exp(\mathbf{x}'\theta)\}^2}\theta_j \\ &= \Lambda(\mathbf{x}'\theta)\{1-\Lambda(\mathbf{x}'\theta)\}\theta_j\end{aligned}$$

where $\Lambda(\mathbf{x}'\theta)\{1-\Lambda(\mathbf{x}'\theta)\}$ is known as the scale factor. As usual, we need to evaluate this at particular values such as the mean values of x i.e.

$$\left.\frac{\partial \mathrm{E}[y|\mathbf{x}]}{\partial x_j}\right|_{\bar{\mathbf{x}}} = \Lambda(\bar{\mathbf{x}}'\theta)\{1-\Lambda(\bar{x}'\theta)\}\theta_j$$

Both probit and logit models specify a nonlinear conditional expectation and so nonlinear estimation methods to estimate the unknown θ is needed. However

- NLLS is not efficient in this context because heteroskedasticity is not accounted for
- So, the ML estimator which is based on the distribution of y given $\mathbf{x}$ is more efficient, precisely because it takes into account heteroskedasticity

In general, let (y_i, x_i) be a random sample size n with conditional density

$$\mathrm{P}_\theta(y_i|\mathbf{x}_i) = [G(\mathbf{x}_i'\theta)]^{y_i}[1-G(\mathbf{x}_i'\theta)]^{1-y_i}, \quad \text{where } y_i = 0, 1$$

The joint density as usual for $y_1, \ldots, y_n$ given $\mathbf{x}_1, \ldots, \mathbf{x}_n$ is then

$$\mathrm{P}_\theta(y_1,\ldots,y_n|\mathbf{x}_1,\ldots,\mathbf{x}_n) = \prod_{i=1}^n [G(\mathbf{x}_i'\theta)]^{y_i}[1-G(\mathbf{x}_i'\theta)]^{1-y_i}$$

The log-likelihood is

$$\ell_{y_1,\ldots,y_n|\mathbf{x}_1,\ldots,\mathbf{x}_n}(\theta) = \log\mathrm{P}_\theta(y_1,\ldots,y_n|\mathbf{x}_1,\ldots,\mathbf{x}_n)$$

and using the properties of log

$$\ell_{y_1,\ldots,y_n|\mathbf{x}_1,\ldots,\mathbf{x}_n}(\theta) = \sum_{i=1}^n y_i \ln[G(\mathbf{x}_i'\theta)] + \sum_{i=1}^n (1-y_i)\ln[1-G(\mathbf{x}_i'\theta)]$$

The maximum likelihood estimator is

$$\hat{\theta}_{ML} = \underset{\theta\in\mathbb{R}^K}{\operatorname{argmax}} \ell_{y_1,\ldots,y_n|\mathbf{x}_1,\ldots,\mathbf{x}_n}(\theta)$$

where our FOCs are

$$\frac{\partial \ell_{y_1,\ldots,y_n|\mathbf{x}_1,\ldots,\mathbf{x}_n}(\theta)}{\partial\theta} = \sum_{i=1}^n y_i \frac{g(\mathbf{x}_i'\theta)\mathbf{x}_i}{G(\mathbf{x}_i\theta)} - \sum_{i=1}^n (1-y_i)\frac{g(\mathbf{x}_i'\theta)\mathbf{x}_i}{1-G(\mathbf{x}_i'\theta)} = \mathbf{0}$$

which can be rewritten as

$$\frac{\partial \ell_{y_1,\ldots,y_n|\mathbf{x}_1,\ldots,\mathbf{x}_n}(\theta)}{\partial\theta} = \sum_{i=1}^n \left[y_i \frac{g(\mathbf{x}_i'\theta)}{G(\mathbf{x}_i'\theta)} - (1-y_i)\frac{g(\mathbf{x}_i'\theta)}{1-G(\mathbf{x}_i\theta)}\mathbf{x}_i\right] = 0$$

Proposition 6.7 (Probit ML Estimator). Consider the Probit model

$$G(\mathbf{x}_i'\theta) = \Phi(\mathbf{x}_i'\theta)$$

The FOC is

$$\frac{\partial \ell_{y_1,\ldots,y_n|\mathbf{x}_1,\ldots,\mathbf{x}_n}(\theta)}{\partial \theta} = \sum_{i=1}^{n} w_i\{y_i - \Phi(\mathbf{x}_i'\theta)\}\mathbf{x}_i = \mathbf{0}$$

with $w_i = \phi(\mathbf{x}_i'\theta)/[\Phi(\mathbf{x}_i'\theta)\{1-\Phi(\mathbf{x}_i'\theta)\}]$.
We may solve for $\hat{\theta}$ via numerical approximation.

Proof. Differentiating gives us

$$\frac{\partial \ell_{y_1,\ldots,y_n|\mathbf{x}_1,\ldots,\mathbf{x}_n}(\theta)}{\partial \theta} = \sum_{i=1}^{n} \left\{ y_i \frac{\phi(\mathbf{x}_i'\theta)}{\Phi(\mathbf{x}_i'\theta)} - (1-y_i)\frac{\phi(\mathbf{x}_i'\theta)}{1-\Phi(\mathbf{x}_i'\theta)} \right\} \mathbf{x}_i = \mathbf{0}$$

which simplifies to

$$\frac{\partial \ell_{y_1,\ldots,y_n|\mathbf{x}_1,\ldots,\mathbf{x}_n}(\theta)}{\partial \theta} = \sum_{i=1}^{n} w_i\{y_i - \Phi(\mathbf{x}_i'\theta)\}\mathbf{x}_i = \mathbf{0}$$

with $w_i = \phi(\mathbf{x}_i'\theta)/[\Phi(\mathbf{x}_i'\theta)\{1-\Phi(\mathbf{x}_i'\theta)\}]$. ∎

Proposition 6.8 (Logit ML Estimator). Consider the logit model

$$G(\mathbf{x}_i'\theta) = \Lambda(\mathbf{x}_i'\theta) = e^{\mathbf{x}_i'\theta}/(1+e^{\mathbf{x}_i'\theta})$$

The FOC is

$$\frac{\partial \ell_{y_1,\ldots,y_n|\mathbf{x}_1,\ldots,\mathbf{x}_n}(\theta)}{\partial \theta} = \sum_{i=1}^{n} \{y_i - \Lambda(\mathbf{x}_i'\theta)\}\mathbf{x}_i = \mathbf{0}$$

We may solve for $\hat{\theta}$ via numerical approximation.

Proof. Differentiating gives us

$$\frac{\partial \ell_{y_1,\ldots,y_n|\mathbf{x}_1,\ldots,\mathbf{x}_n}(\theta)}{\partial \theta} = \sum_{i=1}^{n} \left\{ y_i \frac{e^{\mathbf{x}'\theta}/(1+e^{\mathbf{x}_i\theta})^2}{e^{\mathbf{x}'\theta}/(1+e^{\mathbf{x}_i\theta})} - (1-y_i)\frac{e^{\mathbf{x}'\theta}/(1+e^{\mathbf{x}_i\theta})^2}{e^{\mathbf{x}'\theta}/(1+e^{\mathbf{x}_i\theta})} \right\} \mathbf{x}_i = \mathbf{0}$$

which simplifies to

$$\frac{\partial \ell_{y_1,\ldots,y_n|\mathbf{x}_1,\ldots,\mathbf{x}_n}(\theta)}{\partial \theta} = \sum_{i=1}^{n} \{y_i - \Lambda(\mathbf{x}_i'\theta)\}\mathbf{x}_i = \mathbf{0}$$

∎

Just like linear models, we want to understand how well the model performs and thus we need to develop measures of fit.

Definition 6.9 (McFadden's Pseudo-R^2). We define

$$R_M^2 = 1 - \frac{\ell_{y_1,\ldots,y_n|\mathbf{x}_1,\ldots,\mathbf{x}_n}(\hat{\theta}_{ML})}{\ell_{y_1,\ldots,y_n|\mathbf{x}_1,\ldots,\mathbf{x}_n}(\bar{y}}$$

More precisely, we have

$$R_M^2 = 1 - \frac{\sum_{i=1}^{n}\{y_i \ln \hat{p}_i + (1-y_i)\ln(1-\hat{p}_i)\}}{n\{\bar{y}\ln\bar{y} + (1-\bar{y})\ln(1-\bar{y})\}}$$

where $\hat{p}_i = G(\mathbf{x}_i'\hat{\theta}_{ML})$ and $\bar{y} = n^{-1}\sum_{i=1}^{n} y_i$

The intuition is that it compares the log-likelihood function with all regressors to the log-likelihood function with no regressors. We may interpret the R^2_M similarly to the standard R^2 where if the model is not good, then

$$\ell_{y_1,\dots,y_n|\mathbf{x}_1,\dots,\mathbf{x}_n}(\hat{\theta}_{ML}) \approx \ell_{y_1,\dots,y_n|\mathbf{x}_1,\dots,\mathbf{x}_n}(\bar{y})$$

and R^2_M will be close to 0.

Proposition 6.10. Consider

$$R^2_M = 1 - \frac{\ell_{y_1,\dots,y_n|\mathbf{x}_1,\dots,\mathbf{x}_n}(\hat{\theta}_{ML})}{\ell_{y_1,\dots,y_n|\mathbf{x}_1,\dots,\mathbf{x}_n}(\bar{y}}$$

Then, $R^2_M \in [0,1]$

The intuition for this is that since $\bar{y}$ and $\hat{p}_i$ are in [0,1], it follows that $\ell_{\mathbf{y}|\mathbf{x}}(\hat{\theta})$ and $\ell_{\mathbf{y}|\mathbf{x}}(\bar{y})$ are negative. Furthermore, $\ell_{\mathbf{y}|\mathbf{x}}(\bar{y}) \leq \ell_{\mathbf{y}|\mathbf{x}}(\hat{\theta})$ since the model with no regressors will have a smaller (more negative) or equal likelihood value. Hence, the magnitude of $\ell_{\mathbf{y}|\mathbf{x}}(\hat{\theta})$ is smaller than $\ell_{\mathbf{y}|\mathbf{x}}(\bar{y})$ and so $R^2_M \in [0,1]$
Asymptotic Properties
We may refer to **Theorem 5.14** to see conditions under which the ML estimator is consistent.

Proposition 6.11 (Asymptotic Variance). Suppose Assumptions (A)-(C) hold from **Theorem 5.14**. Then, we can find the asymptotic variance of $\hat{\theta}_{ML}$ by estimating

$$\widehat{\text{AVar}}(\hat{\theta}_{ML}) = \left(\sum_{i=1}^{n} \frac{\{g(\mathbf{x}_i'\hat{\theta}_{ML}\}^2 \mathbf{x}_i \mathbf{x}_i'}{G(\mathbf{x}_i'\hat{\theta}_{ML})\{1 - G(\mathbf{x}_i'\hat{\theta}_{ML})\}} \right)$$

for a $K \times K$ matrix

Proof. To be followed ■

Inference
For a single hypothesis such as

$$H_0 : \theta_j = 0, \quad H_1 : \theta_j \neq 0$$

we can test using the t-statistic

$$t = \frac{\hat{\theta}_j}{s.e.(\hat{\theta}_j)}$$

where $s.e.(\hat{\theta}_j)$ is computed using $\widehat{\text{AVar}(\hat{\theta}_{ML})}$. The asymptotic distribution is $\mathcal{N}(0,1)$ so the decision rule is as usual.
We are interested in testing the following set of J hypotheses where

$$H_0 : c(\theta) = \mathbf{q}, \quad H_1 : c(\theta) \neq \mathbf{q}$$

where $c(\theta)$ is possibly a nonlinear transformation of θ. We have three tests to do so.

1. Wald Test: Use the statistic

$$W = \{c(\hat{\theta}) - \mathbf{q}\}'[\widehat{\text{AVar}}\{c(\hat{\theta}) - \mathbf{q}\}]^{-1}\{c(\hat{\theta}) - \mathbf{q}\}$$

If the restriction in H_0 holds, then $c(\hat{\theta}) - \mathbf{q}$ should be close to zero. Moreover, W is self-normalised since $\{c(\hat{\theta}) - \mathbf{q}\}$ is divided by its standard deviation. Hence,

$$W \overset{a}{\sim} \chi^2_J$$

For a chosen significance level α and corresponding critical value, say cv_α, the decision rule is to reject the null if $W > cv_\alpha$

2. Likelihood Ratio Test: we use the likelihood ratio test statistic, which is based on estimates of both the unrestricted and restricted models i.e.

$$LR = -2(\ln\hat{L}_R - \ln\hat{L}_{UR}) = -2\ln\left(\frac{\hat{L}_R}{\hat{L}_{UR}}\right)$$

   where $\hat{L}_R$ and $\hat{L}_{UR}$ are the values of the likelihood function evaluated at the restricted and unrestricted models respectively. If the restrictions in H_0 hold, then $\hat{L}_R$ and $\hat{L}_{UR}$ would be close (and its ratio close to 1) and LR would be close to 0.
   It can be shown that

$$LR \overset{a}{\sim} \chi^2_J$$

   For a chosen significance level α and corresponding critical value, say cv_α, the decision rule is to reject the null if $LR > cv_\alpha$

3. Lagrange Multiplier Test: based on estimates of the restricted model i.e.

$$LM = \left\{\frac{\partial \ln L(\hat{\theta}_R)}{\partial \hat{\theta}_R}\right\}' \{I(\hat{\theta}_R)\}^{-1} \left\{\frac{\partial \ln L(\hat{\theta}_R)}{\partial \hat{\theta}_R}\right\}$$

   where $I(\hat{\theta}_R)$ is the information matrix (expected Hessian matrix of second derivatives). If the restrictions in H_0 hold, then $\hat{\theta}_R$ will not be far from the point that maximises the log-likelihood and the score $\partial L(\hat{\theta}_R)/\partial\hat{\theta}_R$ will be close to zero.
   It can be shown that

$$LM \overset{a}{\sim} \chi^2_J$$

   For a chosen significance level α and corresponding critical value, say cv_α, the decision rule is to reject the null if $LM > cv_\alpha$

Theorem 6.12. The test procedures and statistics W, LR and LM are asymptotic equivalent.

Remark. They differ in finite samples

6.2 Count Data

The variable of intertest is a count of the number of occurrences of a certain event such as number of times someone is arrested or number of patents applied for by a firm in a year. The dependent variable y takes on values $0, 1, 2, ...$ The linear probability model is not good for obvious reasons.

Definition 6.13 (Poisson Model). The Poisson regression model specifies

$$\mathrm{E}[y|\mathbf{x}] = \exp(\theta_1 + \theta_2 x_2 + \theta_3 x_3 + \cdots + \theta_K x_K)$$

The Poisson regression models ensures that the predicted values of y are also positive, since the exponential function is always positive.

Definition 6.14 (Marginal Effects for Poisson model). For a binary regressor x_2, the marginal effect of going for 0 to 1 is

$$\mathrm{E}[y|x_2 = 1, \ldots, x_k] - \mathrm{E}[y|x_2 = 0, \ldots, x_k]$$

and hence

$$\exp(\theta_1 + \theta_2 + \theta_3 x_3 + \cdots + \theta_K x_K) - \exp(\theta_1 + \theta_3 x_3 + \cdots + \theta_K x_K)$$

For a continuous regressor x_j,

$$\frac{\partial \mathrm{E}[y|\mathbf{x}]}{\partial x_j} = \exp(\theta_1 + \theta_2 x_2 + \theta_3 x_3 + \cdots + \theta_K x_K)\theta_j$$

As usual, these marginal effects depend on all other regressors and need to be evaluated for some representative individual.

Coefficient Interpretation

The Poisson regression model is nonlinear and so

$$\mathrm{E}[y|\mathbf{x}] = \exp(\theta_1 + \theta_2 x_2 + \theta_3 x_3 + \cdots + \theta_K x_K)$$

so that the logarithm of $\mathrm{E}[y|\mathbf{x}]$ is linear, that is,

$$\ln\mathrm{E}[y|\mathbf{x}] = \theta_1 + \theta_2 x_2 + \theta_3 x_3 + \cdots + \theta_K x_K$$

Notice that

$$\ln\mathrm{E}(y|x_1, \ldots, x_K) - \ln\mathrm{E}(y|x_1, \ldots, \tilde{x}_K) = \theta_K(x_K - \tilde{x}_K) = \theta_K \Delta x_K$$

where $\Delta x_K = x_K - \tilde{x}_K$. Then $100 \times \theta_K$ is approximately the percentage change in $\mathrm{E}[y|\mathbf{x}]$ is given one-unit increase in x_K. Furthermore,

$$\Delta_K \mathrm{E}(y|\mathbf{x}) = \frac{\mathrm{E}(y|x_1, \ldots, x_K) - \mathrm{E}(y|x_1, \ldots, \tilde{x}_K)}{\mathrm{E}(y|x_1, \ldots, \tilde{x}_K)}$$

so that

$$\Delta_K \mathrm{E}(y|\mathbf{x}) = \frac{\exp(\theta_1 + \theta_2 x_2 + \cdots + \theta_K x_K) - \exp(\theta_1 + \theta_2 x_2 + \cdots + \theta_K \tilde{x}_K)}{\exp(\theta_1 + \theta_2 x_2 + \cdots + \theta_K \tilde{x}_K)}$$

We can write this as

$$\Delta_K \mathrm{E}(y|\mathbf{x}) = \frac{\exp(\theta_1 + \theta_2 x_2 + \cdots + \theta_K x_K)}{\exp(\theta_1 + \theta_2 x_2 + \cdots + \theta_K \tilde{x}_K)} - 1$$

which simplifies as

$$\Delta_K \mathrm{E}(y|\mathbf{x}) = \exp(\theta_K \Delta x_K) - 1$$

Using a first order Taylor expansion of $\exp(\theta_K \Delta x_K)$ around zero,

$$\exp(\theta_K \Delta x_K) \approx \exp(0) + \frac{\exp(0)}{1!}(\theta_K \Delta x_K - 0) = 1 + \Delta_K \Delta x_K$$

Inserting this approximation into $\Delta_K \mathrm{E}(y|\mathbf{x})$, we get

$$\Delta_K \mathrm{E}(y|\mathbf{x}) \approx \theta_K \Delta x_K$$

Therefore, for a unit increase in x_K, so that $\Delta x_K = 1$, the coefficient θ_K represents (approximately) the change in the conditional expectation $\Delta_K \mathrm{E}(y|\mathbf{x})$.

Proposition 6.15 (ML Poisson Estimator). The ML estimator is defined as

$$\hat{\beta}_{ML} = \underset{\beta \in \mathbb{R}^n}{\operatorname{argmin}} \ell_{y_1,\ldots,y_n|x_1,\ldots,x_n}(\beta)$$

The first order condition is

$$\frac{\partial \ell_{y_1,\ldots,y_n|x_1,\ldots,x_n}(\beta)}{\partial \beta} = \frac{1}{n}\sum_{i=1}^{n}(y_i - e^{\mathbf{x}_i'\beta})\mathbf{x}_i = \mathbf{0}$$

since this is a system of non-linear equations with no closed form solution. Therefore, iterative methods, such as Gauss-Newton need to be used in order to approximate the solution.

Proof. The pdf for the model is

$$f_{\theta_i(\beta)}(y_i|\mathbf{x}_i) = \frac{e^{-\theta_i(\beta)}\{\theta_i(\beta)\}^{y_i}}{y_I!}, \quad y_i = 0, 1, 2, \ldots$$

where $\theta_i(\beta) = e^{\mathbf{x}_i'\beta}$. Hence, in this setting

$$\mathrm{E}[y_i|\mathbf{x}_i] = \mathrm{Var}(y_i|\mathbf{x}_i) = \theta_i(\beta) = e^{\mathbf{x}_i'\beta}$$

and we can write

$$y_i = e^{\mathbf{x}_i'\theta} + u_i$$

By the i.i.d. assumption, we can write the conditional density function as

$$f_{\theta_i(\beta)}(\mathbf{y}|\mathbf{x}_1, \ldots, \mathbf{x}_n) = \prod_{i=1}^{n} \frac{e^{-\theta(\beta)}\{\theta_i(\beta)\}^{y_i}}{y_i!}$$

The likelihood function is

$$L_{y_1,\ldots,y_n|\mathbf{x}_1,\ldots,\mathbf{x}_n}(\beta) = f_{\theta_i(\beta)}(\mathbf{y}|\mathbf{x}_1, \ldots, \mathbf{x}_n) = \prod_{i=1}^{n} \frac{e^{-\theta(\beta)}\{\theta_i(\beta)\}^{y_i}}{y_i!}$$

where $\theta_i(\beta) = e^{\mathbf{x}_i\beta}$. The conditional log-likelihood function is

$$\ell_{y_1,\ldots,y_n|\mathbf{x}_1,\ldots,\mathbf{x}_n} = \ln \prod_{i=1}^{n} \frac{e^{-\theta_i(\beta)}\{\theta_i(\beta)\}^{y_i}}{y_i!} = \sum_{i=1}^{n} \ln \frac{e^{-\theta_i(\beta)}\{\theta_i(\beta)\}^{y_i}}{y_i!}$$

Using the properties of log, we can write

$$\ell_{y_1,\ldots,y_n|\mathbf{x}_1,\ldots,\mathbf{x}_n} = \sum_{i=1}^{n} \{-e^{\mathbf{x}_i'\beta}\ln(e) + y_i\ln(e^{\mathbf{x}_i'\beta}) - \ln(y_i!)\}$$

which simplifies as

$$\ell_{y_1,\ldots,y_n|\mathbf{x}_1,\ldots,\mathbf{x}_n}(\beta) = \sum_{i=1}^{n} \{-e^{\mathbf{x}_i'\beta} + y_i\mathbf{x}_i'\beta - \ln(y_i!)\}$$

Hence, our FOC is

$$\frac{\partial \ell_{y_1,\ldots,y_n|x_1,\ldots,x_n}(\beta)}{\partial \beta} = \frac{1}{n}\sum_{i=1}^{n} (y_i - e^{\mathbf{x}_i'\beta})\mathbf{x}_i = \mathbf{0}$$

■

Asymptotics
Refer to **Theorem 5.14** to see that the MLE is consistent, asymptotically normal and efficient.

Proposition 6.16. The asymptotic variance can be consistently estimated by the $K \times K$ matrix

$$\widehat{\mathrm{AVar}}(\hat{\beta}) = \left\{\sum_{i=1}^{n} \exp(\mathbf{x}_i'\hat{\beta})\mathbf{x}_i\mathbf{x}_i'\right\}^{-1}$$

Inferences on a single parameter β_k can be conducted using $\widehat{\mathrm{AVar}}(\hat{\beta})$.

Proof. To be followed ■

Inference
We can use

- Wald
- Likelihood ratio
- Lagrange multiplier

tests to conduct inferences in a general framework.

7 Appendix

7.1 Basic Distribution Theory

Discrete

Definition 7.1 (Bernoulli Distribution). An r.v. X is said to have the Bernoulli distribution with parameter p if $\mathrm{P}(X = 1) = p$ and $\mathrm{P}(X = 0) = 1 - p$, where $0 < p < 1$. We write this as $X \sim \mathrm{Bern}(p)$.

Definition 7.2 (Binomial Distribution). Suppose n independent Bernoulli trials are performed, each with the same success probability p. Let X be the number of successes. The distribution of X is called the Binomial distribution with parameters n and p. We write $X \sim \mathrm{Bin}(n, p)$ to mean that X has the Binomial distribution with parameters n and p where n is a positive integer and $0 < p < 1$

Theorem 7.3 (Binomial PMF, mean and variance). If $X \sim \mathrm{Bin}(n, p)$, then the PMF of X is

$$\mathrm{P}(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

for $k = 0, 1, \dots, n$ (and $\mathrm{P}(X = k) = 0$ otherwise).
Moments:

- $\mathrm{E}[X] = p$
- $\mathrm{Var}(X) = np(1 - p)$

Remark. The Bernoulli distribution is a special case of the Binomial where $n = 1$ and $k \leq 1$

Definition 7.4 (Poisson Distribution). An r.v. X has the Poisson distribution with parameter λ if the PMF of X is

$$\mathrm{P}(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}, \quad k = 0, 1, 2, \dots$$

We write this as $X \sim \mathrm{Pois}(\lambda)$

Theorem 7.5 (Poisson Dist Moments). Let $X \sim \mathrm{Pois}(\lambda)$. Then,

- $\mathrm{E}[X] = \lambda$
- $\mathrm{Var}(X) = \lambda$

We usually use the Poisson distribution in situations where we are counting the number of successes in a particular region or interval of time and there are a large number of trials, each with a small probability of success. λ is interpreted as the rate of occurrence of these rare events.

Theorem 7.6 (Sum of Independent Poissons). If $X \sim \mathrm{Pois}(\lambda_1)$ and $Y \sim \mathrm{Pois}(\lambda_2)$ and X is independent of Y, then $X + Y \sim \mathrm{Pois}(\lambda_1 + \lambda_2)$

Theorem 7.7 (Poisson given a sum of Poissons). If $X \sim \mathrm{Pois}(\lambda_1)$ and $Y \sim \mathrm{Pois}(\lambda_2)$ and X is independent of Y, then the conditional distribution of X given $X + Y = n$ is $\mathrm{Bin}(n, \lambda_1/(\lambda_1 + \lambda_2))$

Definition 7.8 (Multinomial Distribution). Each of n objects is independently placed into one of k categories. An object is placed into category j with probability p_j, where the p_j are nonnegative and $\sum_{j=1}^{k} p_j = 1$. Let X_1 be the number of objects in category 1, X_2 the number of objects in category 2 etc., so that $X_1 + \dots + X_k = n$. Then let $\mathbf{X} = (X_1, \dots, X_k)$ is said to have the Multinomial distribution with parameters n and $\mathbf{p} = (p_1, \dots, p_k)$. We write this as $\mathbf{X} \sim \mathrm{Mult}_k(n, \mathbf{p})$

Theorem 7.9 (Multinomial joint and marginal PMFs, moments). If $\mathbf{X} \sim \mathrm{Mult}_k(n, \mathbf{p})$, then the joint PMF of $\mathbf{X}$ is

$$\mathrm{P}(X_1 = n_1, \dots, X_k = n_k) = \frac{n!}{n_1! n_2! \dots n_k!} \cdot p_1^{n_1} p_2^{n_2} \dots p_k^{n_k}$$

for $n_1, \dots, n_k$ satisfying $n_1 + n_2 + \dots + n_k = n$.
Furthermore, $X_j \sim \mathrm{Bin}(n, p_j)$.
The moments are

- $\mathrm{E}[X_j] = np_j$, for a particular outcome X_j
- $\mathrm{Var}(X_j) = np_j(1-p_j)$
- $\mathrm{Cov}(X_j, X_i) = -np_ip_j$ for $i \neq j$

Theorem 7.10 (Multinomial Properties). We have two properties:

- Multinomial Lumping: If $\mathbf{X} \sim \mathrm{Mult}_k(n, \mathbf{p})$, then for any distinct i and j, $X_i + X_j \sim \mathrm{Bin}(n, p_i + p_j)$. The random vector of counts obtained from merging categories i and j is still Multinomial (but we must update the probabilities by taking p_j divided by the total probabilities after removing the category excluded)
- Multinomial Conditioning: If $\mathbf{X} \sim \mathrm{Mult}_k(n, \mathbf{p})$, then

$$(X_2, \ldots, X_k)|X_1 = n_1 \sim \mathrm{Mult}_{k-1}(n - n_1, (p_2', \cdots, p_k')$$

where $p_j' = p_j/(p_2 + \cdots + p_k)$

Continuous

Definition 7.11 (Uniform Distribution). A continuous r.v. U is said to have the Uniform Distribution on the interval (a, b) if its PDF is

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{if } a < x < b \\ 0 & \text{otherwise} \end{cases}$$

We denote this by $U \sim \mathrm{Unif}(a, b)$

Theorem 7.12 (Moments for Uniform Distribution). For $U \sim \mathrm{Unif}(a, b)$, we have

- $\mathrm{E}(U) = \frac{a+b}{2}$
- $\mathrm{Var}(U) = \frac{(b-a)^2}{12}$

Definition 7.13 (Normal Distribution). A continuous r.v. Z is said to have the standard Normal distribution if its PDF φ is given by

$$\varphi(z) = \frac{1}{\sqrt{2\prod}} e^{-z^2/2}, \quad -\infty < z < \infty$$

We denote the CDF of Z as $\Phi(z)$. If $Z \sim \mathcal{N}(0, 1)$, then X is said to have the normal distribution, denoted $X \sim \mathcal{N}(\mu, \sigma)$ if $X = \mu + \sigma Z$

Theorem 7.14 (Properties of the Normal Distribution). We consider properties for the standard Normal for ease of notation, but it applies to all Normal distributions

- Symmetry of PDF: φ satisfies $\varphi(z) = \varphi(-z)$
- Symmetry of tail areas: For the CDF Φ and for all z,

$$\Phi(z) = 1 - \Phi(-z)$$

- Symmetry of Z and $-Z$: if $Z \sim \mathcal{N}(0, 1)$, then $-Z \sim \mathcal{N}(0, 1)$ as well. Similarly, if $X \sim \mathcal{N}(\mu, \sigma^2)$, then $-X \sim \mathcal{N}(-\mu, \sigma^2)$

Definition 7.15 (Exponential Distribution). A continuous r.v. X is said to have the Exponential distribution with parameter λ if its PDF is

$$f(x) = \lambda e^{-\lambda x}, \quad x > 0$$

We denote this by $X \sim \mathrm{Expo}(\lambda)$. The corresponding CDF is

$$F(x) = 1 - e^{-\lambda x}. \quad x > 0$$

Note that $Y \sim \text{Expo}(\lambda)$ iff $\lambda Y \sim \text{Expo}(1)$.

Theorem 7.16 (Moments for Exponential Distribution). For $X \sim \text{Expo}(1)$,

- $\text{E}(X) = 1$
- $\text{Var}(X) = 1$

For $Y = X/\lambda \sim \text{Expo}(\lambda)$,

- $\text{E}(Y) = \frac{1}{\lambda}$
- $\text{Var}(Y) = \frac{1}{\lambda^2}$

Theorem 7.17 (Memoryless Property). A distribution is said to have the memoryless property if a random variable X from that distribution satisfies

$$\text{P}(X \geq s + t | X \geq s) = \text{P}(X \geq t)$$

If $X \sim \text{Expo}(\lambda)$, then X has this property.

Definition 7.18 (Multivariate Normal Distribution). A random vector $\mathbf{X} = (X_1, \ldots, X_k)$ is said to have a Multivariate Normal distribution if every linear combination of X_j has a Normal distribution. That is, we require $t_1 X_1 + \cdots + t_k X_k$ to have a Normal distribution for any choice of constants $t_1, \ldots, t_k$. If $t_1 X_1 + \cdots + t_k X_k$ is a constant (such as when $t_i = 0$ for all i), we still consider it to have a Normal distribution, albeit a degenerate one with variance 0. We write $\mathbf{X} \sim \text{MVN}(\mu, \Sigma)$ where

- μ is the mean vector $\mu = (\mu_1, \ldots, \mu_k)$ where $\text{E}(X_j) = \mu_j$
- the covariance matrix Σ which is the $k \times k$ matrix of covariances between components. The diagonal entries are precisely $\text{Var}(X_j)$ and the off-diagonal entries are $\text{Cov}(X_i, X_j)$ for any ij-th entry where $i \neq j$.

The general form PDF is

$$f(\mathbf{X}) = (2\pi)^{-n/2} |\mathbf{\Sigma}|^{-1/2} e^{(-1/2)(\mathbf{X} - \mu' \mathbf{\Sigma}^{-1}(\mathbf{X} - \mu)}$$

If $(X_1, \ldots, X_k)$ is MVN, then the marginal distribution of X_j is Normal since we can take $t_j = 1$ and all else to be 0.

Theorem 7.19 (Properties of Multivariate Normal). We have the following properties:

- If $(X_1, \ldots, X_k)$ is a Multivariate Normal, then so is any subvector
- If $\mathbf{X} = (X_1, \ldots, X_n)$ and $\mathbf{Y} = (Y_1, \ldots, Y_m)$ are MVN vectors with $\mathbf{X}$ independent of $\mathbf{Y}$, then the concatenated random vector $\mathbf{W} = (X_1, \ldots, X_n, Y_1, \ldots, Y_m)$ is Multivariate Normal
- Within an MVN random vector, uncorrelatedness implies independence. That is, if $\mathbf{X} \sim \text{MVN}(\mu, \Sigma)$ can be written as $\mathbf{X} = (\mathbf{X}_1, \mathbf{X}_2)$, where $\mathbf{X}_1$ and $\mathbf{X}_2$ are subvectors, and every component of $\mathbf{X}_1$ is uncorrelated with every component of $\mathbf{X}_2$, then $\mathbf{X}_1$ and $\mathbf{X}_2$ are independent

Definition 7.20 (χ^2_n distribution). Let $V = Z_1^2 + \cdots + Z_n^2$ where $Z_1, \ldots, Z_n$ are i.i.d. $\mathcal{N}(0,1)$. Then V is said to have the χ^2_n distribution (read: Chi-square distribution with n degrees of freedom). We write this as $V \sim \chi^2_n$. Furthermore, consider a multivariate normal $\mathbf{Z} \sim \text{MVN}(\mathbf{0}, \mathbf{V})$. Then, $\mathbf{Z}'\mathbf{V}^{-1}\mathbf{Z} \sim \chi^2_J$

Theorem 7.21 (Property of χ^2_n distribution). Suppose $Z \sim \text{MVN}(\mathbf{0}, \mathbf{I}_n)$. Then, for an nonrandom, idempotent, symmetric $n \times n$ matrix $\mathbf{A}$, we have

$$\mathbf{Z}'\mathbf{A}\mathbf{Z} \sim \chi^2_r$$

where $r = rank(\mathbf{A})$ That is, the quadratic form in $\mathbf{A}$, the degrees of freedom for the χ^2_n distribution is precisely the rank of $\mathbf{A}$. Note that there is a special case

$$\mathbf{Z}'\mathbf{I}_n\mathbf{Z} = \mathbf{Z}'\mathbf{Z} \sim \chi^2_n$$

Proposition 7.22 (Sum of indep. χ^2). Consider $Z_i \sim \text{i.i.d.}\mathcal{N}(0,1)$. Then,

$$\chi_n^2 + \chi_m^2 = \chi_{n+m}^2$$

if they are independent. Furthermore,

$$\chi_n^2 - \chi_m^2 = \chi_{n-m}^2$$

for $n > m$

Definition 7.23 (Student-t distribution). Let $V \sim \chi_n^2$ and $Z \sim \mathcal{N}(0,1)$ and Z is independent of V. Let

$$T = \frac{Z}{\sqrt{V/n}}$$

T is said to have the Student-t distribution with n degrees of freedom. We denote this as $T \sim t_n$.

Theorem 7.24 (Properties of Student-t). The Student-t distribution has the following properties:

- Symmetry: if $T \sim t_n$, then $-T \sim t_n$
- Cauchy as special case: The t_1 distribution is the same as the Cauchy distribution
- Convergence to Normal: as $n \to \infty$, the t_n distribution approaches the standard Normal Distribution

7.2 Basic Probability Theory

Theorem 7.25 (Markov's Inequality). For any r.v. X and constant $a > 0$,

$$\mathrm{P}(|X| \geq a) \leq \frac{\mathrm{E}|X|}{a}$$

Theorem 7.26 (Chebyshev's Inequality). Let X have mean μ and variance σ^2. Then for any $a > 0$,

$$\mathrm{P}(|X - \mu| \geq a) \leq \frac{\sigma^2}{a^2}$$

Theorem 7.27 (Cauchy Schwarz Inequality). For any r.v.s X and Y with finite variances

$$|\mathrm{E}(XY)| \leq \sqrt{\mathrm{E}(X^2)\mathrm{E}(Y^2)}$$

Theorem 7.28 (Jensen's Inequality). Let X be a random variable. If g is a convex function, then $\mathrm{E}(g(X)) \leq g(\mathrm{E}(X))$. If g is a concave function, then $\mathrm{E}(g(X)) \leq g(\mathrm{E}(X))$. In both cases, the only way that equality can hold is if there are constants a and b such that $g(X) = a + bX$ with probability 1

Theorem 7.29 (Law of Iterated Expectations). For any r.v.s X, Y

$$\mathrm{E}\{\mathrm{E}[Y|X]\} = \mathrm{E}[Y]$$

Definition 7.30 (Conditional Variance). The conditional variance of Y given X is

$$\mathrm{Var}(Y|X) = \mathrm{E}\{(Y - \mathrm{E}[Y|X])^2|X\}$$

or equivalently

$$\mathrm{Var}(Y|X) = \mathrm{E}(Y^2|X) - (\mathrm{E}(Y|X))^2$$

Theorem 7.31 (Eve's Law). For any r.v.s. X, Y

$$\mathrm{Var}(Y) = \mathrm{E}(\mathrm{Var}(Y|X)) + \mathrm{Var}(\mathrm{E}(Y|X))$$

7.3 Random Proofs